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Measurement of inclusive $B \rightarrow \Lambda_c$ branching fractions using Belle data and hadronic Full Event Interpretation

Leonardo Benjamin Rizzuto¹.

¹*Institute Jožef Stefan, Ljubljana, Slovenia*

Abstract

Inclusive $B \rightarrow \Lambda_c$ branching fractions were measured most recently by BaBar collaboration. However, the measurement still presented a poor accuracy. A more precise measurement of inclusive $B \rightarrow \Lambda_c$ branching fraction could be useful to gain a better confidence on B meson weak decays treatment. With help of the Full Event Interpretation algorithm, it is possible to perform a more precise measurement of inclusive $B \rightarrow \Lambda_c$ branching fractions using Belle data set.

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Chapter 1

Introduction

Inclusive B meson baryonic decays with a Λ_c baryon in the final state are the most abundant, due to a relatively large V_{cb} element of the CKM matrix. The *BaBar* experiment measured their branching fractions to be around the percent level (see ref. [1]). However, the branching fractions were determined with big uncertainties: nearly 50% on the measured values or, in the case of the $B^0 \rightarrow \Lambda_c^+$ decay, only an upper limit could be established. A more precise measurement of inclusive $B \rightarrow \Lambda_c$ branching fractions may shed light on the appropriateness of B meson weak decays treatment, particularly of strong interaction effects modelling. Predictions for inclusive branching fractions are given, for example, in ref. [2] or in [3] for $B \rightarrow \Lambda_c p$ decays.

Exploiting the Full Event Interpretation (FEI) algorithm, developed for the Belle II experiment, it may be possible to perform a more precise measurement of inclusive $B \rightarrow \Lambda_c$ branching fractions, using the full Belle data set. A more precise measurement may also trigger further research on currently scarce theory predictions for B meson decays to charm baryons.

1.1 Analysis Setup

The reconstruction is performed with BASF2 release 05-02-03 together with the `b2bii` package in order to convert the *Belle* MDST files (BASF data format) to *Belle II* MDST files (BASF2 data format). The FEI version used is `FEI_B2BII_light-2012-minos`.

1.2 Datasets

The Belle detector acquired a dataset of about $L_0 \approx 710 fb^{-1}$ of integrated luminosity in its lifetime at the $\Upsilon(4S)$ energy of 10.58 GeV, which corresponds to about $771 \times 10^6 B\bar{B}$ meson pairs. Additionally, several streams of Monte-Carlo (MC) samples were produced, where each stream of MC corresponds to the same amount of data that was taken with the detector. No specific signal MC was used: instead of producing dedicated signal MC

27 samples, the samples were obtained by filtering the decays of interest from the generic
28 on-resonance MC samples. The following samples were used in this analysis:

- 29 • data
- 30 • MC - 6 streams of B^+B^- and $B^0\bar{B}^0$ (denoted as **charged** and **mixed**) for signal
31 decays and backgrounds (if more of the in total existing 10 streams is used it is
32 explicitly specified throughout this note).
 - 33 - 6 streams of $q\bar{q}$ produced at $\Upsilon(4S)$ resonance energy
 - 34 - 6 streams of $q\bar{q}$ produced at 60 MeV below $\Upsilon(4S)$ resonance energy, where each
35 stream corresponds to $1/10 \times L_0$.

36

Chapter 2

Event selection and reconstruction

In this chapter the procedure for reconstruction of the events where one B meson decays inclusively to a Λ_c baryon and the accompanying B meson decays hadronically is illustrated.

2.1 B_{tag} reconstruction

The FEI is an exclusive tagging algorithm that uses machine learning to reconstruct B meson decay chains and calculates the probability that these decay chains correctly describe the true process. In this analysis only hadronically reconstructed decay chains are considered. The training called `FEI_B2BII_light-2012-minos` is used. Tag-side B meson candidates are required to have a beam-constrained mass greater than $5.22 \text{ GeV}/c^2$ and $-0.15 < \Delta E < 0.07 \text{ GeV}$.

In the case of multiple candidates in the same event, the candidate with the highest SignalProbability (the signal probability calculated by FEI using FastBDT) is chosen. To suppress the background consisting of B^0 events misreconstructed as B^+ (and vice-versa) from neutral (charged) decays also a B^0 (B^+) candidate is reconstructed with FEI and if its SignalProbability is higher than the charged (neutral) reconstructed B meson, the event is discarded. This constitutes a sort of crossfeed-veto, rejecting part of events belonging to the other typology of decays of interest: for example in the case one is interested in reconstructing $B^{+/-}$ decays and the event actually contains $B^0/\bar{B}^0$ decays, the FEI reconstructed neutral B meson candidate most likely presents a higher SignalProbability than the charged FEI reconstructed candidate.

2.2 Λ_c reconstruction

In the *rest of event* (ROE) of the reconstructed B_{tag} meson, to select $\Lambda_c \rightarrow pK\pi$ signal candidates, the following event selection criteria are applied (same PID cuts were used for example in the Belle Note 1521 https://belle.kek.jp/secured/belle_note/gn1521/BN_v1.pdf). Charged tracks with the impact parameters perpendicular to and along the nominal interaction point (IP) are required to be less than 2 cm and 4 cm respectively

64 ($dr < 2$ cm and $|dz| < 4$ cm).
 65 The pion tracks are required to be identified with $\frac{\mathcal{L}_\pi}{\mathcal{L}_K + \mathcal{L}_\pi} > 0.6$. The kaon tracks are
 66 required to be identified with $\frac{\mathcal{L}_K}{\mathcal{L}_K + \mathcal{L}_\pi} > 0.6$, and the proton/anti-proton tracks are
 67 required to be identified with $\frac{\mathcal{L}_{p/\bar{p}}}{\mathcal{L}_K + \mathcal{L}_{p/\bar{p}}} > 0.6$ and $\frac{\mathcal{L}_{p/\bar{p}}}{\mathcal{L}_\pi + \mathcal{L}_{p/\bar{p}}} > 0.6$, where the $\mathcal{L}_{\pi,K,p/\bar{p}}$ are the
 68 likelihoods for pion, kaon, proton/anti-proton, respectively, determined using the ratio of
 69 the energy deposit in the ECL to the momentum measured in the SVD and CDC, the
 70 shower shape in the ECL, the matching between the position of charged track trajectory
 71 and the cluster position in the ECL, the hit information from the ACC and the dE/dx
 72 information in the CDC.
 73 For the Λ_c candidates a vertex fit is performed with **TreeFitter**, requiring it to converge.
 74 If there are more than one Λ_c combination, then the best candidate based on the χ^2
 75 probability is chosen. The Λ_c signal region is defined to be $|M_{\Lambda_c} - m_{\Lambda_c}| < 20$ MeV/ c^2 ($\sim$
 76 3σ), here m_{Λ_c} is the nominal mass of m_{Λ_c} .
 77

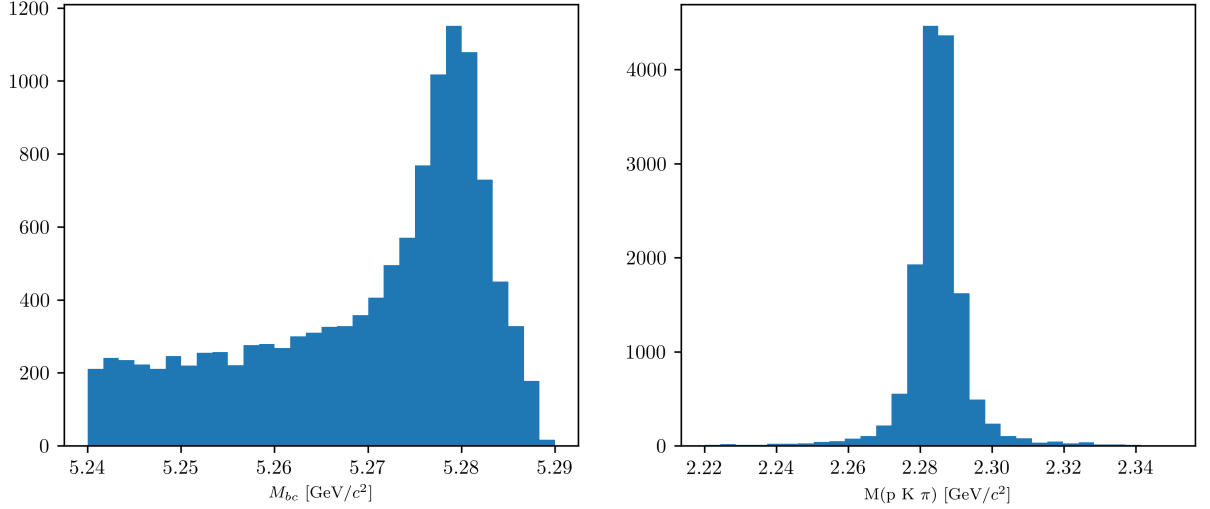


Figure (2.1) M_{bc} and $M(pK\pi)$ distributions of B_{tag} and Λ_c candidates reconstructed in the signal sample.

78 **2.3 Wrongly reconstructed B_{tag} candidates**

79 In the case of the signal sample the distributions for the beam-constrained mass M_{bc} and
 80 for the correctly reconstructed Λ_c candidates, look like in Fig. 2.1. If one then investigates
 81 the M_{bc} distribution of the B_{tag} candidates reconstructed with FEI, it can be seen that
 82 there is a peaking structure for wrongly reconstructed B mesons (as in Fig. 2.2), according
 83 to the BASF2 internal truth matching variable **isSignal**. It is obvious from this that the
 84 BASF2 internal truth matching variable cannot be used to separate properly the signal
 85 events in correctly and wrongly reconstructed B mesons. In the study BELLE2-NOTE-TE-

2021-026 <https://docs.belle2.org/record/2711/files/BELLE2-NOTE-TE-2021-026.pdf> a possible solution was found developing new variables that can be used for an improved truth matching for the FEI (those variables were added to a newer BASF2 release than the one used for this study). In the present study instead a more "traditional" approach was adopted: fitting the M_{bc} distribution with a sum of PDFs that account for the flat (background) component and the peaking (signal) component. The first component represents the combinatorial background, i.e. B mesons that were mis-reconstructed, and therefore those events are denoted from now on as "**misreconstructed signal**". The peaking component represents the correctly reconstructed signal events in M_{bc} and therefore denoted from now on as "**reconstructed signal**". Only the second one is then considered for the signal yield, while the first is counted as a background. To validate this method a control decay study was performed on the flavor correlated $B^+ \rightarrow \bar{D}^0$ channel.

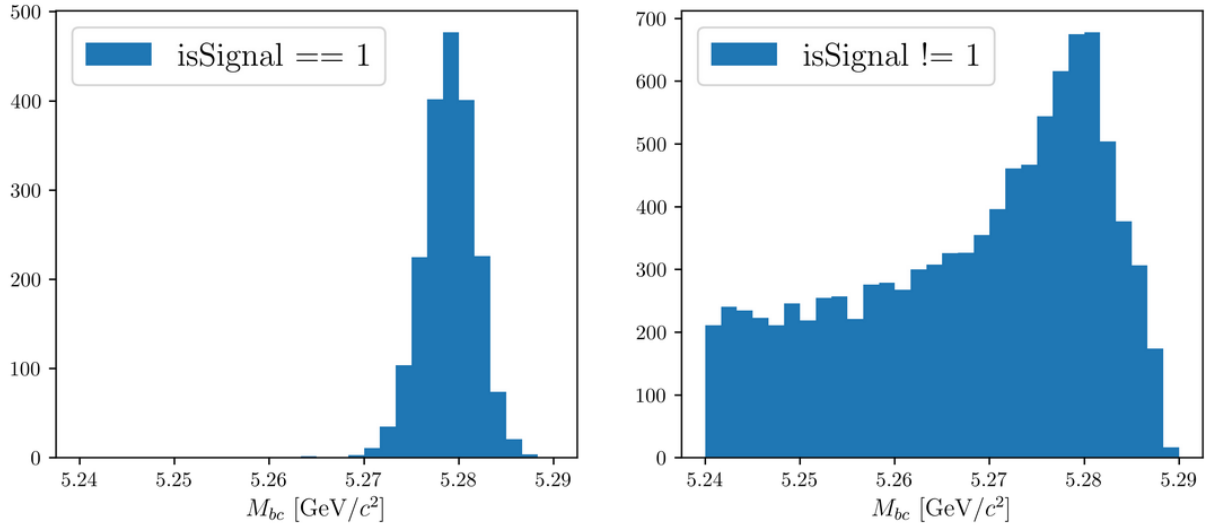


Figure (2.2) M_{bc} distribution of B_{tag} candidates reconstructed in the signal sample, truth-matched (on the left) and not (on the right).

2.4 Signal selection optimization

To further enhance the purity of the signal decays, an optimization procedure is adopted to determine optimal cuts for a set of variables for each decay mode under investigation by this study. The cuts on the following variables are optimized:

- *foxWolframR2*: the event based ratio of the 2-nd to the 0-th order Fox-Wolfram moments
- *SignalProbability*: the already mentioned signal probability calculated by FEI using FastBDT
- $p_{CMS}^{\Lambda_c}$: momentum of the Λ_c candidates in the center of mass system

The optimization is based on the Figure Of Merit (FOM): $FOM = \frac{S}{\sqrt{S+B}}$

Where S and B are respectively signal and background events in the signal region: $M_{bc} > 5.27 \text{ GeV}/c^2$, $2.2665 < M(pK\pi) < 2.3065 \text{ GeV}/c^2$.

Due to the issue reported in Sec. ??, to separate signal events that peak in M_{bc} from the ones that are not (which are then categorized as background events), the events reconstructed in the signal sample are fitted with a sum of Crystal Ball function and Argus for each cut value on the corresponding variable to optimize (as in Fig. 2.3).

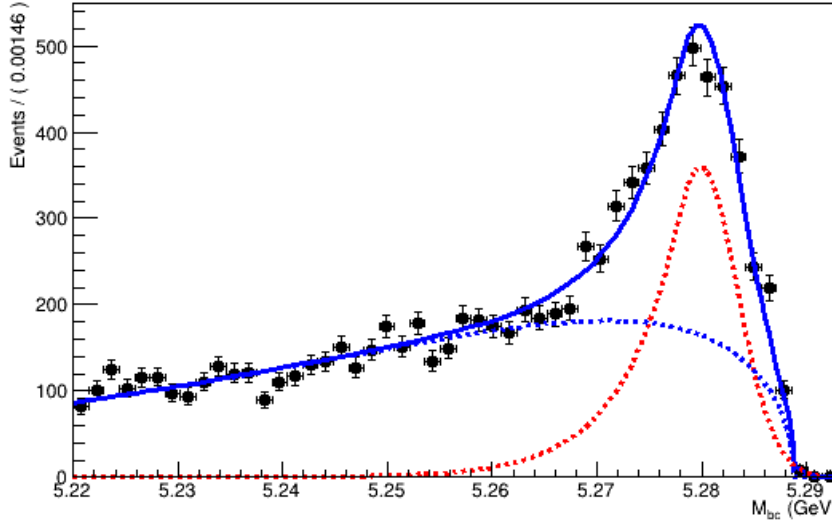


Figure (2.3) Example of a fit used to separate the correctly reconstructed B mesons (described by the red dotted Crystal Ball function) from the wrongly reconstructed ones (described by the blue dotted Argus function).

The next sections illustrate the procedure for each of the four decay channels.

115 2.4.1 $B^- \rightarrow \Lambda_c^+$ decays

116 Here below the procedure of optimized signal selection of charged correlated decays is
 117 presented.

118 First, in order to suppress the continuum background the cut on *foxWolframR2* is
 119 optimized. Fig. 2.4 shows the *foxWolframR2* distributions for signal and continuum
 120 events.

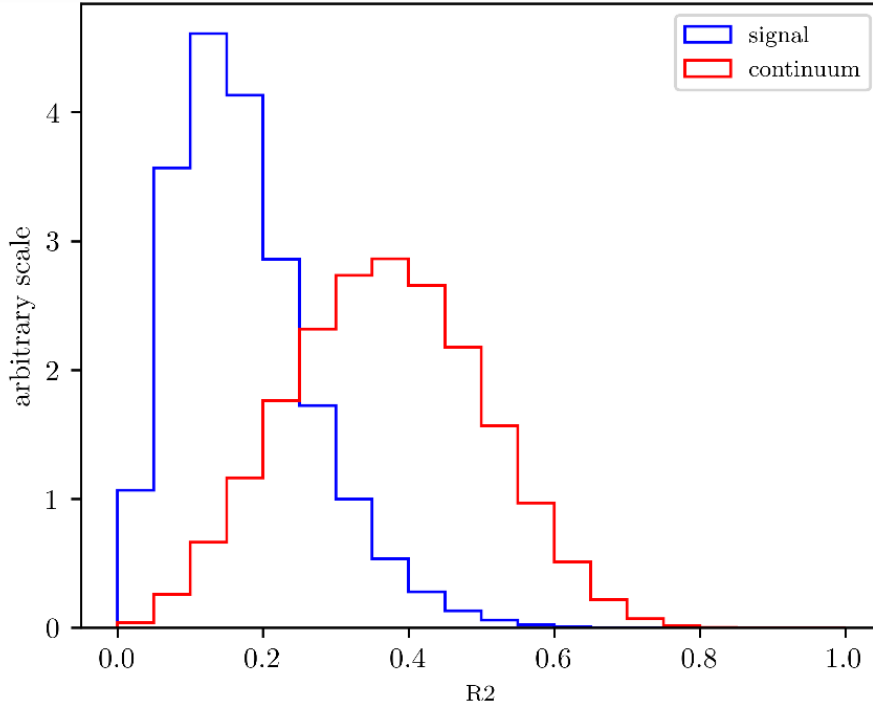


Figure (2.4) Distribution of the *foxWolframR2* variable for signal and continuum background events.

121 With the optimized cut $foxWolframR2 < 0.27$ (corresponding to the maximum of
 122 the FOM curve shown in Fig. 2.5), the cut on SignalProbability is optimized in the same
 123 way (see Fig. 2.6).

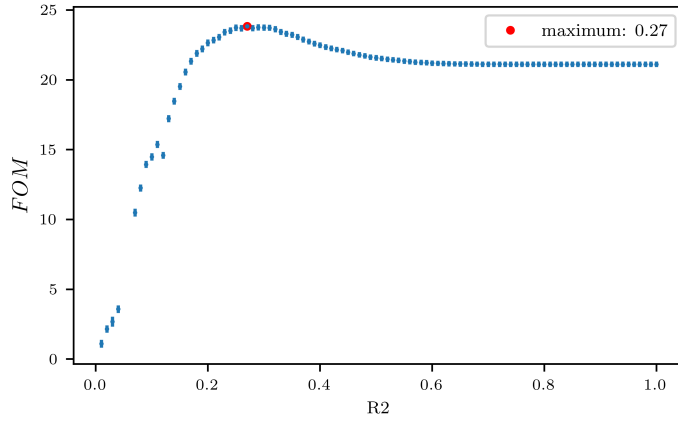


Figure (2.5) Figure of Merit values calculated at several cuts on the *foxWolframR2* variable

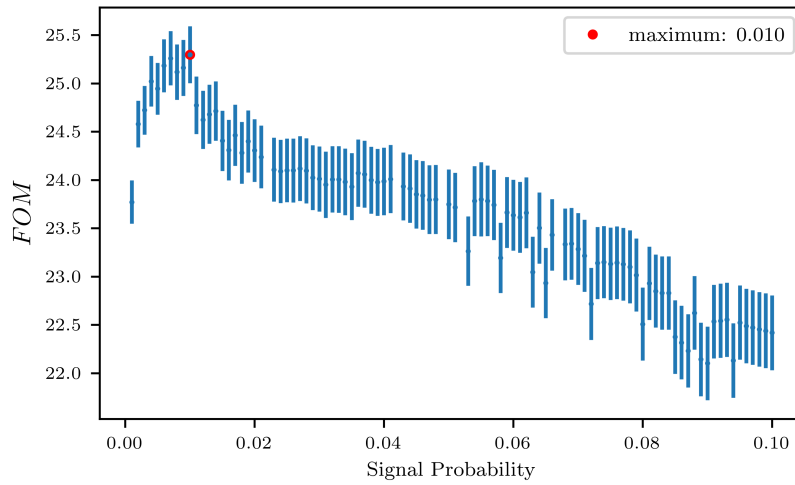


Figure (2.6) Figure of Merit values calculated at several cuts on the SignalProbability variable

124 With the optimized cut $\text{SignalProbability} > 0.01$, the cut on *foxWolframR2* variable is
 125 rechecked (Fig. 2.7). Being the maximum values fluctuating around $\text{foxWolframR2} < 0.3$,
 126 this cut is the one finally chosen for this variable.

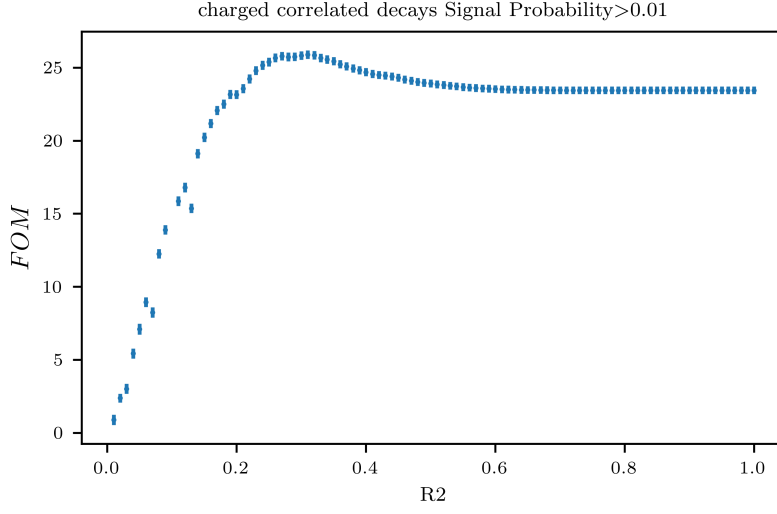


Figure (2.7) Figure of Merit values calculated at several cuts on the *foxWolframR2* variable

127 With the optimized cuts on SignalProbability and *foxWolframR2* variable, the cut
 128 on $p_{CMS}^{\Lambda_c}$ is optimized
 129

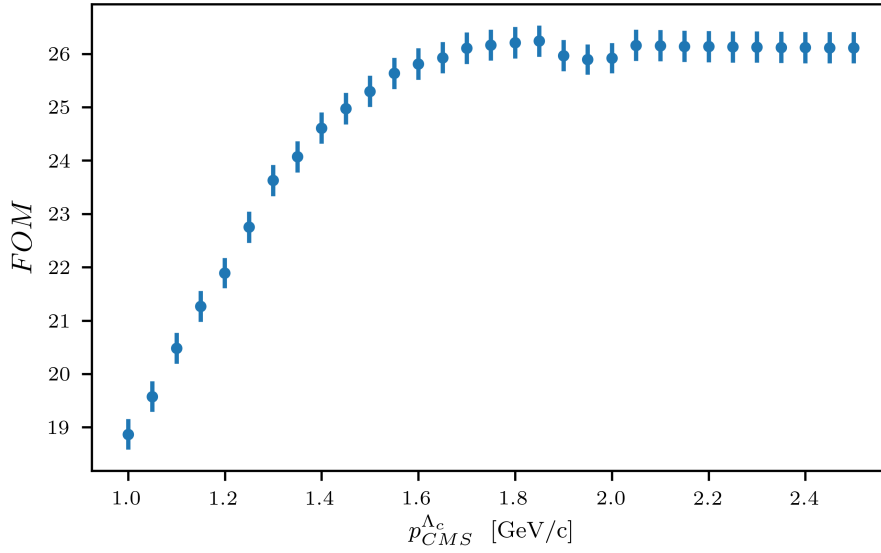


Figure (2.8) Figure of Merit values calculated at several cuts on the momentum of the Λ_c candidates in the center of mass system

130 From Fig. 2.8 one can see that with values of the cut above $p_{CMS}^{\Lambda_c} < 1.8 \text{ GeV}/c^2$ a
 131 plateau of maximum FOM values is reached. But such a cut would still be useful to reject

132 some background events as one can see from Fig. 2.9.
 133 Finally the optimized selection cuts are:

- 134 • $foxWolframR2 < 0.3$
- 135 • $SignalProbability > 0.01$
- 136 • $p_{CMS}^{\Lambda_c} < 1.8 \text{ GeV}/c$

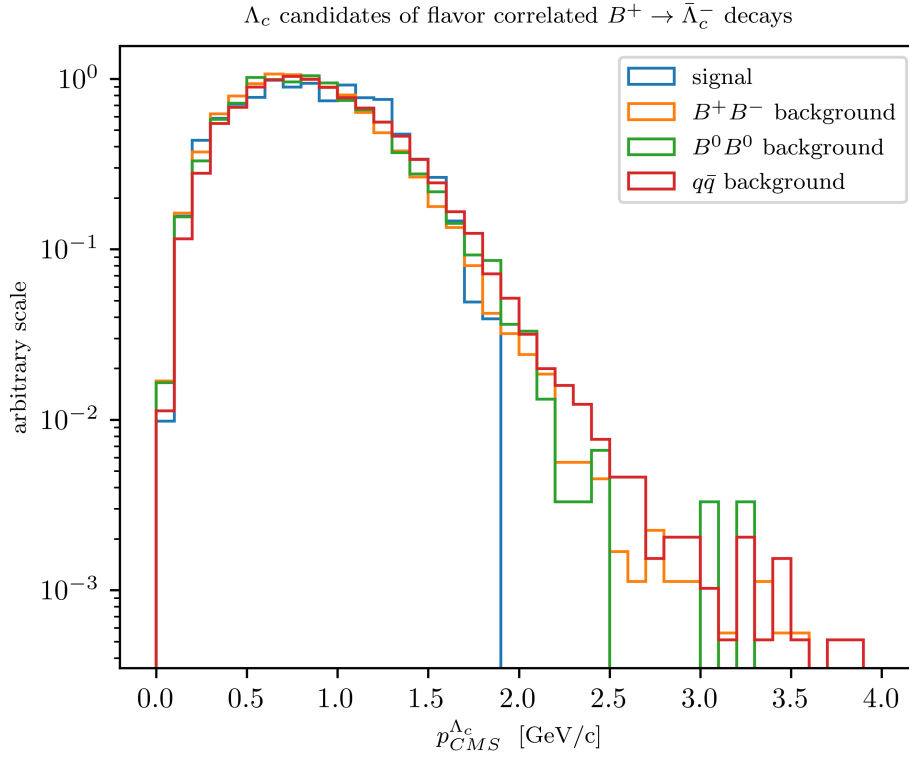


Figure (2.9) Distribution of Λ_c candidates momenta in the center of mass system

137 2.4.2 $B^+ \rightarrow \Lambda_c^+$ decays

138 For anticorrelated decays, the same $foxWolframR2$ cut is defined after the optimization
 139 procedure (see Fig. 2.13).

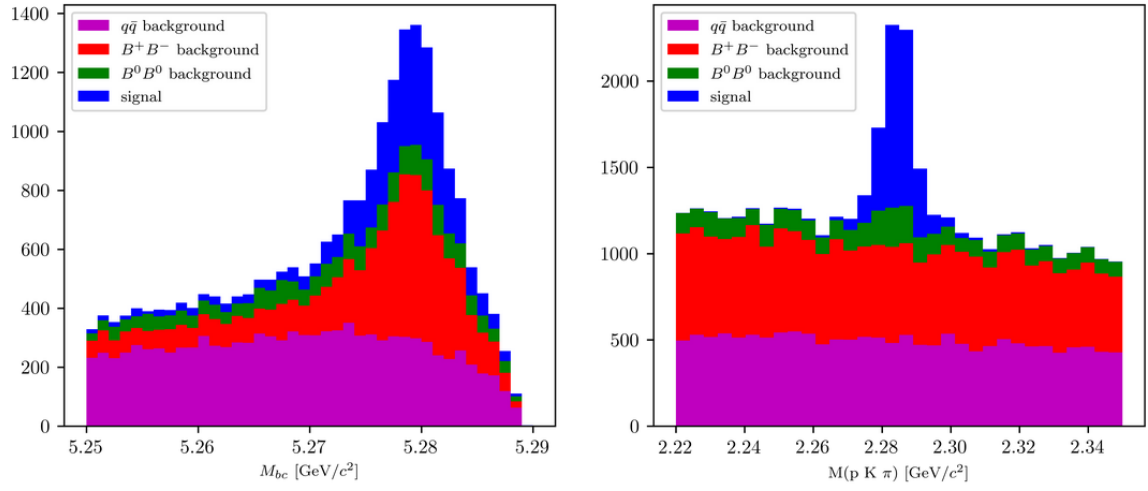


Figure (2.10) Distribution of M_{bc} (left) and invariant mass of charged correlated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

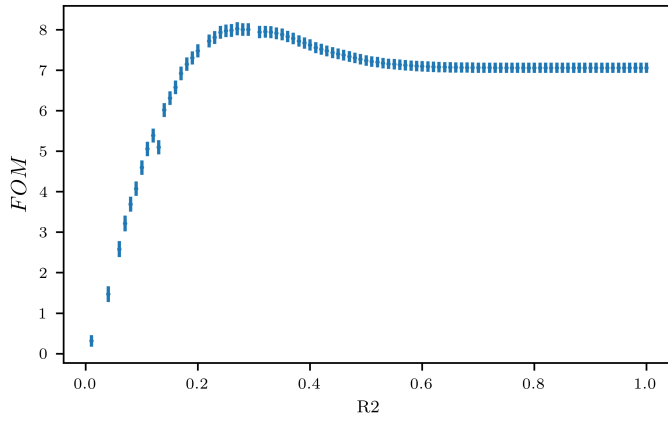


Figure (2.11) Figure of Merit values calculated at several cuts on the *foxWolframR2* variable

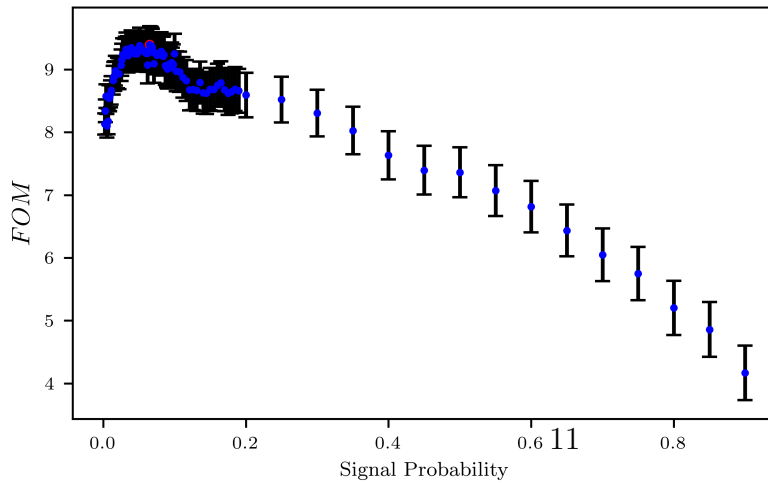


Figure (2.12) Figure of Merit values calculated at several cuts on the SignalProbability variable

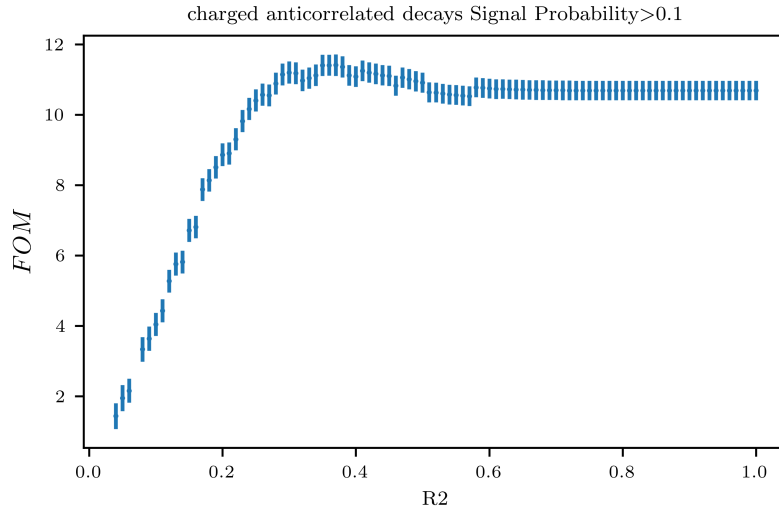


Figure (2.13) Figure of Merit values calculated at several cuts on the *foxWolframR2* variable

140 With the optimized cuts on SignalProbability and *foxWolframR2* variable, the cut
 141 on $p_{CMS}^{\Lambda_c}$ is selected.
 142

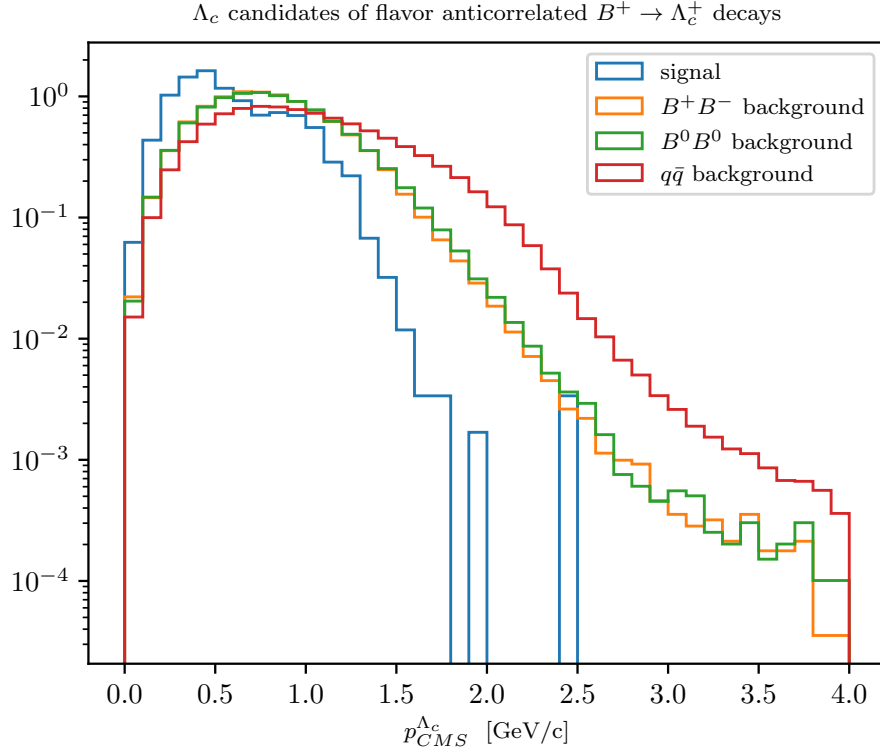


Figure (2.14) Distribution of Λ_c candidates momenta in the center of mass system

143 The final optimized selection cuts are:

- 144 • $foxWolframR2 < 0.3$
- 145 • $SignalProbability > 0.1$
- 146 • $p_{CMS}^{\Lambda_c} < 1.5 \text{ GeV}/c$

147 Fig. 2.15 shows the projections of the M_{bc} and $M(pK\pi)$ distributions.

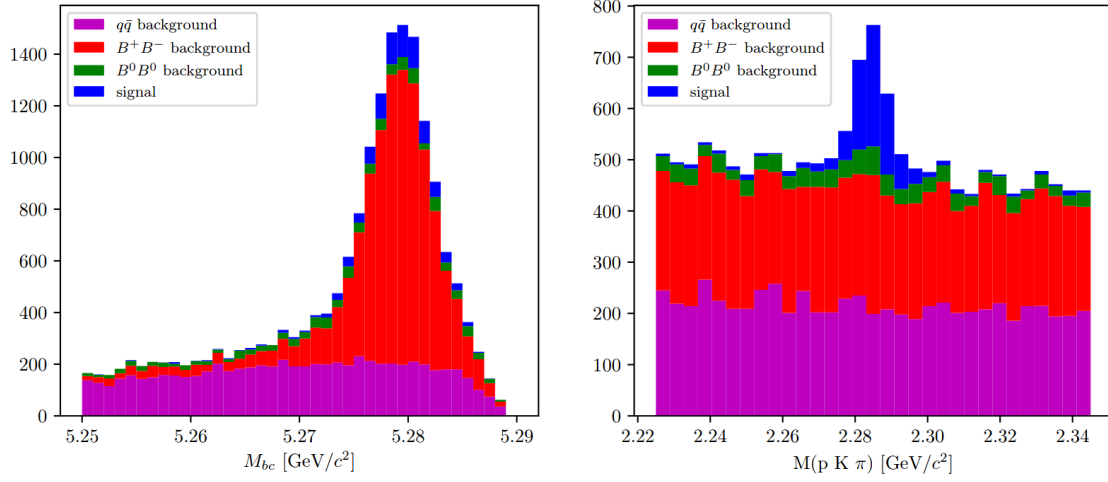


Figure (2.15) Distribution of M_{bc} (left) and invariant mass of charged correlated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

148 2.4.3 $\bar{B}^0 \rightarrow \Lambda_c^+$ decays

149 Also for neutral correlated decays, first the cut on *foxWolframR2* is optimized.

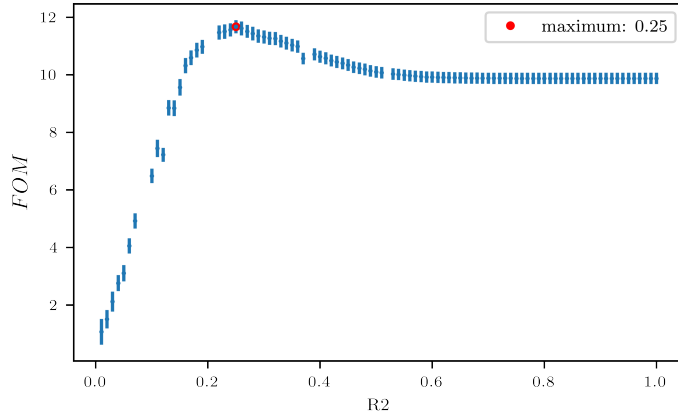


Figure (2.16) Figure of Merit values calculated at several cuts on the *foxWolframR2* variable

150 The cut suggested by Fig. 2.16, $\text{foxWolframR2} < 0.27$, is used to define the cut on
 151 the SignalProbability variable. From Fig. 2.17, it seems the optimal cut maximizing the
 152 *FOM* would be around 0.05. If one zooms like in Fig. 2.18 one can see that there is a
 153 sort of plateau starting around 0.03 and ending after values around 0.11, where the values
 154 fluctuate within statistical uncertainties around $FOM = 13$. In this case, a legitimate
 155 choice is to use the most stringent cut ($\text{SignalProbability} > 0.11$) being the *FOM* same
 156 but rejecting more background events.

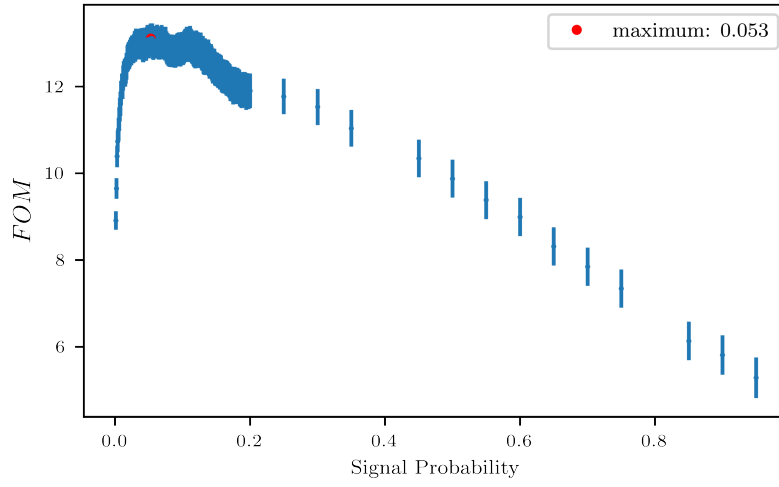


Figure (2.17) Figure of Merit values calculated at several cuts on the SignalProbability variable

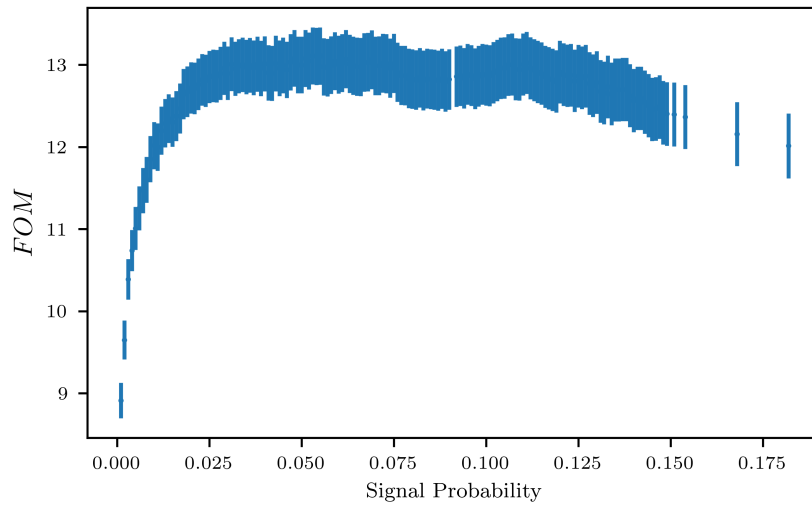


Figure (2.18) Figure of Merit values calculated at several cuts on the SignalProbability variable

157 The *FOM* curve for the *foxWolframR2* variable is rechecked applying the chosen
 158 cut on SignalProbability (Fig. 2.19). As done in the other cases the final cut chosen is
 159 *foxWolframR2* < 0.3

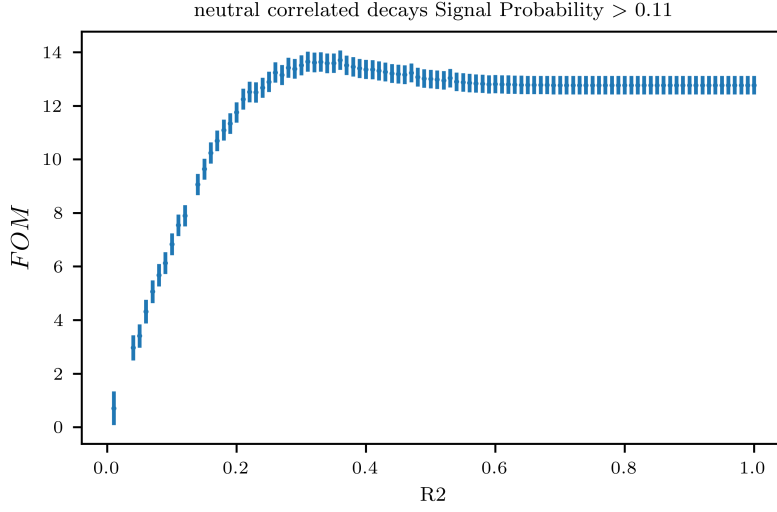


Figure (2.19) Figure of Merit values calculated at several cuts on the *foxWolframR2* variable

160 Using now the two optimized cuts to check the *FOM* curve for the momenta a plateau
 161 appears for cuts starting from values around $p_{CMS}^{\Lambda_c} < 1.7 \text{ GeV}/c$ (see Fig. 2.20). In fact,
 162 around that value the level of background becomes significantly higher compared to the
 163 amount o signal events as one can see in Fig. 2.21

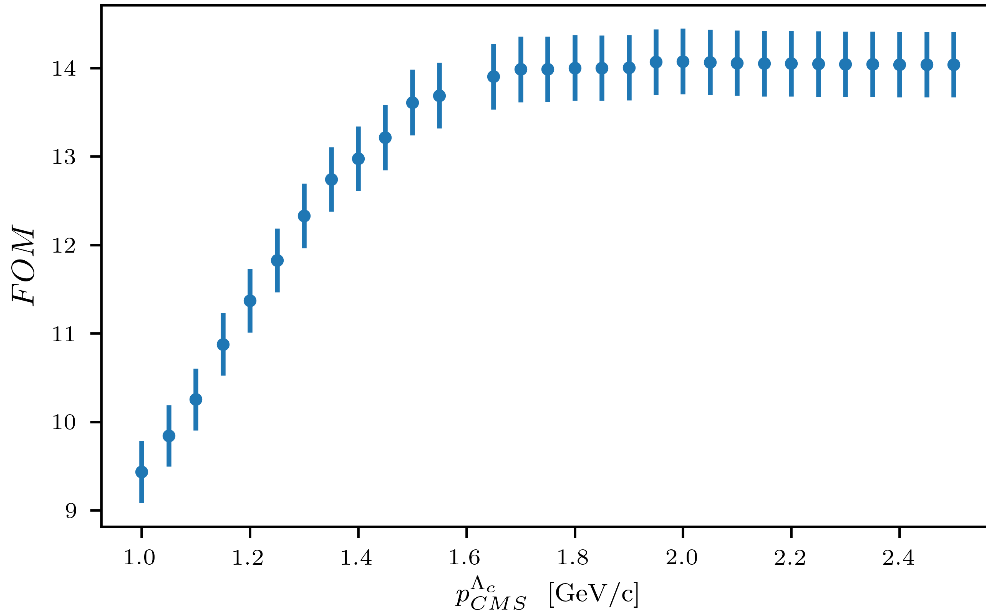


Figure (2.20) Figure of Merit values calculated at several cuts on the momentum of the Λ_c candidates in the center of mass system

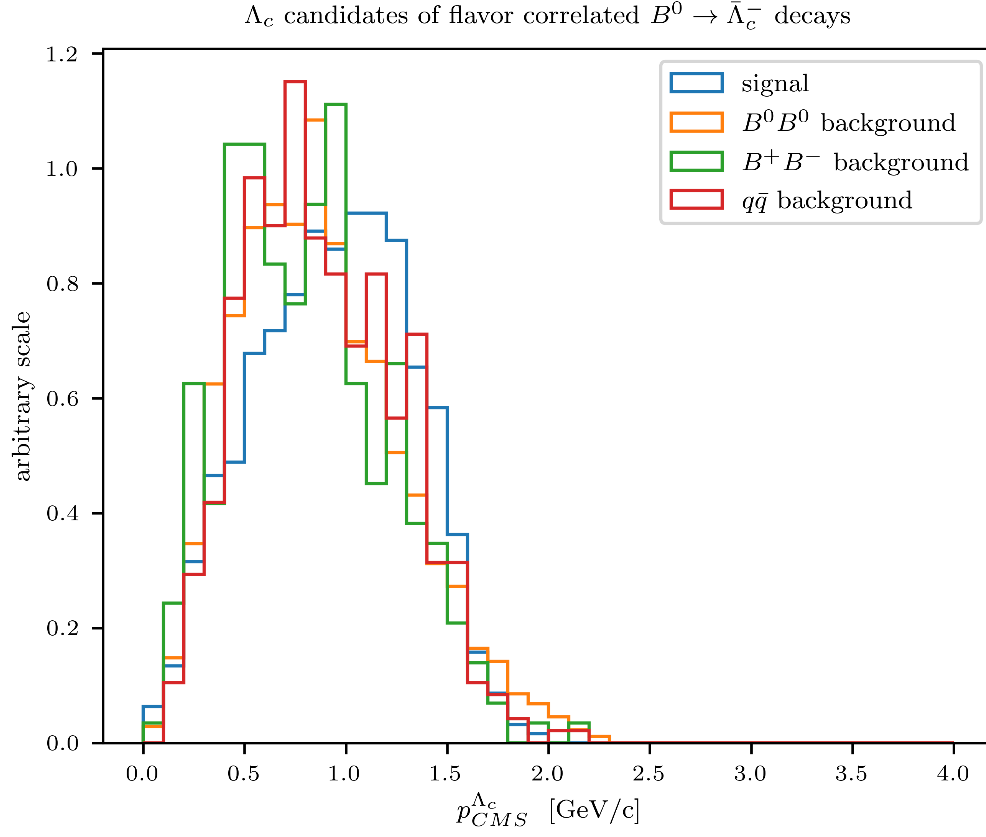


Figure (2.21) Distribution of Λ_c candidates momenta in the center of mass system

164 Fig. 2.22 shows the M_{bc} and $M(pK\pi)$ distributions after applying the following set of
 165 cuts:

- 166 • $foxWolframR2 > 0.3$
- 167 • $SignalProbability > 0.11$
- 168 • $p_{CMS}^{\Lambda_c} < 1.7 \text{ GeV}/c$

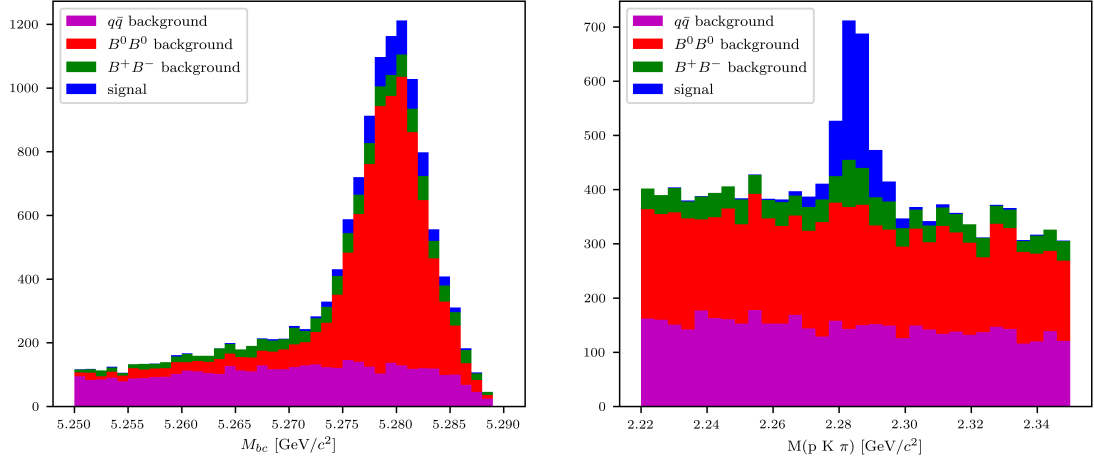


Figure (2.22) Distribution of M_{bc} (left) and invariant mass of neutral correlated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

2.4.4 $B^0 \rightarrow \Lambda_c^+$ decays

Finally same procedure is applied also to the neutral anticorrelated decays. The final selections for the variables of *foxWolframR2*, *SignalProbability* and the momentum of the Λ_c candidates in the center of mass system:

- $foxWolframR2 < 0.3$
- $SignalProbability > 0.15$
- $p_{CMS}^{\Lambda_c} < 1.4 \text{ GeV}/c$

Fig. 2.26 shows the M_{bc} and $M(pK\pi)$ distributions after applying these cuts.

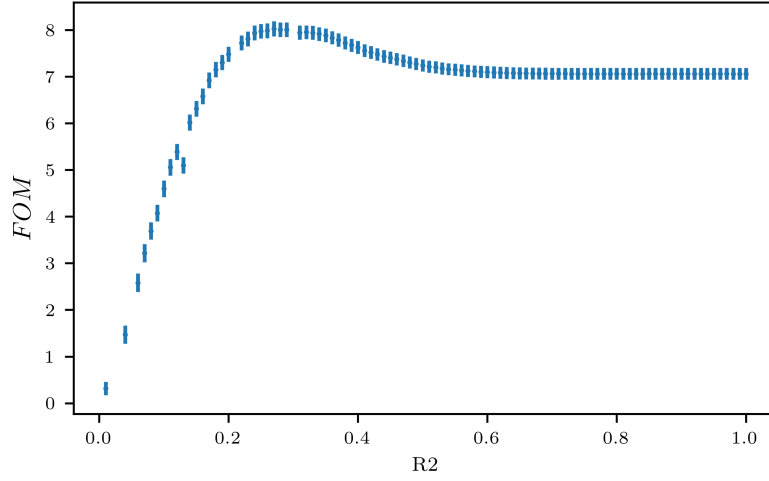


Figure (2.23) Figure of Merit values calculated at several cuts on the *foxWolframR2* variable

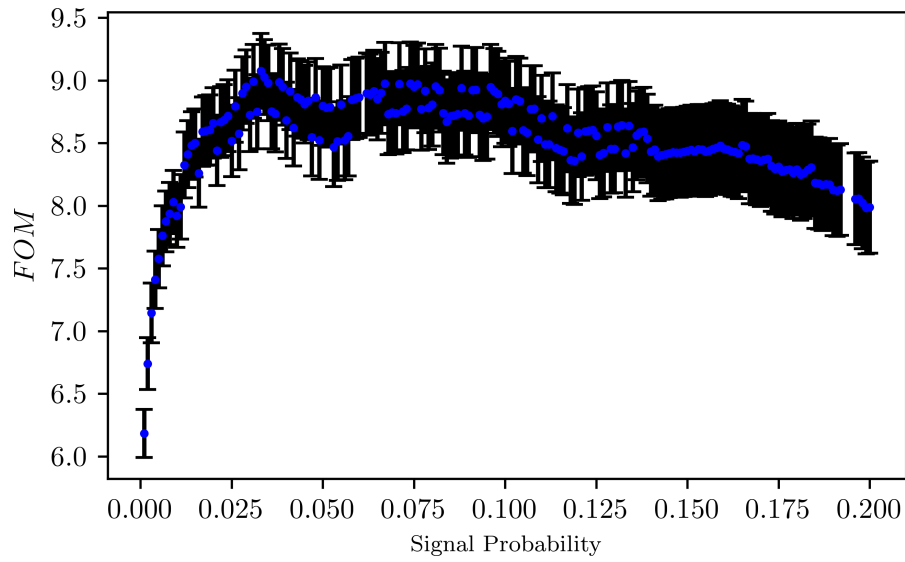


Figure (2.24) Figure of Merit values calculated at several cuts on the SignalProbability variable

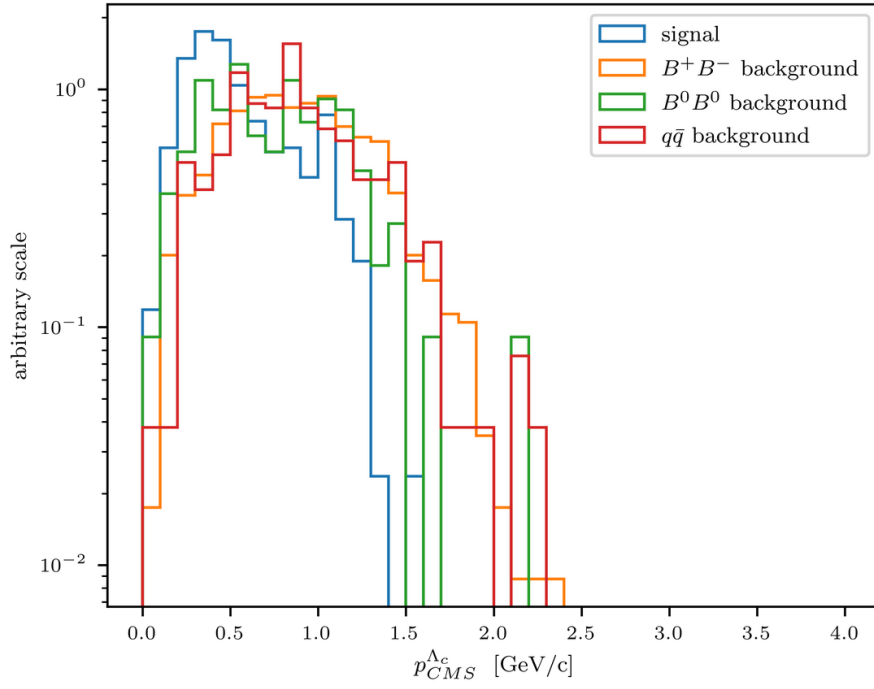


Figure (2.25) Distribution of Λ_c candidates momenta in the center of mass system

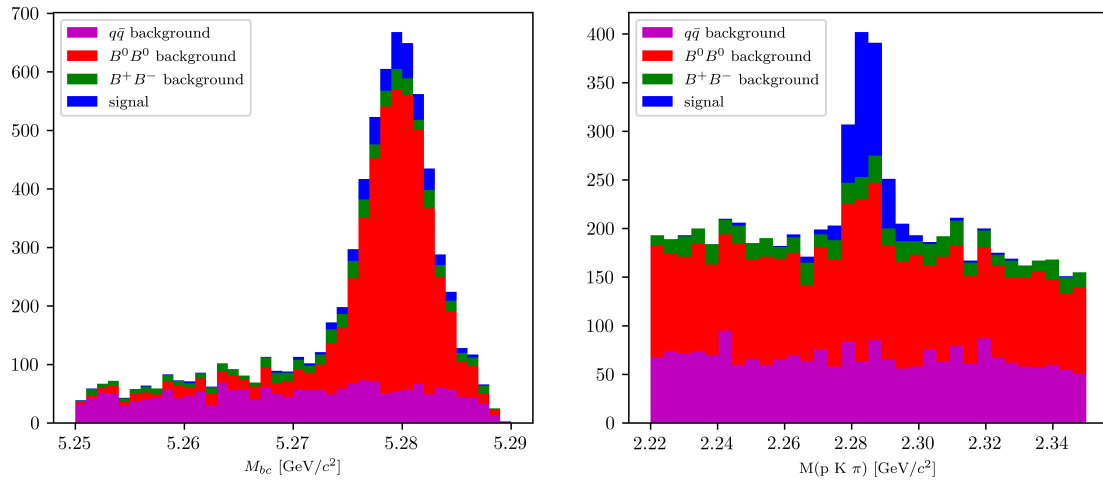


Figure (2.26) Distribution of M_{bc} (left) and invariant mass of neutral anticorrelated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

Chapter 3

2D simultaneous fit

3.1 Probability Density Functions (PDFs) for the two dimensional fit

The reconstructed events can be categorized as follows:

- peaking in both M_{bc} and $M(pK\pi)$
- peaking in M_{bc} but not in $M(pK\pi)$
- peaking in $M(pK\pi)$ but not in M_{bc}
- flat in both M_{bc} and $M(pK\pi)$

The first category is represented by the reconstructed signal: signal events which are correctly reconstructed. The signal events which are misreconstructed fall into the third category. The sum of the two is the so called "total signal". The PDFs used to describe the total signal distributions are discussed first.

3.1.1 Total Signal fits

For all the decays, the final sample of total signal events presents a peak around the expected B meson mass and a tail at low M_{bc} values.

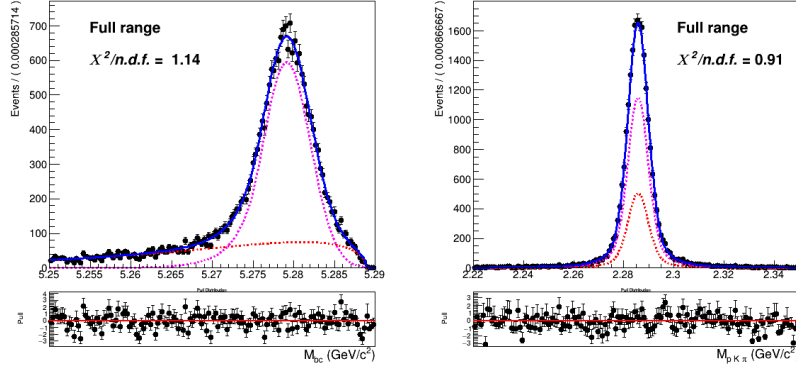


Figure (3.1) Two dimensional fit of charged correlated total signal events in M_{bc} and $M(pK\pi)$

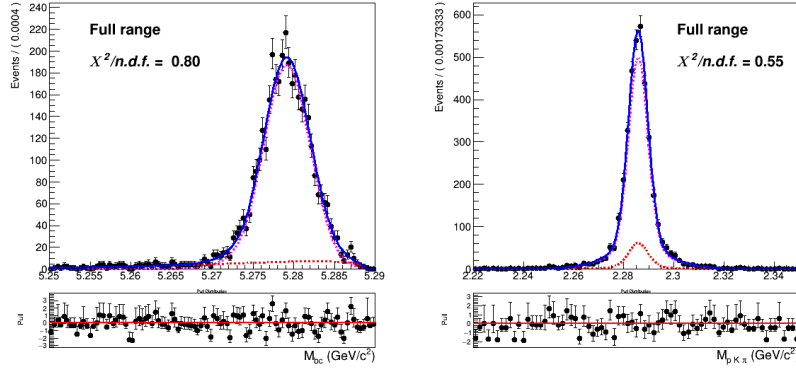


Figure (3.2) Two dimensional fit of charged anticorrelated total signal events in M_{bc} and $M(pK\pi)$

The 2D fits shown above are performed on five streams of signal MC with a sum of the following probability density functions:

$$P_{B,\Lambda_c}^{recSig}(M_{bc}, M(pK\pi)) = \Gamma_{CB}(M_{bc}) \times \rho_G(M(pK\pi)) \quad (3.1)$$

$$P_{B,\Lambda_c}^{misSig}(M_{bc}, M(pK\pi)) = \Gamma_{ARG}(M_{bc}) \times \rho_G(M(pK\pi)) \quad (3.2)$$

The first is used to fit the reconstructed signal and $\Gamma_{CB}(M_{bc})$ is a Crystal Ball function. The second is used to model the misreconstructed signal and $\Gamma_{ARG}(M_{bc})$ is an Argus function. In both cases a sum of three Gaussian functions $\rho_G(M(pK\pi))$ describes the mass of the Λ_c baryon.

As already said, only the events of reconstructed signal are considered as signal, while the misreconstructed signal is considered as background. ¹

¹in the case of neutral decays, the fits shown are containing unmixed events only, the extracted parameters are then the same used to extract signal yields also when mixing is included.

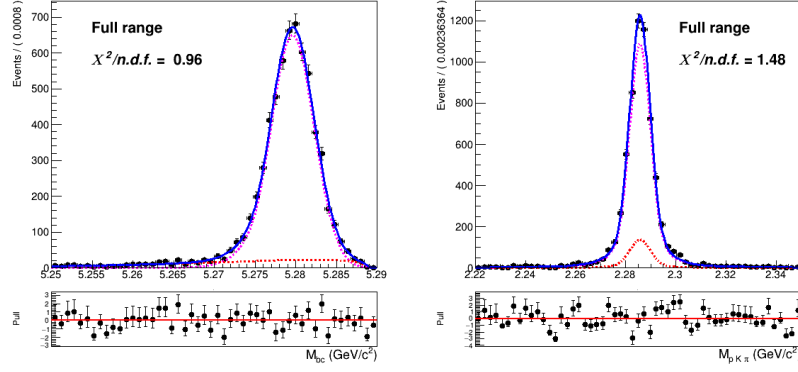


Figure (3.3) Two dimensional fit of neutral correlated total signal events in M_{bc} and $M(pK\pi)$

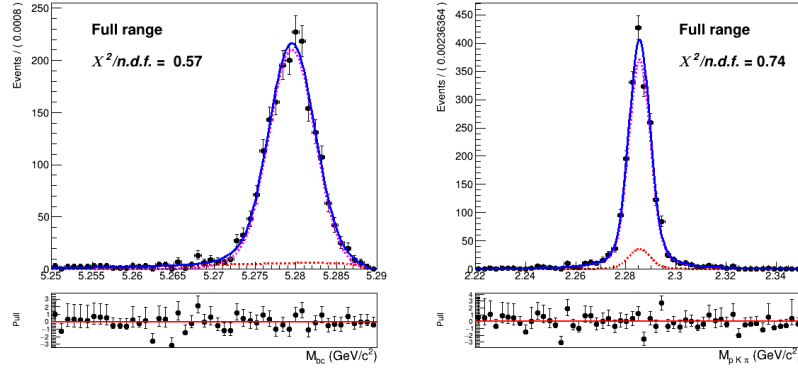


Figure (3.4) Two dimensional fit of neutral anticorrelated total signal events in M_{bc} and $M(pK\pi)$

3.1.2 M_{bc} peaking and flat background

The background composed of $B\bar{B}$ events where no Λ_c baryon is produced, is flat in its invariant mass, but can be distinguished in the following two categories:

- peaking in M_{bc} but not in $M(pK\pi)$
- flat in both M_{bc} and $M(pK\pi)$

as one can see in the following plots.

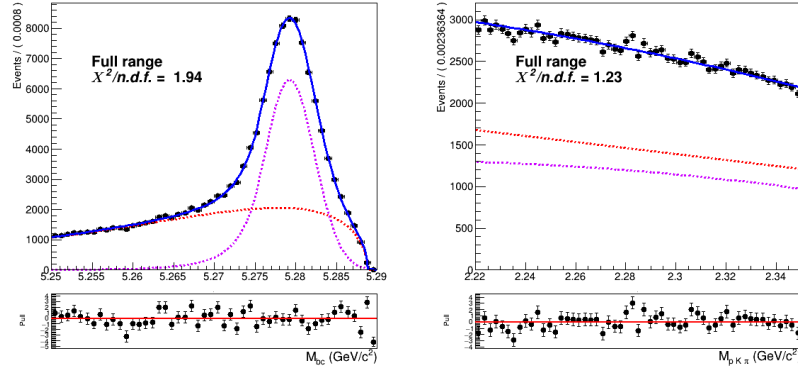


Figure (3.5) Two dimensional fit of charged correlated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

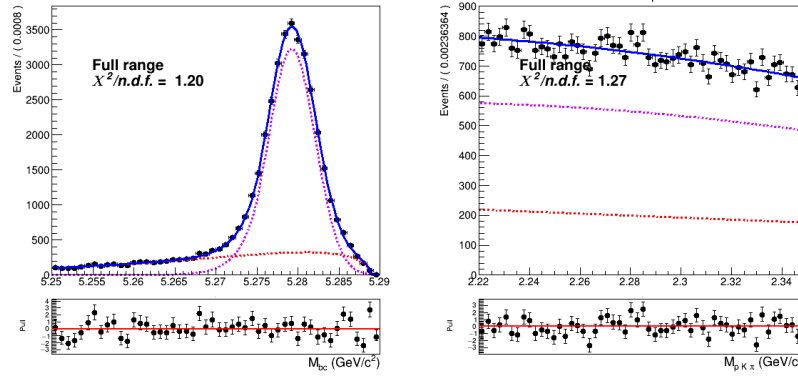


Figure (3.6) Two dimensional fit of charged anticorrelated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

209 This background presents a similar shape of the distribution in M_{bc} : the probability
 210 density functions used for it are again a Crystal Ball and an Argus.
 211 The two types of background (peaking/flat in M_{bc}) are described by:

$$P_{B,\Lambda_c}^{peaking}(M_{bc}, M(pK\pi)) = \Gamma_{CB}(M_{bc}) \times \rho_{Chab(a0,a1)}(M(pK\pi)) \quad (3.3)$$

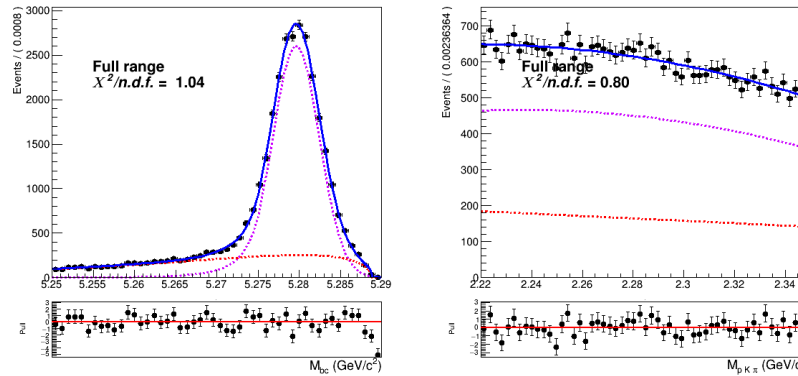


Figure (3.7) Two dimensional fit of neutral correlated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

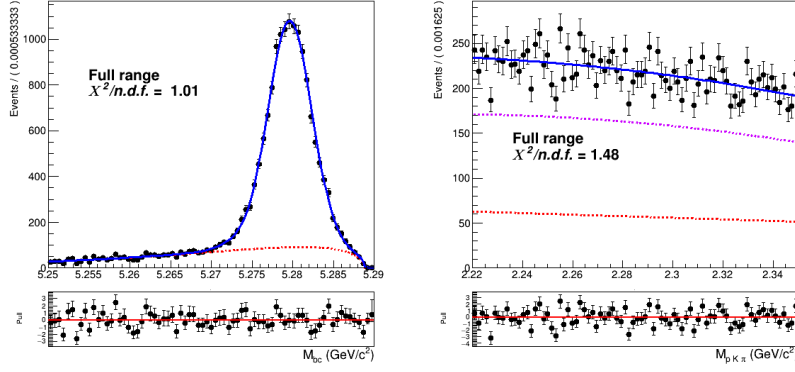


Figure (3.8) Two dimensional fit of neutral anticorrelated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

$$P_{B,\Lambda_c}^{flat}(M_{bc}, M(pK\pi)) = \Gamma_{ARG}(M_{bc}) \times \rho_{Cheb(b0)}(M(pK\pi)) \quad (3.4)$$

212

213 where $\rho_{Cheb(a0,a1)}(M(pK\pi))$ and $\rho_{Cheb(b0)}(M(pK\pi))$ represent a second order and first order
 214 Chebychev polynomial function respectively.

215 3.1.3 Crossfeed background

216 The contamination of misreconstructed $B^0 \rightarrow \Lambda_c$ events in the B^+ signal (and vice-versa)
 217 induces a background peaking in $M(pK\pi)$, but it also slightly peaks near the B meson
 218 mass, as one can see in Fig. 3.9

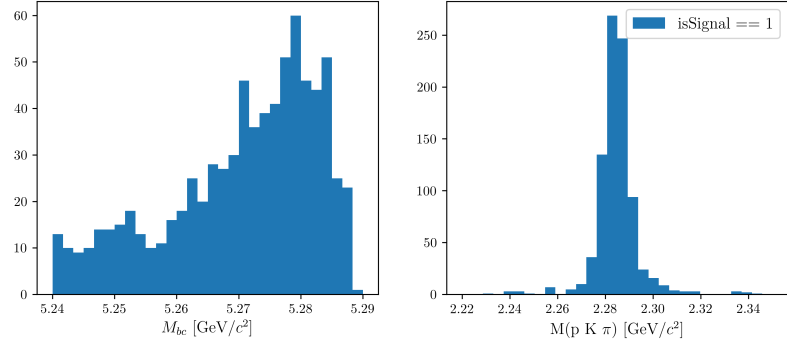


Figure (3.9) Crossfeed distribution in M_{bc} and $M(pK\pi)$

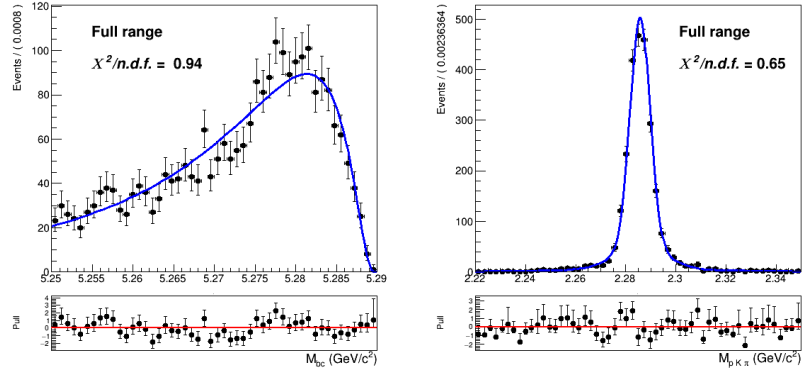


Figure (3.10) Two dimensional fit of crossfeed events in charged correlated channel in M_{bc} and $M(pK\pi)$

219

The 2D fit shown above are performed on five streams of signal MC with a sum of the

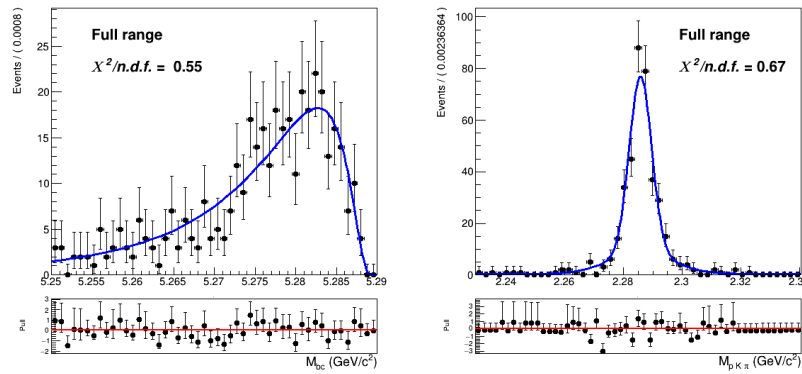


Figure (3.11) Two dimensional fit of crossfeed events in charged anticorrelated channel in M_{bc} and $M(pK\pi)$

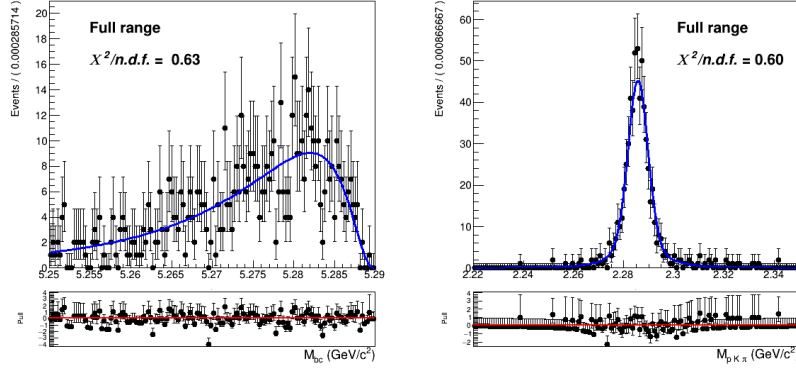


Figure (3.12) Two dimensional fit of crossfeed events in neutral correlated channel in M_{bc} and $M(pK\pi)$

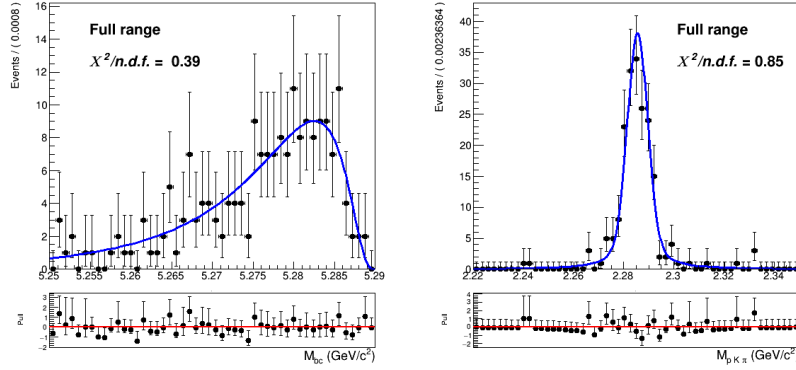


Figure (3.13) Two dimensional fit of crossfeed events in neutral anticorrelated channel in M_{bc} and $M(pK\pi)$

220 following probability density function:

$$P_{B,\Lambda_c}^{Crossfeed}(M_{bc}, M(pK\pi)) = \Gamma_{Novosibirsk}(M_{bc}) \times \rho_G(M(pK\pi)) \quad (3.5)$$

221 where $\Gamma_{Novosibirsk}(M_{bc})$ is a Novosibirsk function and the mass of the Λ_c baryon is
 222 described by the same sum of three Gaussian functions $\rho_G(M(pK\pi))$ as in Eq.3.1.

223 3.1.4 Continuum background

224 Besides the dataset recorded at the energy of the $\Upsilon(4S)$ resonance ($E_{CMS}^{on-res} = 10.58$ GeV),
 225 the *Belle* experiment recorded a sample of 89.4 fb^{-1} at an energy 60 MeV below the
 226 nominal $\Upsilon(4S)$ resonance ($E_{CMS}^{off-res} = 10.52$ GeV). The dataset allows to check for an
 227 appropriate modeling of the continuum MC simulation. Using the official tables
 228 (<https://belle.kek.jp/secured/nbb/nbb.html>) the off-resonance sample is scaled by

$$\frac{\mathcal{L}^{on-res}}{\mathcal{L}^{off-res}} \left(\frac{E_{CMS}^{off-res}}{E_{CMS}^{on-res}} \right)^2 \quad (3.6)$$

229 taking into account the difference in luminosity and in E_{CMS} (Energy in center of mass
230 system).

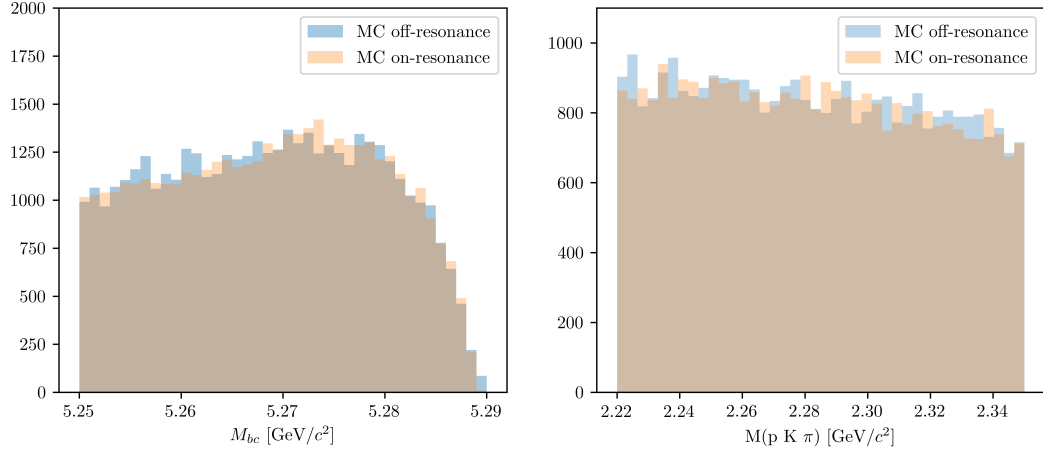


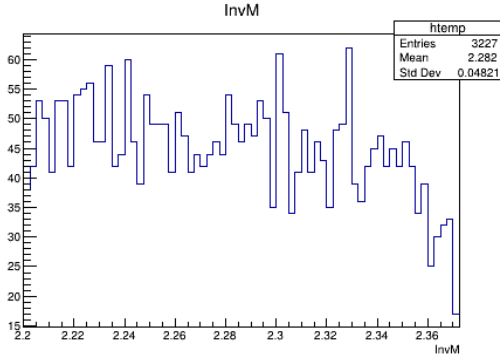
Figure (3.14) M_{bc} and $M(pK\pi)$ comparison between on-/off-resonance (scaled) Monte Carlo simulated continuum. The scaling is applied according to Eq. (3.6) and shifting the M_{bc} distribution by $E_{CMS}^{on-res} - E_{CMS}^{off-res}$.

231 The plot in Fig.3.14 shows the M_{bc} and $M(pK\pi)$ distributions in the MC on-/off-resonance
232 continuum after the scaling².

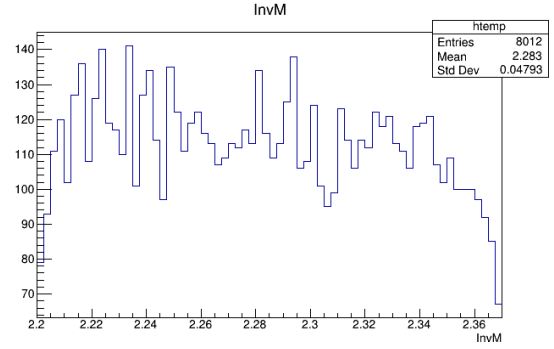
233 Ideally, provided that there's a good agreement between MC and data for the off-
234 resonance sample and also between the MC on-/off-resonance continuum after the scaling,
235 one could directly use the scaled off-resonance data to describe the continuum background
236 in the fit on data. There are two reasons that prevent this very straightforward approach:

- 237 • First, since the off-resonance MC (and data) present very low statistics (Fig. 3.15a
238 shows the Λ_c invariant mass in off-resonance data), scaling them with all the applied
239 selection cuts would cause the PDF describing the continuum to be very much
240 affected by statistical fluctuations.
- 241 • Secondly, the B meson candidates are reconstructed in both on-resonance and off-
242 resonance events for values of $M_{bc} \geq 5.22 \text{ GeV}/c^2$, but the E_{CMS} differs: there can be
243 effects of correlations between the applied *SignalProbability* cut and the M_{bc} variable
244 that one needs to take into account.

²it is obtained with the MC off-resonance sample being composed of 6 streams: the total amount is normalized



(a)



(b)

Figure (3.15) On the left: Λ_c invariant mass in off-resonance data (all nominal cuts applied). On the right: Λ_c invariant mass in off-resonance data after the continuum suppression cut removal.

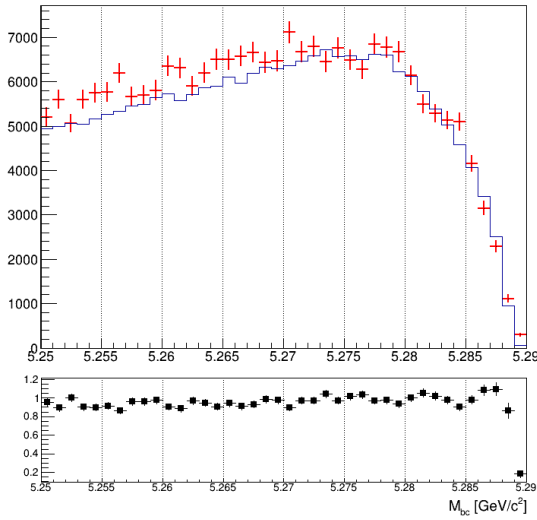
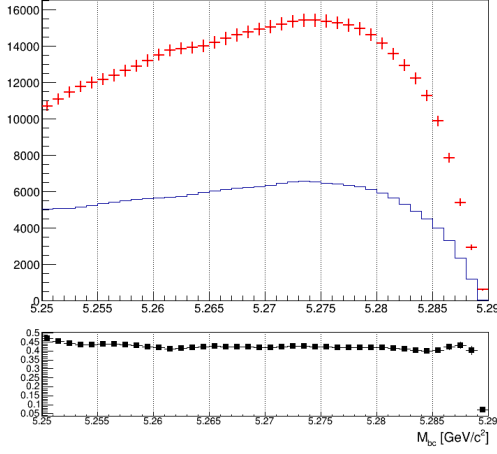
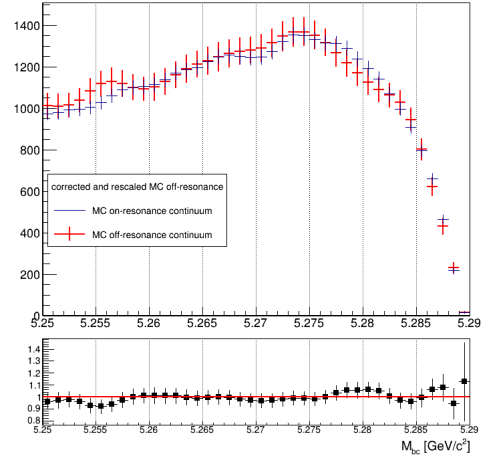


Figure (3.16) M_{bc} distributions of the MC (scaled) off-resonance sample (in red) and on-resonance (in blue) using 5 streams statistics and all nominal selection cuts applied.

245 In Fig. 3.16 one can notice some discrepancy in the shapes, apart from the not negligible
 246 statistical fluctuations in the (scaled) off-resonance distribution.



(a)



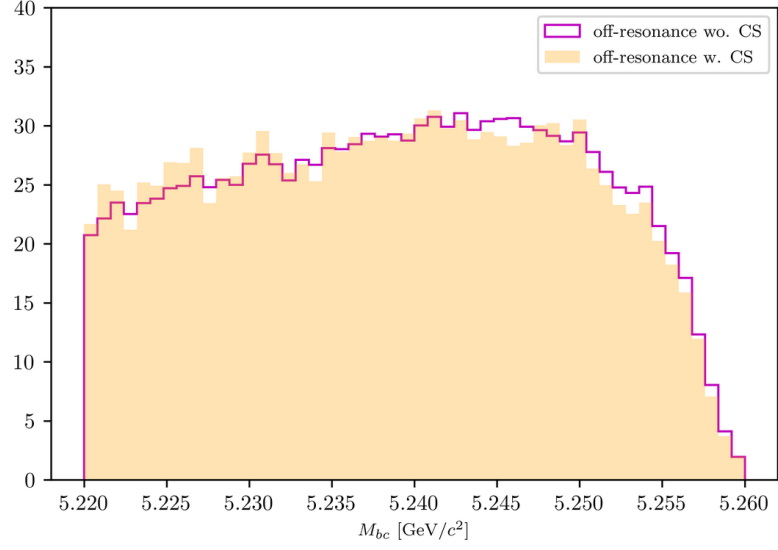
(b)

Figure (3.17) On the left: M_{bc} distributions of the MC off-resonance sample without continuum suppression and the MC continuum sample with applied continuum suppression. On the right: M_{bc} distributions of the corrected scaled MC off-resonance and on-resonance MC continuum.

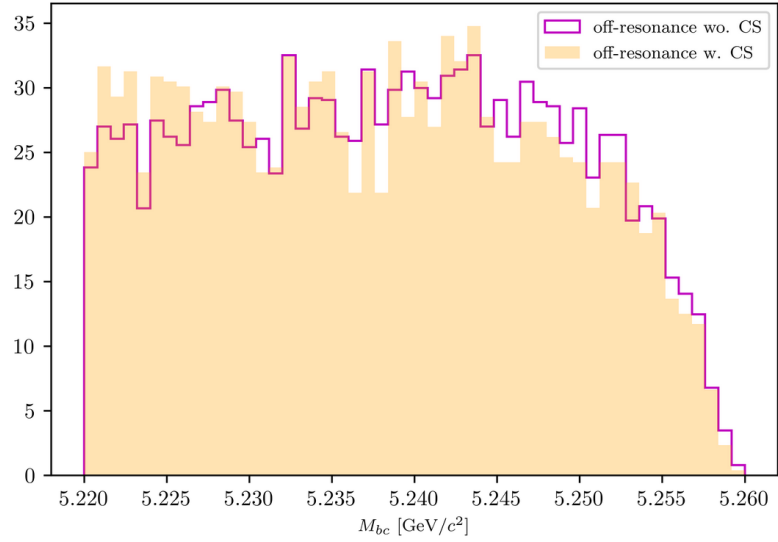
The procedure adopted to obtain the PDF describing the continuum background M_{bc} distribution is the following:

- 5 streams of off-resonance MC were scaled according to Eq. (3.6) without continuum suppression being applied and compared to the distribution of 5 streams of on-resonance continuum
- From a ratio plot, like the one in Fig. 3.17a, the bin-correction is obtained to correct the off-resonance data in the scaling procedure. To obtain the shape that can describe the continuum background M_{bc} distribution on data the continuum suppression is not applied on the off-resonance continuum sample, in order to acquire more statistics.

This procedure is first tested on an independent MC sample (see Fig. 3.17b) to check the result on simulated data before applying it on data.



(a)



(b)

Figure (3.18) Above: M_{bc} distributions of the MC off-resonance sample (5 streams) with and without continuum suppression. Below: M_{bc} distributions on data with and without continuum suppression.

The validity of the method relies on the fact that the difference between on-/off-resonance continuum events are well modeled in MC and that the shape of the M_{bc} distribution doesn't change significantly when removing the continuum suppression cut both on MC and data (as one can see from Figures 3.18a - 3.18b).

Additionally, the continuum suppression cut efficiency should be the same in data and MC in order to have the correct scaling on data with the above mentioned method. Fig.

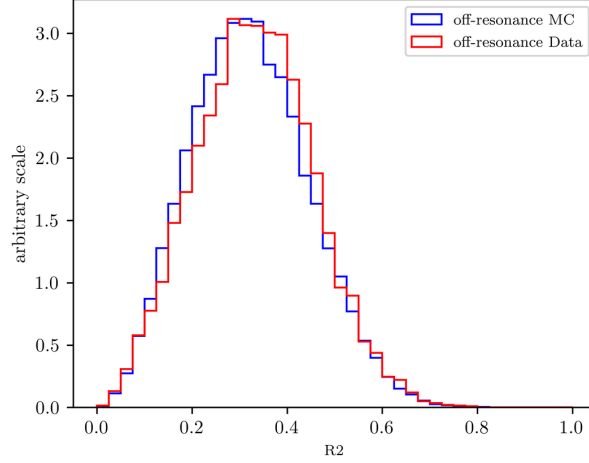


Figure (3.19) Distributions of variable *foxWolframR2* in off-resonance MC and data.

3.19 shows the distribution of the *foxWolframR2* variable in off-resonance MC and data. The slight shift visible in data can cause a different impact on data in terms of rejected continuum background when applying the $foxWolframR2 < 0.3$ cut. It is found to reject about 60% of the continuum background in data, whereas it rejects 55% of the continuum background in MC (56% in on-resonance MC). Therefore in data one can expect about 2.25% less continuum background events. This discrepancy is not statistically significant (the statistical uncertainty for the continuum background events is of the level of $\sim 1\%$), a simple correction to the number of events can be applied on data and its possible systematics can be then taken into account.

The obtained distribution can be then fitted with a Novosibirsk function (see Fig. 3.20). This is the procedure which can be then applied on the off-resonance data to obtain the M_{bc} shape describing the continuum background in data.

In the Λ_c invariant mass one doesn't expect correlation effects, but nevertheless there can be differences due to the limited statistics of the off-resonance sample. In fact, in the case of on-resonance MC for the charged correlated decays some events in which Λ_c candidates survive nominal selection cuts are visible and can be described with a small Gaussian on the top of the flat background (Fig.3.21a). Due to the low statistics one cannot see a similar peak in the off-resonance sample (the Fig.3.21b shows a 5 streams statistics).

The shape describing the Λ_c invariant mass in the case of charged correlated decays is obtained from the simulated on-resonance continuum, again using 5 streams statistics (see Fig. 3.21a). In the other cases no peak is visible and therefore also the Λ_c invariant mass shape is obtained from 5 streams of off-resonance events.

Finally, it is possible to examine the validity of the whole procedure on the independent stream. Fig. 3.22 - 3.25 show the M_{bc} , $M(pK\pi)$ projections of the overlayed two dimensional PDFs obtained with the above described procedure.

The 2D PDF used for Fig. 3.22 can be written as:

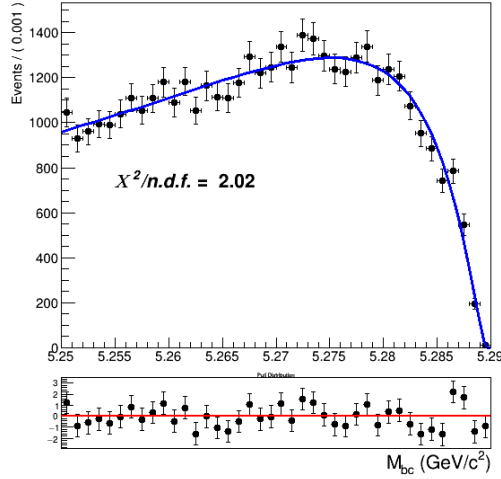


Figure (3.20) Fit of the M_{bc} distribution MC (scaled) off-resonance continuum (one stream).

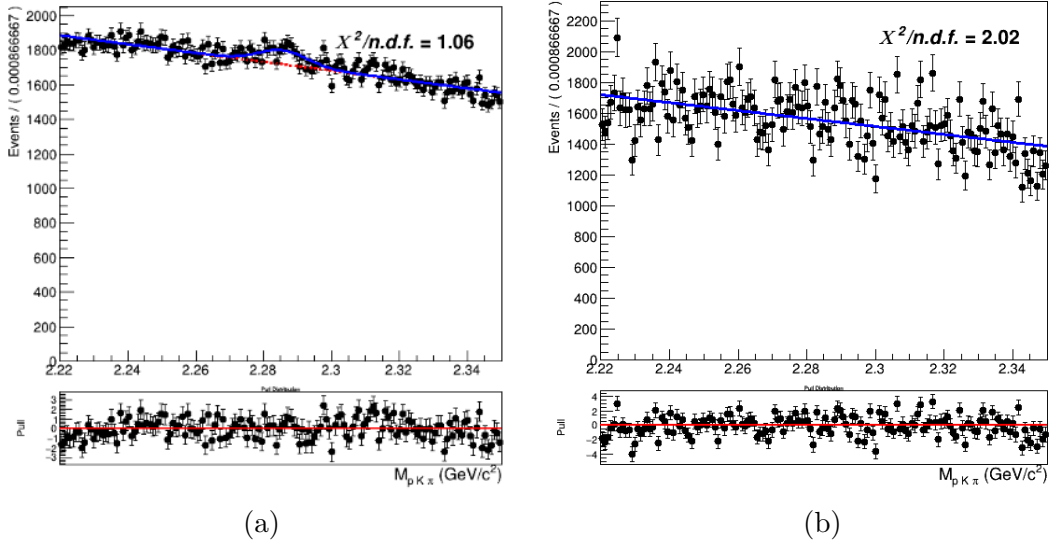


Figure (3.21) Comparison between 5 streams of MC on-resonance continuum 3.21a) and off-resonance (scaled) continuum in $M(pK\pi)$ (3.21b).

292

$$293 \quad P_{B,\Lambda_c}^{Continuum}(M_{bc}, M(pK\pi)) = \Gamma_{Nov}(M_{bc}) \times [\rho_{Cheb1}(M(pK\pi)) + \rho_G(M(pK\pi))]$$

294

295 where, as already anticipated, the invariant mass is described by a sum of a first order
 296 Chebychev polynomial and the peak by the same triple Gaussian PDF adopted for the
 297 signal. The 2D PDF used for Fig. 3.23 and Fig. 3.24 differ only in the invariant mass, not
 298 having the triple gaussian. Whereas the 2D PDF used for Fig. 3.25 can be written as:

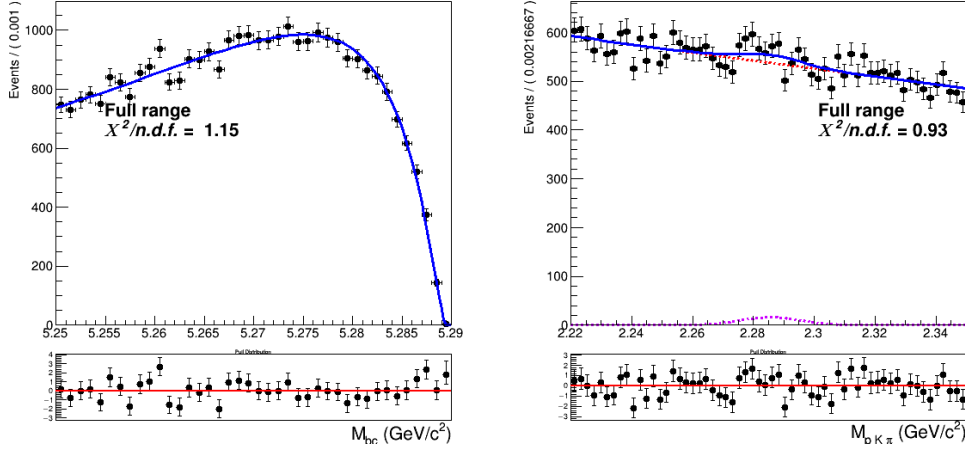


Figure (3.22) Two dimensional fit of charged correlated continuum events (one stream).

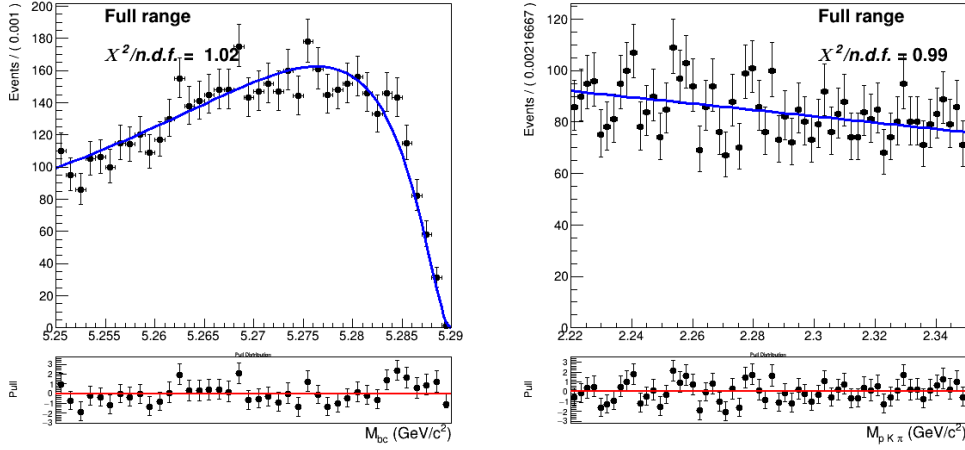


Figure (3.23) Two dimensional fit of charged anticorrelated continuum events (one stream).

299

$$P_{B_s \Lambda_c}^{Continuum}(M_{bc}, M(pK\pi)) = \Gamma_{Argus}(M_{bc}) \times [\rho_{Chet1}(M(pK\pi))]$$

301

302 since the distribution in M_{bc} can be better described by an Argus.

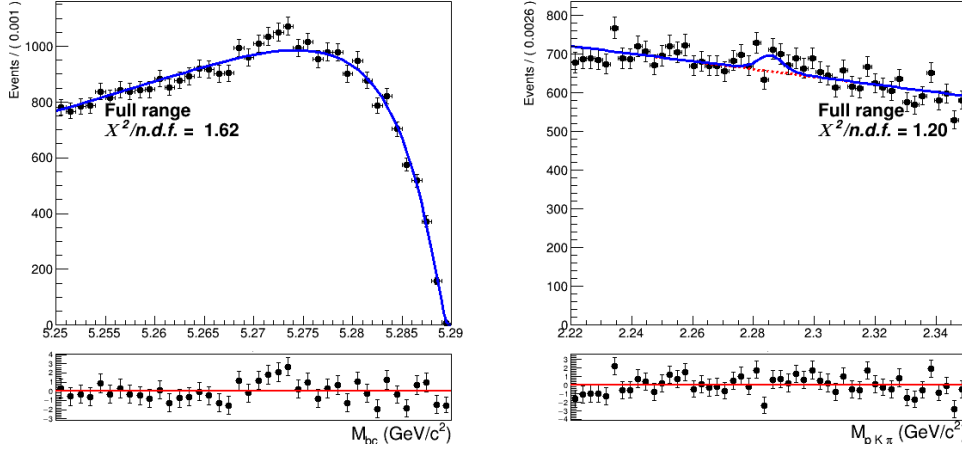


Figure (3.24) Two dimensional fit of continuum events (one stream).

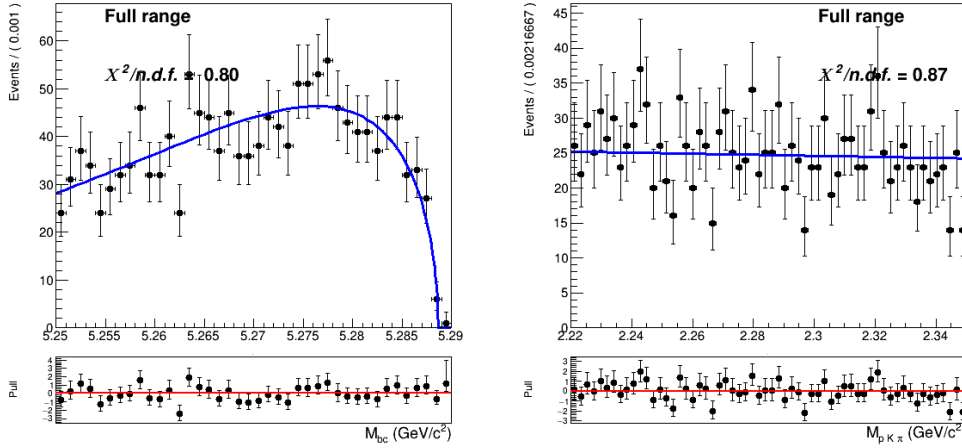


Figure (3.25) Two dimensional fit of continuum events (one stream).

3.2 Two dimensional fit on Monte Carlo

A total of six independent streams were used to construct/validate the fit model: each time five streams were used to construct the already discussed PDFs and the independent stream is used to test if the total PDF enables to extract the signal yield in an unbiased way (a total of six fits are performed on the six different streams of generic MC).

To especially suppress the systematic uncertainties deriving from the amount of crossfeed the samples corresponding to the four different decay channels are fitted simultaneously. In all six fits all the shaping parameters are kept fixed, except:

- σ_{G1} : the width of the main of the three Gaussian functions in $\rho_G(M(pK\pi))$
- σ_{CB} parameter for the Crystal Ball describing the signal peak in M_{bc}

313 In the M_{bc} distribution the σ_{CB} width parameter for the Crystal Ball describing the M_{bc}
 314 peaking background is expressed as function of the signal σ_{CB} with a ratio fixed from MC.
 315 As for the normalizations, mis-/reconstructed signal events and M_{bc} peaking/flat back-
 316 ground events are floated in the two dimensional unbinned maximum likelihood fits.
 317 The continuum background normalization is kept fixed to the value obtained by the
 318 off-resonance scaling procedure.
 319 Instead, the normalization of crossfeed background events is determined by the fit in the
 320 following way:

$$N_{CrossBkg} = N_{recSig}^{corr} \cdot (\epsilon_{cross}/\epsilon_{recSig})^{corr} + N_{recSig}^{anticorr} \cdot (\epsilon_{cross}/\epsilon_{recSig})^{anticorr} \quad (3.7)$$

321 where N_{recSig}^{corr} ($N_{recSig}^{anticorr}$) are the fitted signal yields of the corresponding crossfeeding
 322 decay and $(\epsilon_{cross}/\epsilon_{recSig})$ are the ratios of misreconstruction efficiency and signal recon-
 323 struction efficiency as determined in the Monte Carlo. Since the crossfeed can likely
 324 occur both from correlated and anticorrelated channels, both are considered and summed
 325 together.

326 Exemplary, the distributions of stream 0 overlaid by the fitted PDF are depicted in
 327 Figs. 3.27 to 3.37. In Tables 3.1 to 3.4 the signal yields of the fits (**Reconstructed
 328 Signal**) to the two dimensional distributions for the six streams are listed and compared
 329 to the yields obtained from fits of signal distributions of each individual stream. The
 330 latter are the "expected" yields of reconstructed signal from a fit to the total signal events
 331 in the individual stream as the one plotted on Fig. 3.26 where all the parameters of the
 332 PDFs described in Eq. (3.1) are kept fixed and the corresponding yields are extracted from
 333 the fit. Note that in the case of Tables Table 3.3 - 3.4 the reconstructed signal yields are
 334 contaminated by events from the other neutral B^0 decay channel that experience mixing.
 335 Since those events cannot be analytically distinguished from the true signal events, the
 336 branching fraction calculation for neutral decays needs to take them into account and
 337 correct for their contribution (as will be shown in the next Chapter).

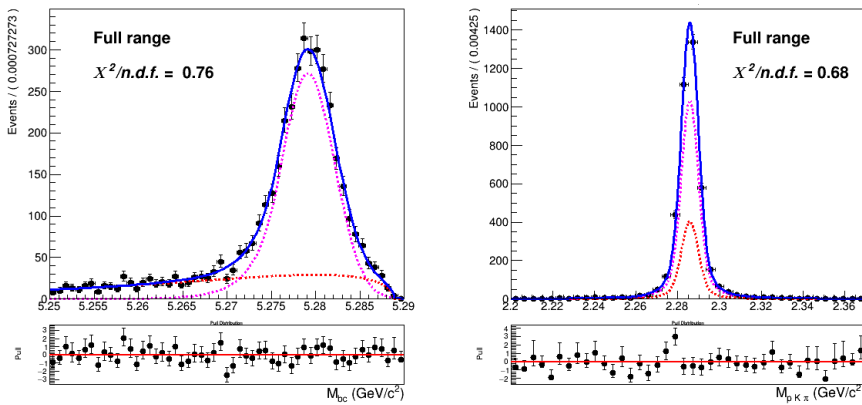


Figure (3.26) Two dimensional fit of Total Signal of stream 0 used to extract the expected reconstructed (corresponding to the PDF colored in magenta) and expected misreconstructed yields (corresponding to the PDF colored in red).

3.2.1 Fit results for $B^- \rightarrow \Lambda_c^+$ decays

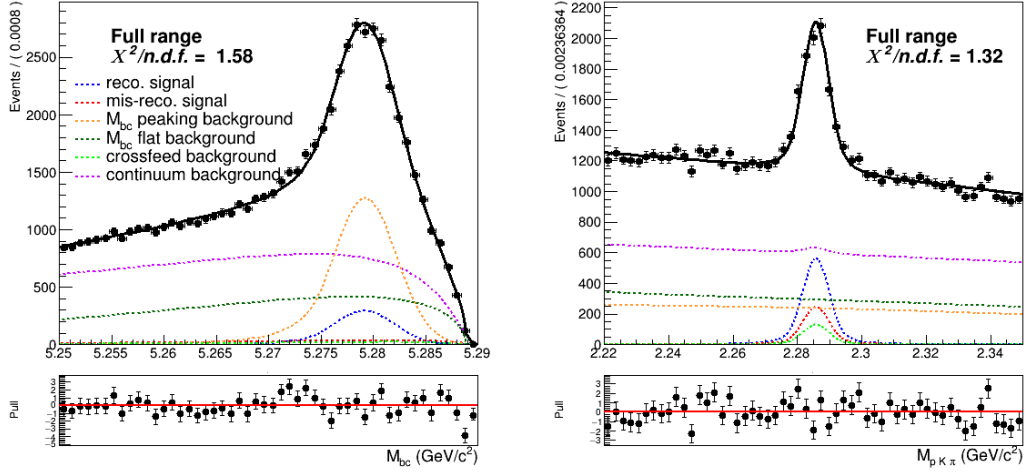


Figure (3.27) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged correlated decays.

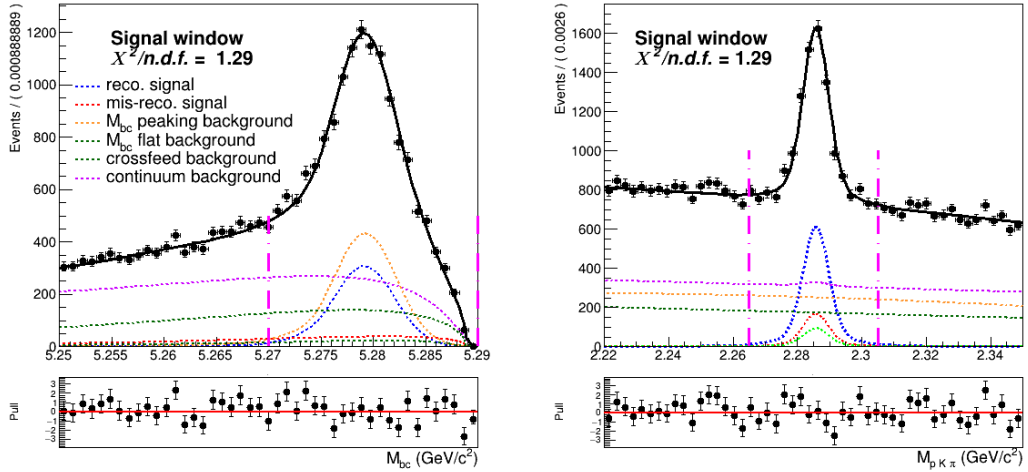


Figure (3.28) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged correlated decays (in the signal window)

	Reconstructed Signal		Total Signal		fit - MC truth	
	fit	expected	fit	MC truth		
stream 0	2829 ± 130	2928 ± 66	4063 ± 157	4061	-2	-0.05%
stream 1	2811 ± 137	2956 ± 65	4095 ± 161	4084	11	0.3%
stream 2	2952 ± 141	2940 ± 65	4345 ± 165	4138	207	5.0%
stream 3	2747 ± 134	2867 ± 66	4267 ± 160	4105	162	3.9%
stream 4	3043 ± 138	3017 ± 67	4148 ± 157	4176	-28	-0.7%
stream 5	2818 ± 136	2816 ± 65	3999 ± 162	4001	-2	-0.05%
sum	17200	17524	24917	24565	348	1.4%

Table (3.1) Charged correlated decays: comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

339 The yields obtained by the fit show a slight tendency of underestimation, but when
340 comparing the sums of them with the sum of expected values one can see the difference is
341 within statistical fluctuations (within 2%).
342 The tendency of underestimation is visible in , but it is just slightly larger than σ

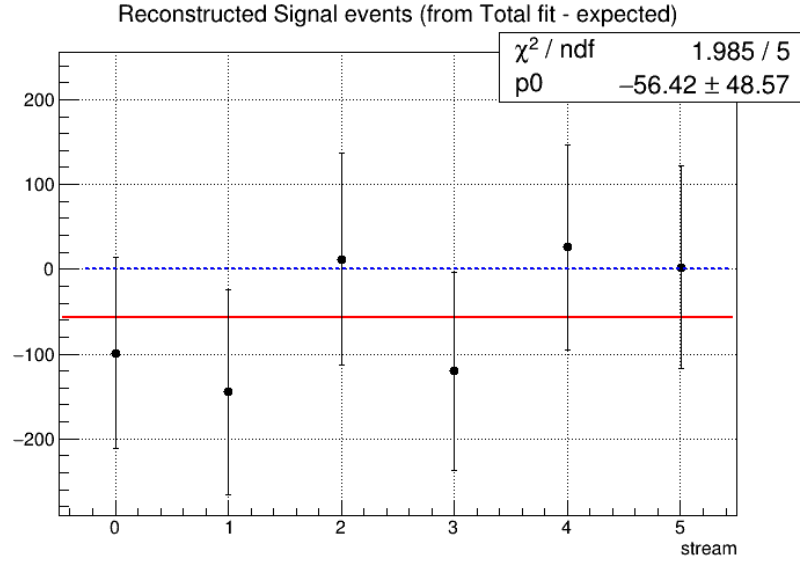


Figure (3.29) Residuals of signal yields (alias reconstructed signal, values reported in the first two columns of the above displayed table).

3.2.2 Fit results for $B^- \rightarrow \bar{\Lambda}_c^-$ decays

For the anticorrelated decays from charged B mesons the slight tendency is in the opposite direction. But the distribution of residuals (see Fig. 3.30) shows this slight overestimation is well within the statistical uncertainty.

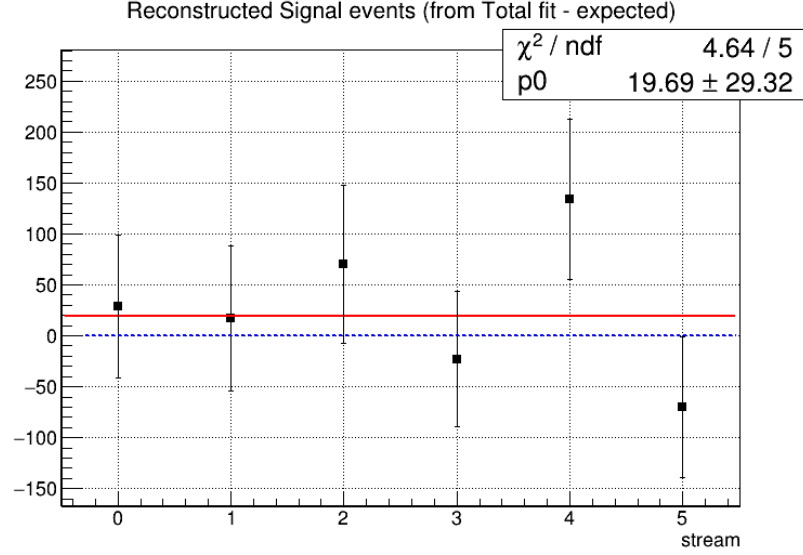


Figure (3.30) Residuals of signal yields (alias reconstructed signal, values reported in the first two columns of the above displayed table).

	Reconstructed Signal		Total Signal			
	fit	expected	fit	MC truth	fit - MC truth	
stream 0	724 ± 64	695 ± 28	813 ± 73	765	48	6.3%
stream 1	726 ± 65	709 ± 29	802 ± 67	785	17	2.2%
stream 2	788 ± 72	718 ± 29	911 ± 72	797	114	14.3%
stream 3	679 ± 60	702 ± 29	768 ± 66	802	-34	4.2%
stream 4	844 ± 73	710 ± 29	982 ± 73	785	197	25.1%
stream 5	605 ± 63	675 ± 29	722 ± 63	760	-38	-5.0%
sum	4366	4209	4998	4694	304	6.5%

Table (3.2) Charged anticorrelated decays: comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

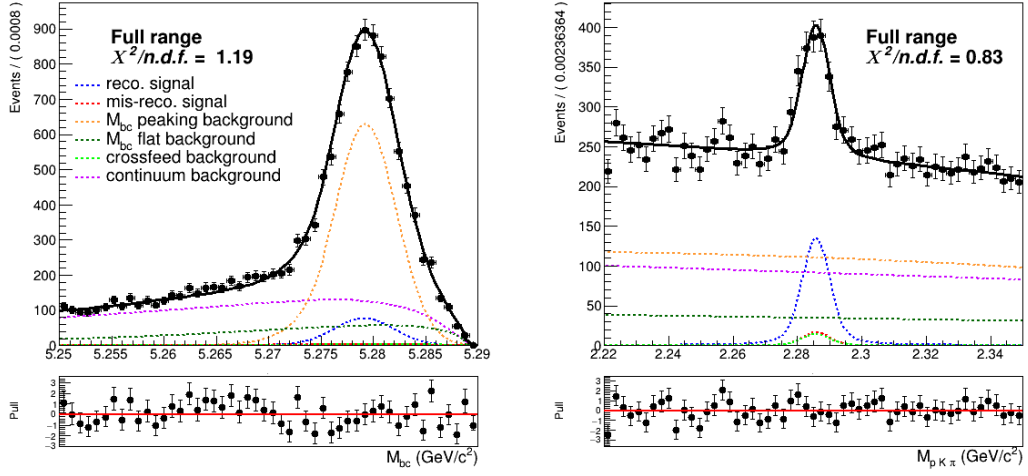


Figure (3.31) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged anticorrelated decays.

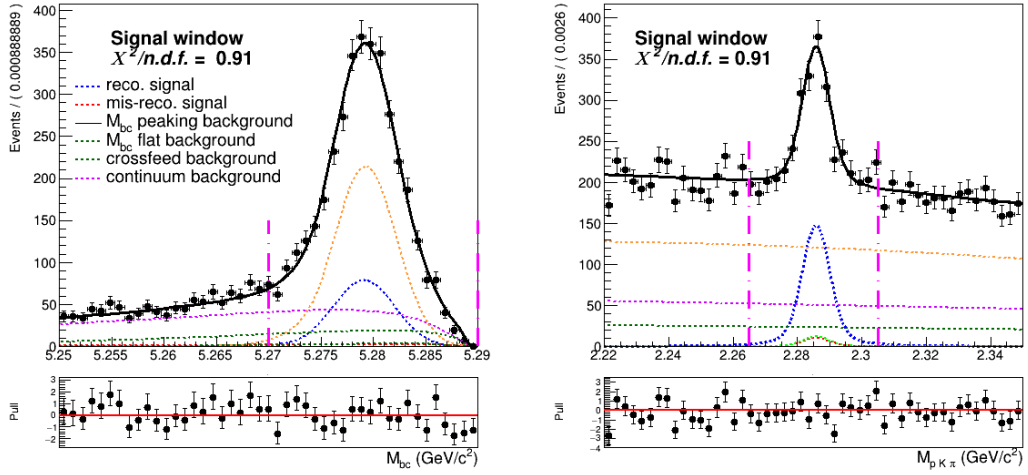


Figure (3.32) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged anticorrelated decays (in the signal window).

3.2.3 Fit results for $\bar{B}^0 \rightarrow \Lambda_c^+$ decays

Also in the case of the neutral correlated one can observe a slight overestimation in the reconstructed signal yields. The residuals (see Fig. 3.33) show that this is not so negligible: the significance is about 1.5σ .

	Reconstructed Signal		Total Signal			
	fit	expected	fit	MC truth	fit - MC truth	
stream 0	1329 ± 69	1240 ± 38	1390 ± 70	1379	11	0.8%
stream 1	1314 ± 68	1327 ± 38	1319 ± 66	1287	14	2.5%
stream 2	1266 ± 68	1254 ± 38	1273 ± 65	1251	22	1.8%
stream 3	1320 ± 67	1248 ± 38	1316 ± 66	1255	35	4.9%
stream 4	1227 ± 67	1214 ± 38	1300 ± 66	1223	77	6.3%
stream 5	1282 ± 70	1233 ± 38	1251 ± 66	1246	5	0.4%
sum	7738	7516	7709	7525	140	1.86%

Table (3.3) Comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

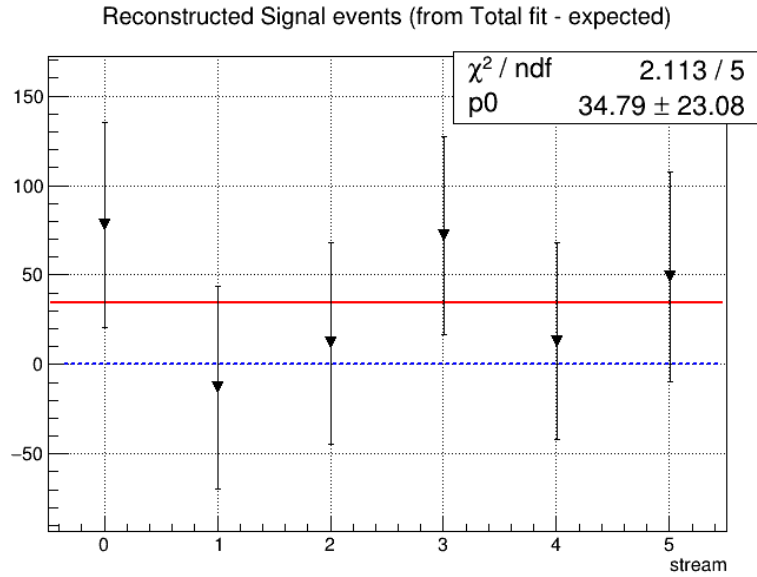


Figure (3.33) Residuals of signal yields for neutral correlated decays.

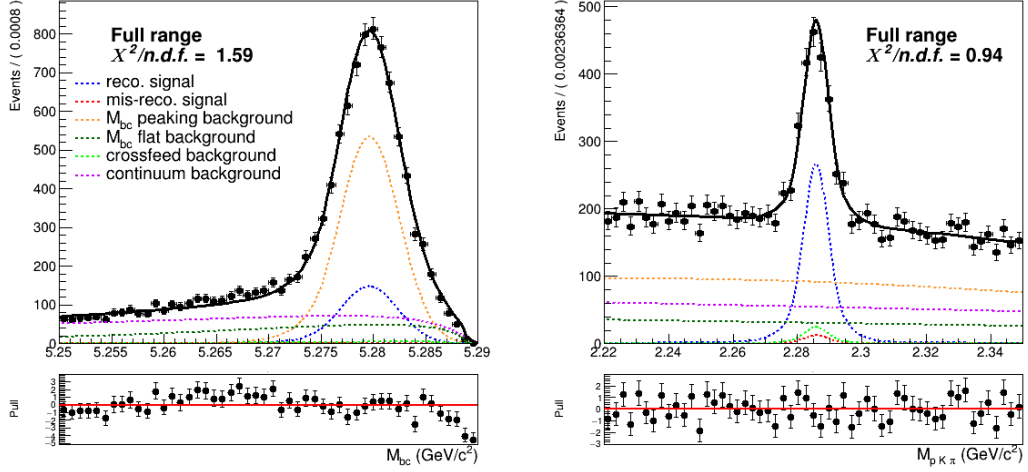


Figure (3.34) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral correlated decays.

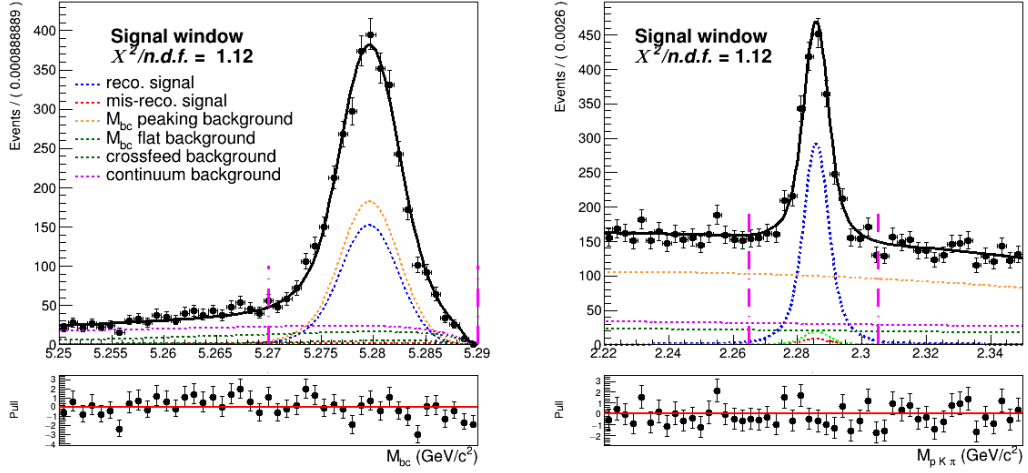


Figure (3.35) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral correlated decays.

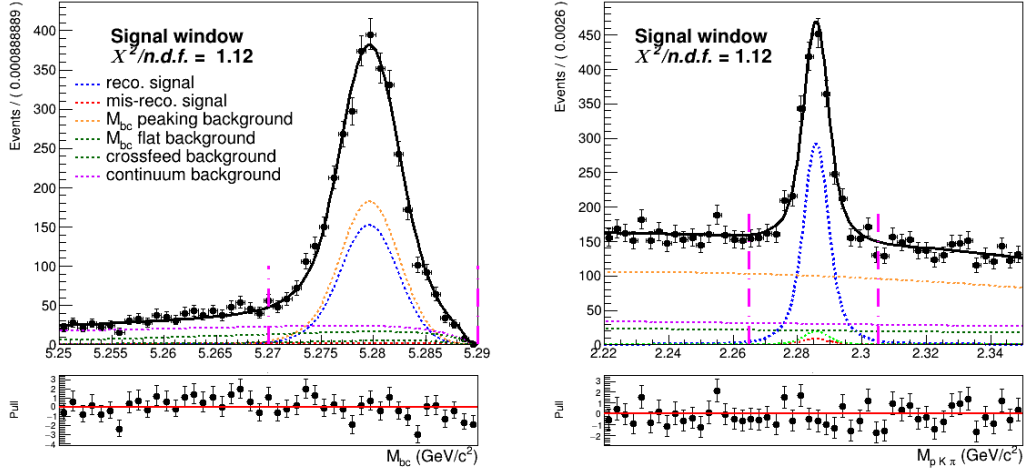


Figure (3.36) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral correlated decays (in the signal window).

3.2.4 Fit results for $\bar{B}^0 \rightarrow \bar{\Lambda}_c^-$ decays

	Reconstructed Signal		Total Signal			
	fit	expected	fit	MC truth	fit - MC truth	
stream 0	567 ± 45	575 ± 26	657 ± 45	668	-11	-1.6%
stream 1	565 ± 46	607 ± 26	660 ± 47	687	-27	-3.9%
stream 2	574 ± 46	593 ± 26	659 ± 45	639	20	3.1%
stream 3	714 ± 49	596 ± 25	792 ± 52	680	112	16.4%
stream 4	603 ± 50	594 ± 25	693 ± 50	638	55	8.6%
stream 5	589 ± 63	595 ± 26	603 ± 44	624	-21	3.4%
sum	3612	3560	4064	3936	170	4.3%

Table (3.4) Neutral anticorrelated decays: comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

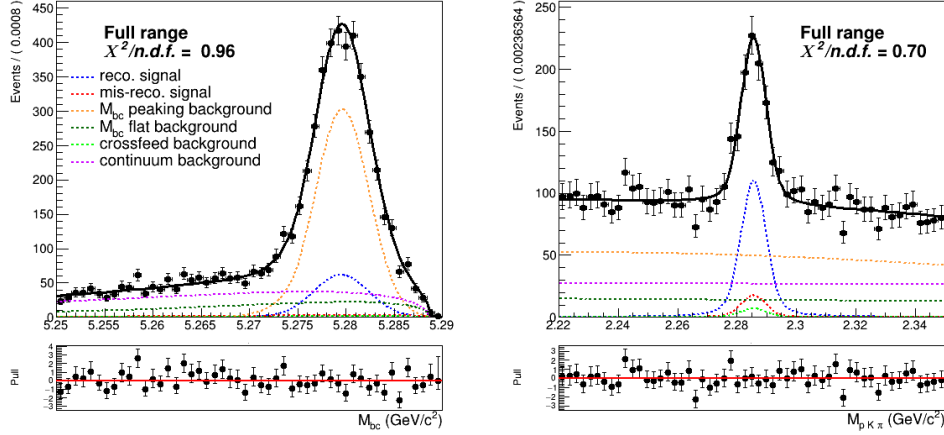


Figure (3.37) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral anticorrelated decays.

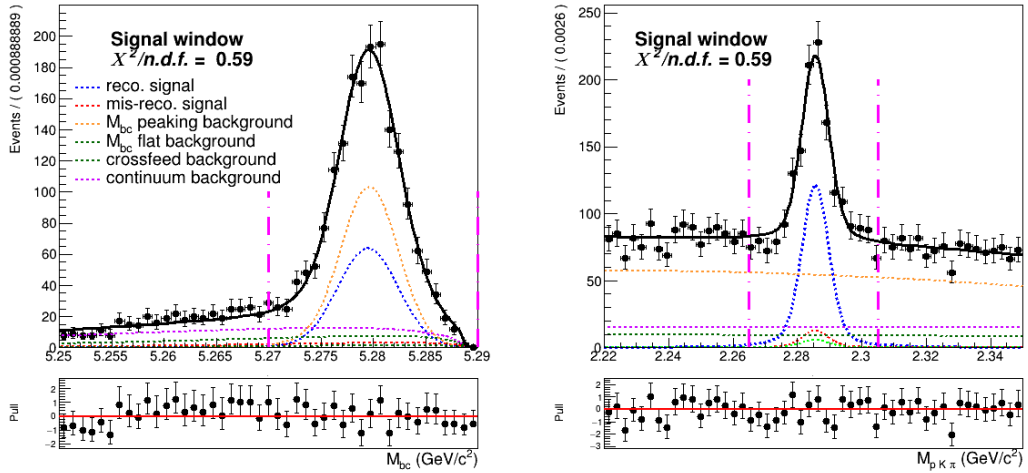


Figure (3.38) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral anticorrelated decays (in the signal window).

3.2.5 Fit residuals

Fig. 3.39 shows the residuals in each fit (stream by stream), calculated as the difference of reconstructed signal yield in the two dimensional fit from the expected value (in the fit of the total signal events). Since the signal events are the same in the two fits, the resulting yields are correlated. Therefore the uncertainties on the residuals are calculated (on a first approximation) as:

$$\sigma_{res} = \sqrt{(\sigma_{tot}^2 - \sigma_{sig}^2)} \quad (3.8)$$

Some of the results may seem not well distributed around 0.

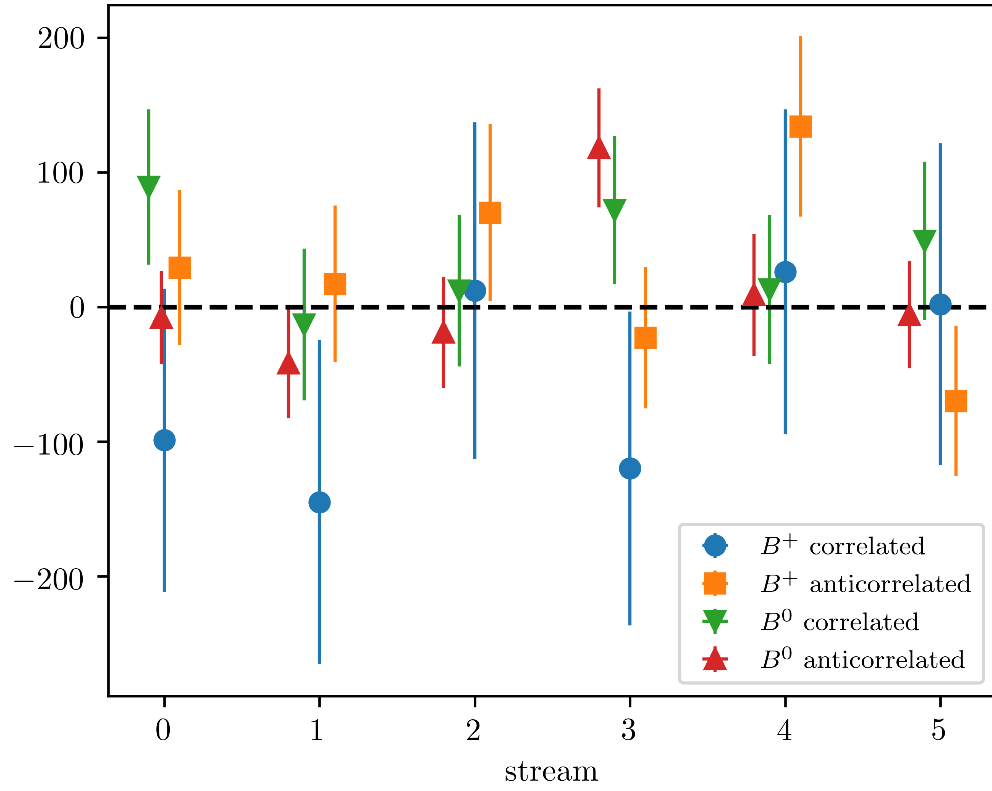


Figure (3.39) Residuals of the fitted signal yields resulting from the simultaneous 2D fit by stream and decay mode.

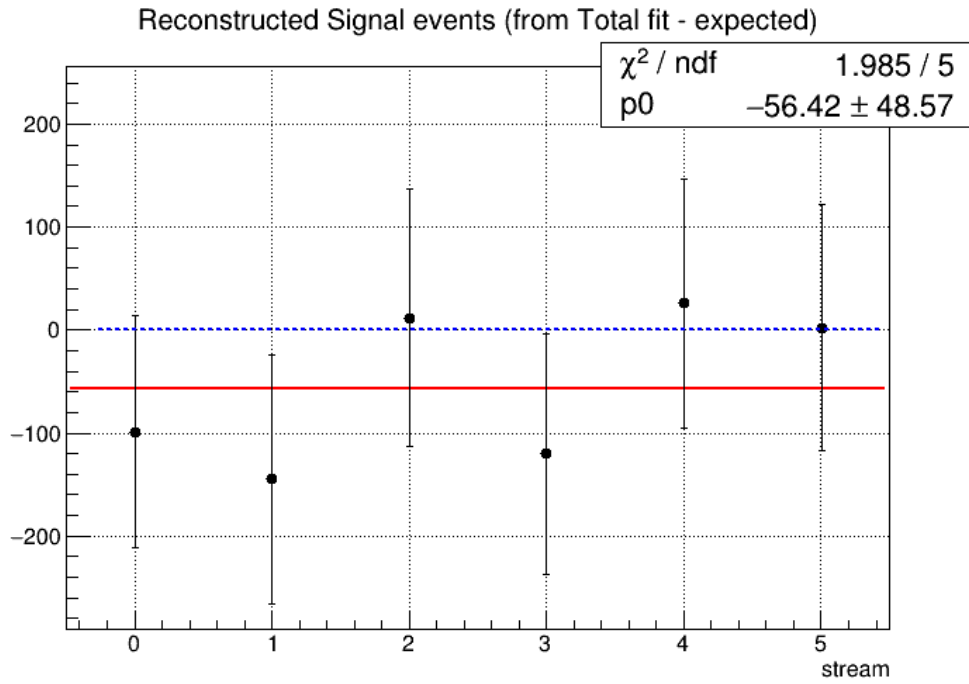


Figure (3.40) Residuals of the fitted signal yields of charged correlated decays.

3.2.6 toy Monte Carlo study

For the fit model also toy MC pseudo-experiments were performed in order to confirm the behavior of the fit setup. With toy MC experiments the yields, errors and the pulls of the fit are studied by generating our own pseudo-datasets, according to the MC (see plots in 3×10^3 pseudo-datasets are constructed, where each dataset was generated with the expected amount of events, distributed according to the Poisson distribution. Then the composition of each toy pseudo-experiment is fitted as if they were data, and the pull-value distributions of the fit results are calculated. From the plots showing the pull distributions of the fitted signal yields one can conclude that no bias is present.

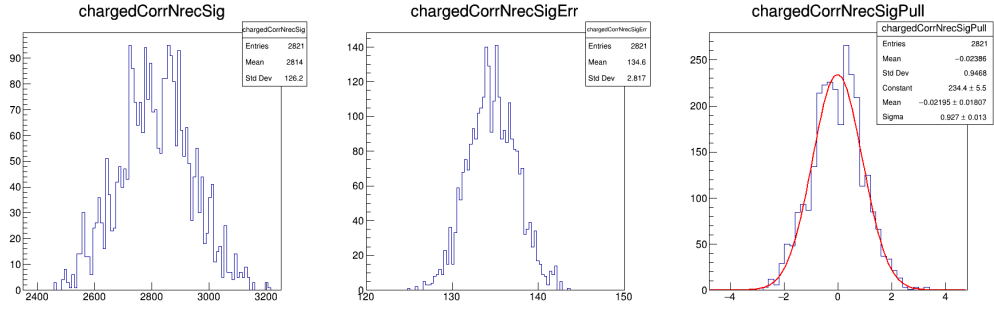


Figure (3.41) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the charged correlated decays.

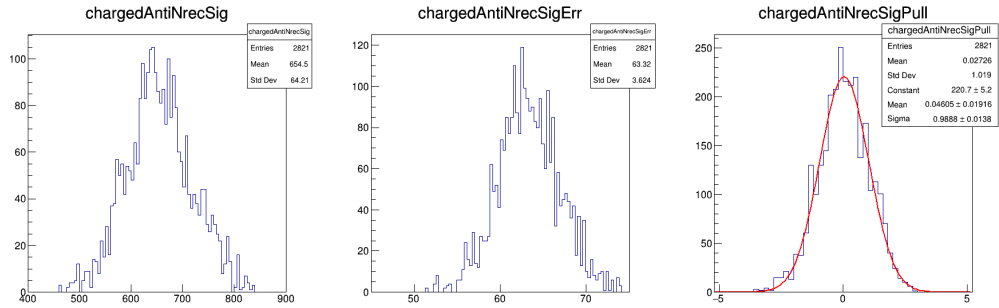


Figure (3.42) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the charged anticorrelated decays.

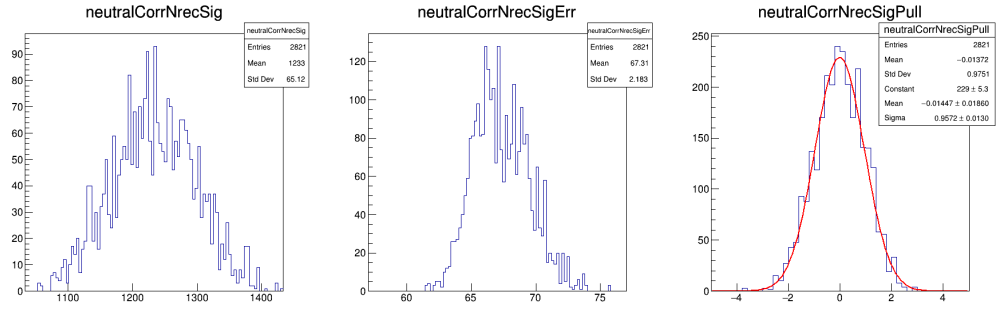


Figure (3.43) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the neutral correlated decays.

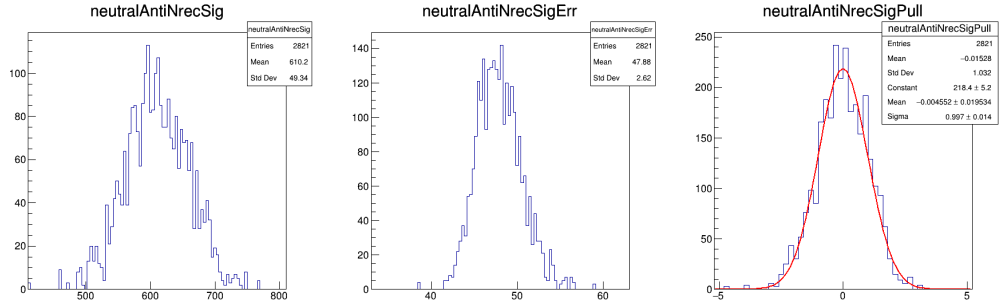


Figure (3.44) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the neutral anticorrelated decays.

Chapter 4

B_{tag} fit

The normalization for the branching fractions is determined from the number of correctly tagged B mesons. This value is again determined by fitting the M_{bc} distribution of all B meson candidates tagged by FEI regardless of the signal side B meson decay. The tagged B meson candidates are selected according to the selection criteria used for the M_{bc} variable already illustrated in subsections Secs. 2.4.1 to 2.4.4.

4.1 Probability Density Functions (PDFs) for the B_{tag}

The reconstructed events can be categorized as follows:

- correctly reconstructed B meson candidates: **reconstructed signal**
- misreconstructed B meson candidates: **misreconstructed signal**
- B^0 mesons misreconstructed as B^+ (and vice-versa): **crossed background**
- and **continuum background**

4.1.1 Total Signal fits

As in the 2D fits the total signal is distinguished in **reconstructed** and **misreconstructed signal** depending on whether it's peaking or not in M_{bc} .

Fig. 4.1 shows the fit on tagged charged B mesons corresponding to the normalization for the branching fractions of $B^- \rightarrow \Lambda_c^+$ decays. The M_{bc} distribution of the tagged B mesons is fitted with a combination of Crystal Ball and Gaussian as for the "peaking" component and the "flat" component is fitted with a Novosibirsk function. For the other decay channels instead an Argus function is used to describe the misreconstructed signal (see Figs. 6.4 to 4.4).

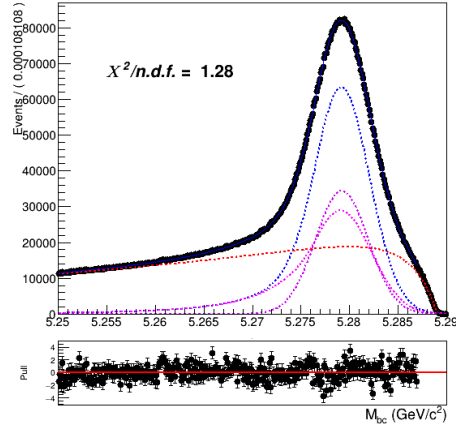


Figure (4.1) Fitted distribution of tagged charged B mesons in the charged correlated decays sample: reconstructed signal events are described by the blue dotted PDF, the misreconstructed with a Novosibirsk function (red dotted).

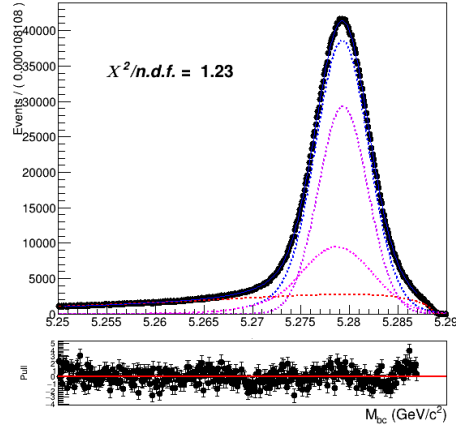


Figure (4.2) Fitted distribution of tagged charged B mesons in the charged anticorrelated decays sample.

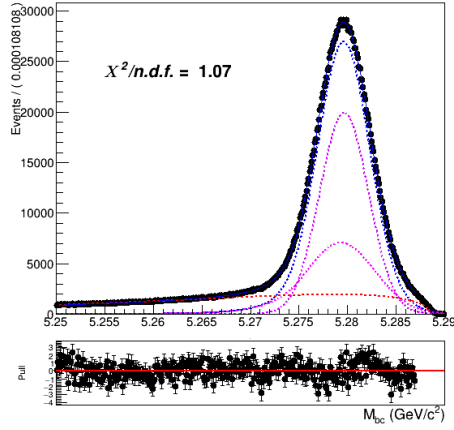


Figure (4.3) Fitted distribution of tagged neutral B mesons reconstructed in neutral correlated decays.

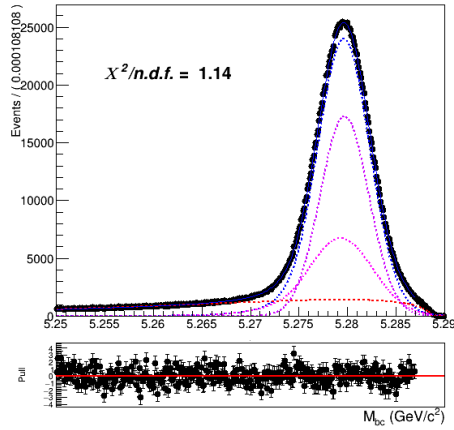


Figure (4.4) Fitted distribution of tagged charged B mesons reconstructed in neutral anticorrelated decays.

4.1.2 Crossfeed PDF

The crossfeed background is always fitted instead with a sum of a Novosibirsk and an asymmetric Gaussian PDF.

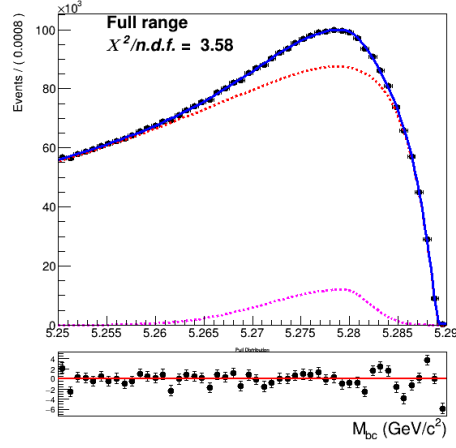


Figure (4.5) Crossfeed distribution of charged correlated decays fitted with a sum of Novosibirsk (red) and asymmetric Gaussian PDF (magenta)

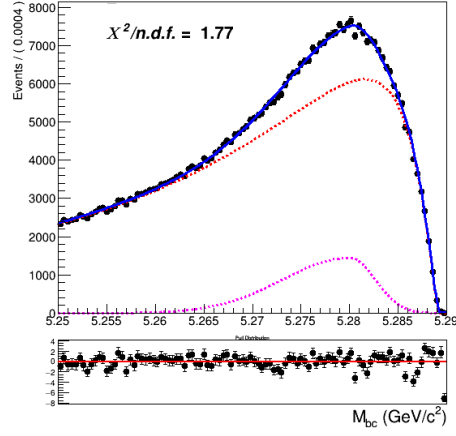


Figure (4.6) Crossfeed distribution of charged anticorrelated decays

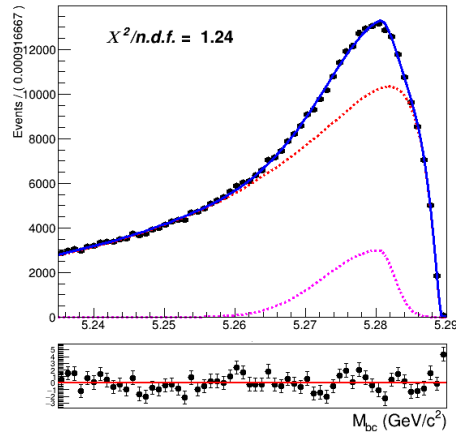


Figure (4.7) Crossfeed distribution of neutral correlated decays

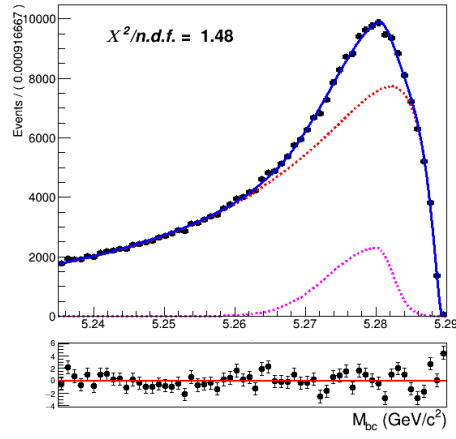


Figure (4.8) Crossfeed distribution of neutral anticorrelated decays

4.1.3 Continuum PDF

As for the continuum background, a similar procedure as the one described already for the two dimensional fit was adopted:

- first the off-resonance sample is scaled accordingly
- the ratio between the scaled off-resonance and the on-resonance in MC is calculated in each bin (see Fig.4.9a)
- the bin-correction is applied on an independent stream and the scaled and bin-corrected M_{bc} distribution is compared with the on-resonance distribution as shown in Fig.4.9b

Being the statistics much larger than in the 2D sample, there's no need to remove the continuum suppression cut on the off-resonance sample.

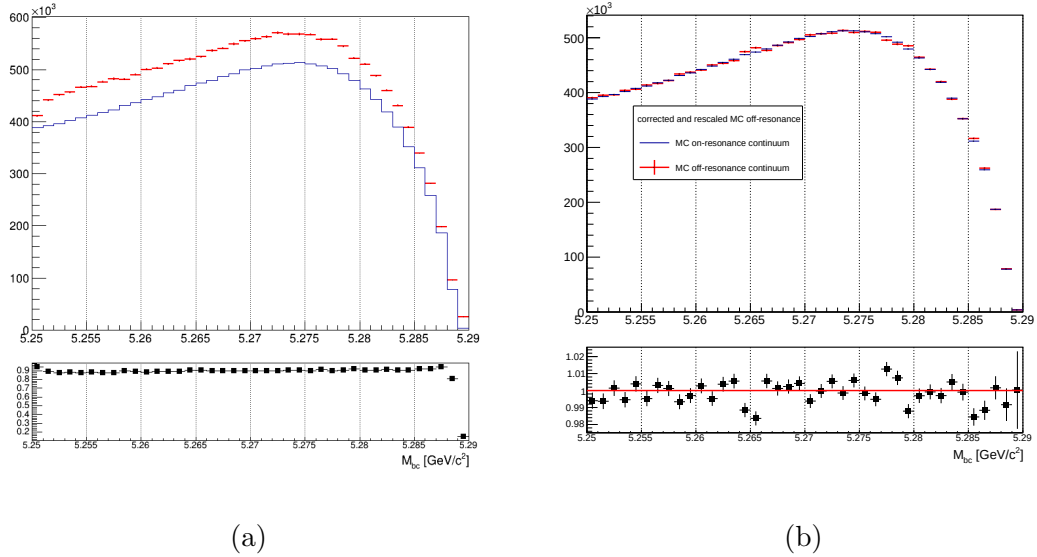


Figure (4.9) On the left: M_{bc} distributions of the MC off-resonance sample and the MC continuum sample with applied continuum suppression. On the right: M_{bc} distributions of the corrected scaled MC off-resonance and on-resonance MC continuum.

4.2 B_{tag} fit

An independent Monte Carlo stream was used to test the total fit model on tagged B meson candidates. As in the 2D fit, the parameter for the width, σ_{CB} , of the Crystal Ball is floated. The ratio between expected crossfeed background events and misreconstructed signal events is fixed from the MC. The misreconstructed signal PDF is also not fully constrained: the parameter describing the tail is free. To avoid introducing significant systematic uncertainties in the fit deriving from the M_{bc} endpoint region, where one has a smearing effect due to variations of the beam energy at the MeV level, the range for the fit is restricted to values between 5.250 and 5.287 GeV/c^2 .

4.2.1 Fit results for $B^- \rightarrow \Lambda_c^+$ decays

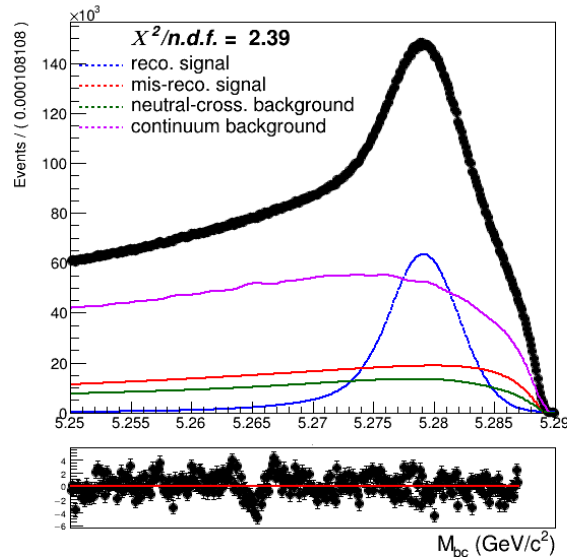


Figure (4.10) Total fit of tagged B mesons on Monte Carlo simulated data.

Yields for the reconstructed and misreconstructed signal are obtained from the fit:

	Fit results	expected
NrecSig	$4.77874 \cdot 10^6 \pm 6748$	$4.7571 \cdot 10^6 \pm 3214$
NmisSig	$5.39870 \cdot 10^6 \pm 5617$	$5.4035 \cdot 10^6 \pm 3376$

The normalization for the branching fraction is given by the NrecSig yields. The yields obtained in the fit differ by $\sim 3.2\sigma$ from the expected ones (obtained from a fit to the Total Signal events only). This discrepancy will impact the final measurement.

420 To check the stability of the fit model a toy MC study was performed with 3×10^3
 421 pseudo-datasets. No evidence for possible biases in the reconstructed signal yields was
 422 found (see Fig. 4.11).

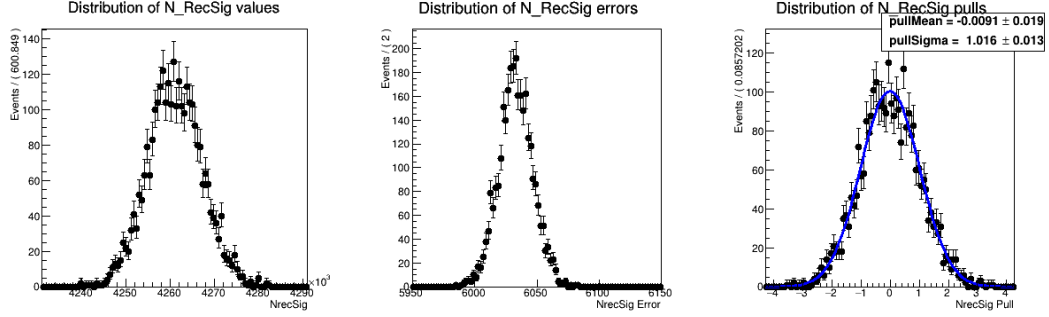


Figure (4.11) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits. (see Appendix .0.7 for the other free parameters' results)

423 4.2.2 Fit results for $B^- \rightarrow \bar{\Lambda}_c^-$ decays

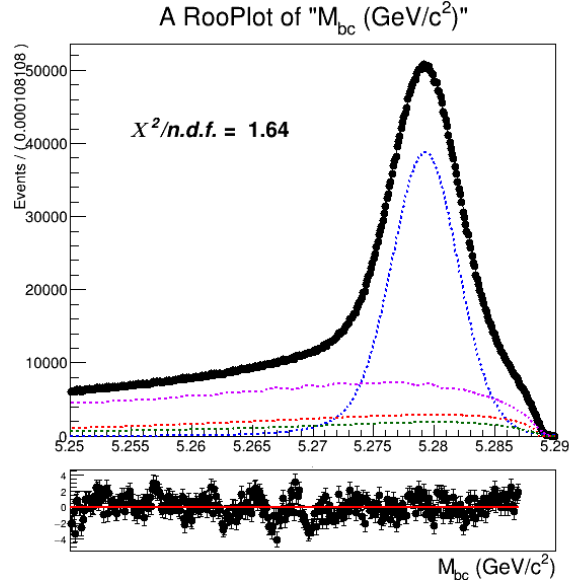


Figure (4.12) Total fit of tagged B mesons on Monte Carlo simulated data.

	Fit results	expected
NrecSig	$2.58006 \cdot 10^6 \pm 11796$	$2.58692 \cdot 10^6 \pm 1858$
NmisSig	$7.22503 \cdot 10^5 \pm 7271$	$7.13865 \cdot 10^5 \pm 1274$

425 The Total Signal (the sum NrecSig+NmisSig) calculated by the fit is 3302559 ± 4914 ,
 426 which is in excellent agreement with the true Monte Carlo value: 3299629.

But when performing a toy Monte Carlo study¹ the result show no bias on the
reconstructed signal yields (see Fig. 4.13)

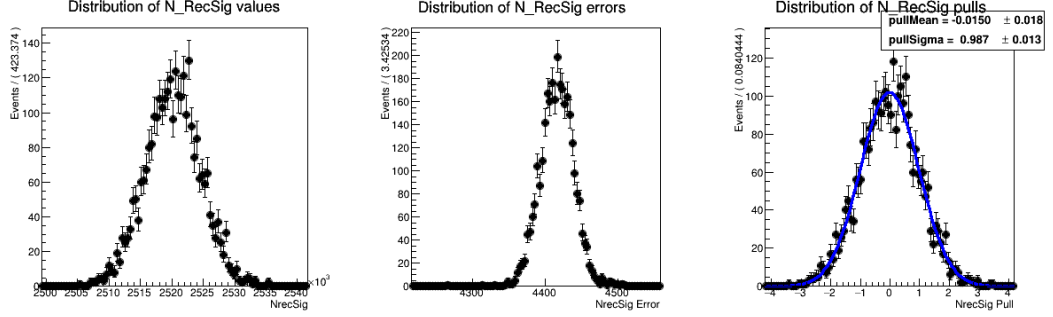


Figure (4.13) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits.

4.2.3 Fit results for $\bar{B}^0 \rightarrow \Lambda_c^+$ decays

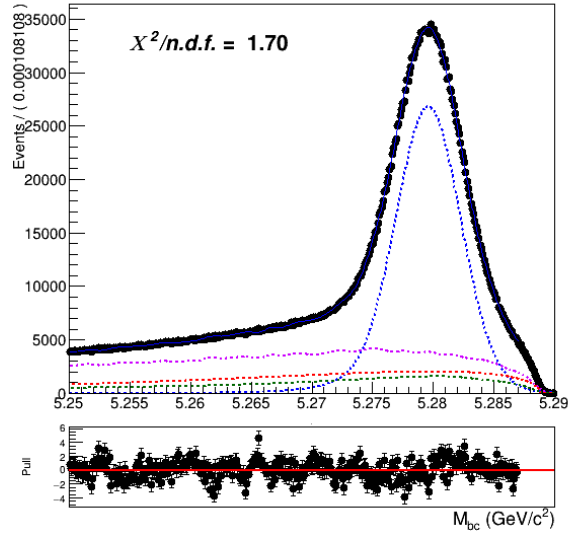


Figure (4.14) Total fit of tagged B^0 mesons on Monte Carlo simulated data.

Reconstructed and misreconstructed signal yields obtained from the fit:

¹as usual performed with 3×10^3 pseudo-datasets

432

	Fit results	expected
NrecSig	$1.7336 \cdot 10^6 \pm 23370$	
NmisSig	$5.3446 \cdot 10^5 \pm 1435$	

433

434

The Total Signal (the sum NrecSig+NmisSig) is 2268012 ± 1747 (to be compared with 2286964 from the Monte Carlo). Also in this case there's a $\sim 3\sigma$ underestimation.

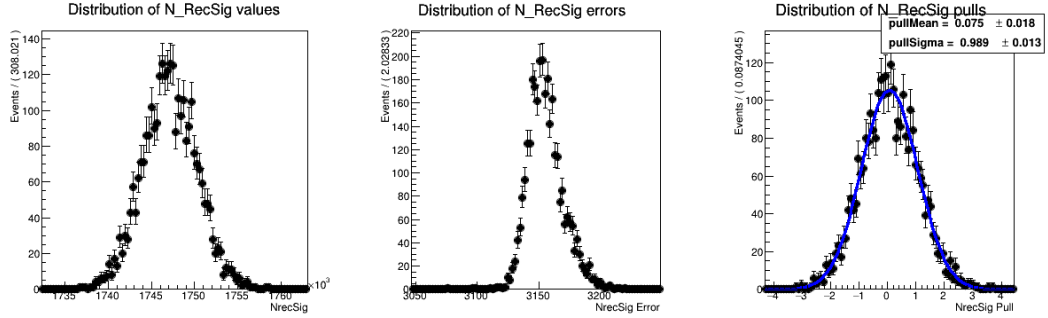


Figure (4.15) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits.

435

4.2.4 Fit results for $B^0 \rightarrow \bar{\Lambda}_c^+$ decays

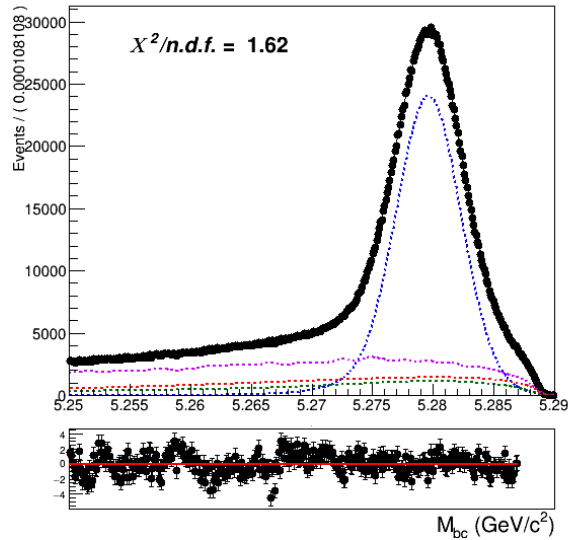


Figure (4.16) Total fit of tagged B mesons on Monte Carlo simulated data.

Reconstructed and misreconstructed signal yields obtained from the fit:

NrecSig	$1.5423 \cdot 10^6 \pm 3269$
NmisSig	$3.7173 \cdot 10^5 \pm 4872$

The Total Signal (the sum NrecSig+NmisSig) is 1914022 ± 2330 (to be compared with 1920156 from the Monte Carlo). Also in this case there is an underestimation of about a $\sim 3.7\sigma$.

The toy MC study result for the signal yields is shown in Fig. 4.17. Also in this case no hints of bias are visible.

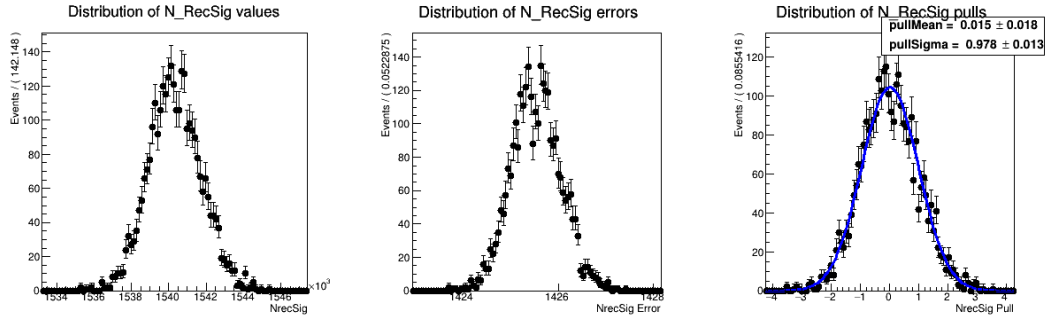


Figure (4.17) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits.

Chapter 5

Efficiencies

Here a decision was made not to rely on the estimated number of B meson pair, as it is usually done, and the absolute FEI efficiency, since the latter shows large discrepancy between MC and data (see i.e. the results reported in the PhD Thesis by M. Gelb [4] and also by J. Schwab [5]) and also it depends strongly on the signal-side (i.e. $\epsilon_{FEI}^+ \neq \epsilon_{FEI,sig}^+$). Instead, to limit the systematics, the branching ratio normalization is obtained using the fitted tagged B mesons and the ratio $\epsilon_{FEI,sig}^+/\epsilon_{FEI}^+$ measured on MC, which one can expect to be better described by the MC than the absolute FEI efficiency.

5.1 FEI Efficiencies

One needs to distinguish two different FEI efficiencies as already anticipated.

The generic FEI efficiency $\epsilon_{FEI} = \frac{n^{of tagged B}}{total n^{of events}}$ is the generic efficiency to tag correctly a B^+B^- or $B^0\bar{B}^0$ event, i.e.: the hadronic decay-chain of the tag-side B meson is correctly reconstructed, independently from signal-side. The denominator is given by all $B\bar{B}$ events present in the considered sample. The generic FEI efficiency is the one that affects the normalization in the branching ratio calculation.

The FEI efficiency that affects the numerator of the branching ratio is called signal-side-dependent efficiency $\epsilon_{FEI,sig}$. In the case one is trying to reconstruct $B^- \rightarrow \Lambda_c^+$ decays, then this efficiency is determined as the following ratio: $\epsilon_{FEI,sig}^+ = \frac{n^{of tagged B^{+/-} of B^- \rightarrow \Lambda_c^+}}{total n^{of B^- \rightarrow \Lambda_c^+ events}}$

5.1.1 Generic FEI efficiency

The number of correctly tagged B mesons is determined from the yields of reconstructed signal events obtained from a fit of the B_{tag} distributions shown already in Sec. 4.1.1. As an example Fig. 5.1 shows the same distributions shown in Fig. 4.1 where all shaping parameters have been fixed and only the normalisations of **reconstructed** and

470 misreconstructed signal were floated.

471

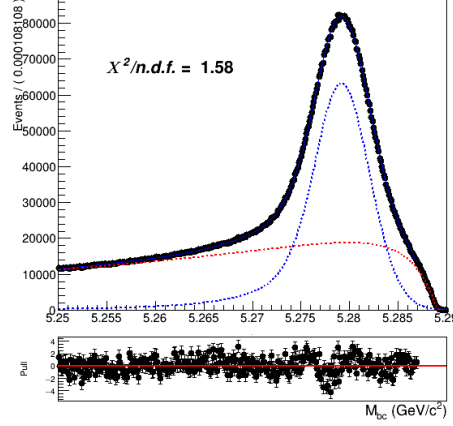


Figure (5.1) Fitted distribution of tagged charged B mesons in the charged correlated decays sample.

472 5.1.2 Signalside-dependent FEI efficiency

473 The FEI tag-side efficiency for signal events is calculated upon 10 streams of on-resonance
 474 Monte Carlo streams of simulated data in order to minimize statistical uncertainties. As
 475 an example Fig. 5.2 shows the fit performed on the tagged B mesons having in the ROE a
 476 companion decaying $B^- \rightarrow \Lambda_c^+$ or $B^+ \rightarrow \Lambda_c^-$ (charged correlated decays). Dividing the
 477 yields obtained by the fit by the total number of charged correlated decays events present
 478 in the 10 Monte Carlo streams it is possible to determine the FEI tag-side efficiency for
 479 signal events.

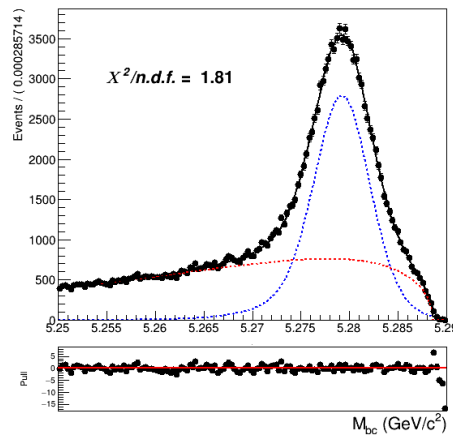


Figure (5.2) Fit of tagged B mesons in the "signal events" sample

5.2 Λ_c efficiency

The efficiency of reconstructing the Λ_c baryon after correctly tagging the B meson, can be estimated from Monte Carlo simulated data. One can again exploit the 10 streams of on-resonance Monte Carlo simulated data for to estimate the Λ_c reconstruction efficiency. The efficiency is evaluated as follows from the fraction:

$$\frac{N_{recSig}(B_{tag}, \Lambda_c)}{N_{recSig}(B_{tag}^{sig})} \quad (5.1)$$

where $N_{recSig}(B_{tag}, \Lambda_c)$ are the yields of reconstructed signal from a two dimensional fit of the corresponding inclusive decay and $N_{recSig}(B_{tag}^{sig})$ are the yields of correctly reconstructed signal from the same fit of tagged B mesons used to calculate the Signalside-dependent FEI efficiency.

5.2.1 $B^- \rightarrow \Lambda_c^+$ decays

$$\begin{aligned} \epsilon_{FEI, sig}^+ &= 0.636 \pm 0.003\% \text{ (calculated with 10 streams)} \\ \epsilon_{FEI}^+ &= 0.6379 \pm 0.0004\% \\ \frac{\epsilon_{FEI, sig}^+}{\epsilon_{FEI}^+} &= 0.997 \pm 0.005 \\ \epsilon_{\Lambda_c} &= 37.78 \pm 0.45 \% \end{aligned}$$

5.2.2 $B^- \rightarrow \bar{\Lambda}_c^-$ decays

$$\begin{aligned} \epsilon_{FEI, sig}^+ &= 0.334 \pm 0.003\% \text{ (calculated with 10 streams)} \\ \epsilon_{FEI}^+ &= 0.3461 \pm 0.0002\% \\ \frac{\epsilon_{FEI, sig}^+}{\epsilon_{FEI}^+} &= 0.965 \pm 0.009 \\ \epsilon_{\Lambda_c} &= 42.39 \pm 0.66 \% \end{aligned}$$

5.2.3 $\bar{B}^0 \rightarrow \Lambda_c^+$ decays

$$\begin{aligned} \text{UNMIXED: } \epsilon_{FEI, sig}^0 &= 0.247 \pm 0.002\% \text{ (calculated with 10 streams)} \\ \epsilon_{FEI}^0 &= 0.234 \pm 0.0002\% \end{aligned}$$

using only unmixed:

$$\frac{\epsilon_{FEI, sig}^0}{\epsilon_{FEI}^0} = 1.056 \pm 0.009$$

Including mixed events: $\epsilon_{FEI, sig}^0 = 0.253 \pm 0.002\%$ (calculated with 10 streams)

$$\epsilon_{FEI}^0 = 0.239 \pm 0.0002\%$$

$$\frac{\epsilon_{FEI, sig}^0}{\epsilon_{FEI}^0} = 1.059 \pm 0.006$$

$\epsilon_{\Lambda_c} = 39.59 \pm 0.49 \%$ (using unmixed events and calculated with 10 streams)

509 **5.2.4 $\bar{B}^0 \rightarrow \Lambda_c^-$ decays**

510 UNMIXED: $\epsilon_{FEI,sig}^0 = 0.210 \pm 0.003\%$ (calculated with 6 streams unmixed)

511 $\epsilon_{FEI}^0 = 0.207 \pm 0.0002\%$

512 using only unmixed:

513 $\frac{\epsilon_{FEI,sig}^0}{\epsilon_{FEI}^0} = 1.017 \pm 0.015$

514

515 Including mixed events: $\epsilon_{FEI,sig}^0 = 0.207 \pm 0.003\%$ (calculated with 6 streams)

516 $\epsilon_{FEI}^0 = 0.208 \pm 0.0002\%$

517 $\frac{\epsilon_{FEI,sig}^+}{\epsilon_{FEI}^+} = 0.995 \pm 0.014$

518

519 $\epsilon_{\Lambda_c} = 41.49 \pm 1.13 \%$

520 (using unmixed events and calculated with 6 streams, unfortunately didn't run on all
521 streams within 2023)

Chapter 6

Branching ratio

To measure the inclusive branching fractions

$$Br(B^{+/-} \rightarrow \Lambda_c^+ X) = \frac{N_{tag, \Lambda_c} \cdot \epsilon_{FEI}^+}{N_{tag} \cdot Br(\Lambda_c^+ \rightarrow pK^-\pi^+) \epsilon_{\Lambda_c} \epsilon_{FEI, sig}^+} \quad (6.1)$$

$$Br(B^0 \rightarrow \Lambda_c^{+/-} X) = \frac{N_{tag, \Lambda_c} \cdot \epsilon_{FEI}^0}{N_{tag} \cdot Br(\Lambda_c^+ \rightarrow pK^-\pi^+) \epsilon_{\Lambda_c} \epsilon_{FEI, sig}^0} \quad (6.2)$$

the following quantities need to be known:

- N_{tag, Λ_c} is the reconstructed signal yield obtained from a two dimensional fit of M_{bc} and $M(pK\pi)$ in the final sample.
- N_{tag} is the reconstructed signal yield obtained from the M_{bc} fit of all the tagged B mesons in the final sample.
- ϵ_{Λ_c} is the Λ_c reconstruction efficiency.
- ϵ_{FEI}^+ (ϵ_{FEI}^0) represent the hadronic tag-side efficiency for generic B^+B^- ($B^0\bar{B}^0$) events.
- $\epsilon_{FEI, sig}^+$ ($\epsilon_{FEI, sig}^0$) represents the hadronic tag-side efficiency for B^+B^- ($B^0\bar{B}^0$) events where the tagged B meson decays hadronically and the accompanying meson decays inclusively into the studied signal channel.
- $Br(\Lambda_c^+ \rightarrow pK^-\pi^+)$: the branching fraction of the decay mode used to reconstruct the Λ_c baryon.

540 6.1 charged decays branching ratios

541 Once all the factors of Ch. 6 are obtained from the MOnTe Carlo, the corresponding
 542 branching ratios can be calculated in each stream (6 streams in total, since there are only
 543 6 available for the off-resonance).

	total fit	signal fit	BELLE MC VALUE
stream 0	$(2.84 \pm 0.13)\%$	$(2.96 \pm 0.07)\%$	$(2.91 \pm 0.03)\%$
stream 1	$(2.82 \pm 0.14)\%$	$(2.99 \pm 0.07)\%$	$(2.91 \pm 0.03)\%$
stream 2	$(2.97 \pm 0.14)\%$	$(2.97 \pm 0.07)\%$	$(2.90 \pm 0.03)\%$
stream 3	$(2.76 \pm 0.14)\%$	$(2.90 \pm 0.07)\%$	$(2.91 \pm 0.03)\%$
stream 4	$(3.06 \pm 0.14)\%$	$(3.05 \pm 0.07)\%$	$(2.90 \pm 0.03)\%$
stream 5	$(2.83 \pm 0.14)\%$	$(2.84 \pm 0.07)\%$	$(2.92 \pm 0.03)\%$
average	$(2.88 \pm 0.06)\%$	$(2.95 \pm 0.03)\%$	$(2.91 \pm 0.01)\%$

Table (6.1) Monte Carlo measured branching fraction values of charged correlated decays

	total fit	signal fit	BELLE MC VALUE
stream 0	$(1.24 \pm 0.11)\%$	$(1.19 \pm 0.05)\%$	$(1.217 \pm 0.002)\%$
stream 1	$(1.24 \pm 0.11)\%$	$(1.21 \pm 0.05)\%$	$(1.218 \pm 0.002)\%$
stream 2	$(1.35 \pm 0.12)\%$	$(1.23 \pm 0.05)\%$	$(1.218 \pm 0.002)\%$
stream 3	$(1.16 \pm 0.10)\%$	$(0.20 \pm 0.05)\%$	$(1.215 \pm 0.002)\%$
stream 4	$(1.44 \pm 0.13)\%$	$(1.21 \pm 0.05)\%$	$(1.218 \pm 0.002)\%$
stream 5	$(1.04 \pm 0.11)\%$	$(1.15 \pm 0.05)\%$	$(1.217 \pm 0.002)\%$
average	$(1.25 \pm 0.05)\%$	$(1.20 \pm 0.02)\%$	$(1.217 \pm 0.001)\%$

Table (6.2) Monte Carlo measured branching fraction values of charged anticorrelated decays

6.2 extraction of neutral decays branching fractions

Due to the mixing phenomenon of neutral B mesons the measurement requires an intermediate step in order to correct the production rates of Λ_c baryons obtained in the recoil of the reconstructed neutral B mesons for the $B^0\bar{B}^0$ mixing.

Being χ_d the mixing parameters, on average, if one looks at the recoil of a correctly reconstructed B^0 ($\bar{B}^0$) meson, $1 - \chi_d$ of the time the reconstructed Λ_c baryon will have decayed from a $\bar{B}^0$ (B^0) and χ_d of the time from B^0 ($\bar{B}^0$), due to mixing.

i.e.:

$$\begin{cases} \mathcal{B}(B_{tag}^0, \Lambda_c^-) = (1 - \chi_d)\mathcal{B}_c + \chi_d\bar{\mathcal{B}}_c \\ \mathcal{B}(\bar{B}_{tag}^0, \Lambda_c^-) = (1 - \chi_d)\bar{\mathcal{B}}_c + \chi_d\mathcal{B}_c \end{cases} \quad (6.3)$$

are the two equations corresponding to the spurious correlated/ anticorrelated branching ratios, with χ_d factor of missing events and χ_d amount of events that belong actually to the other type of decays. Solving those simultaneous equations one obtains the mixing-corrected branching ratios $\mathcal{B}_c$ (correlated decays) and $\bar{\mathcal{B}}$ (anticorrelated decays):

$$\begin{cases} \mathcal{B}_c = \frac{\mathcal{B}(B_{tag}^0, \Lambda_c^-) - \chi_d(\mathcal{B}(B_{tag}^0, \Lambda_c^-) + \mathcal{B}(\bar{B}_{tag}^0, \Lambda_c^-))}{1 - 2\chi_d} \\ \bar{\mathcal{B}}_c = \frac{\mathcal{B}(\bar{B}_{tag}^0, \Lambda_c^-) - \chi_d(\mathcal{B}(\bar{B}_{tag}^0, \Lambda_c^-) + \mathcal{B}(B_{tag}^0, \Lambda_c^-))}{1 - 2\chi_d} \end{cases} \quad (6.4)$$

First a check of the decays that didn't experience mixing is performed. For this check, samples of signal events that didn't experience mixing were used to obtain signal yields (see Figures below).

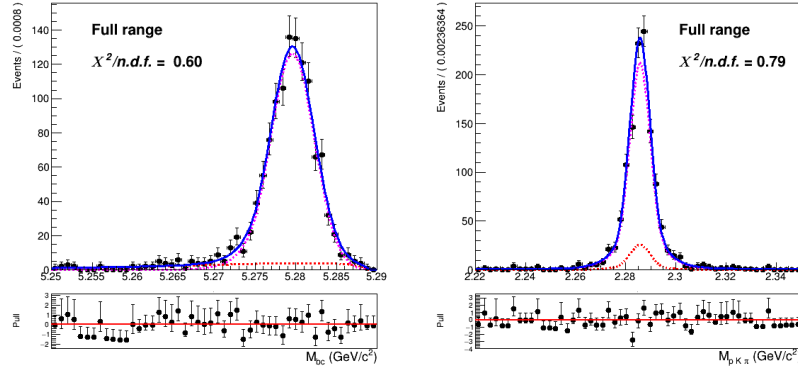


Figure (6.1) Unmixed neutral correlated total signal events in M_{bc} and $M(pK\pi)$

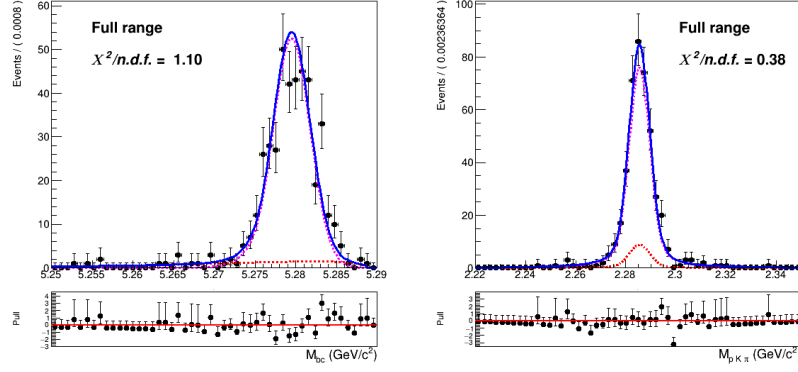


Figure (6.2) Unmixed neutral anticorrelated total signal events in M_{bc} and $M(pK\pi)$

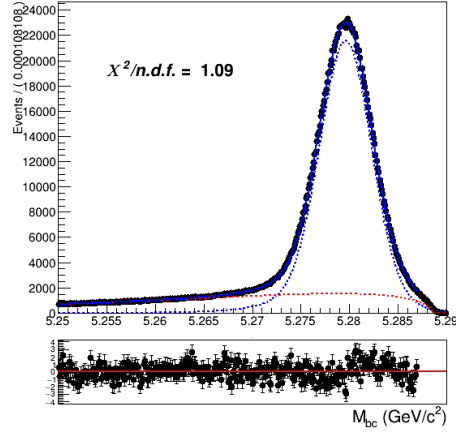


Figure (6.3) Unmixed neutral B mesons in the correlated decay sample.

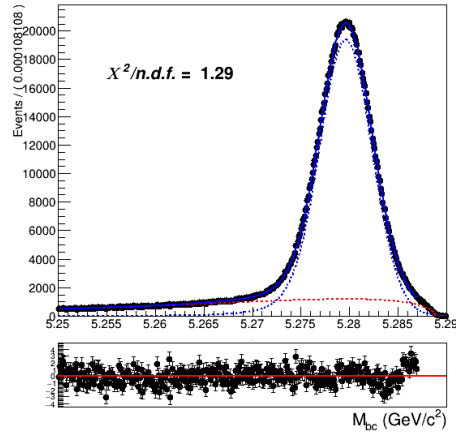


Figure (6.4) Unmixed neutral B mesons in the anticorrelated decay sample.

6.2.1 unmixed neutral decays branching ratios

The following two tables report the values of branching fractions for the neutral decays' events that didn't experience mixing (i.e.: $\mathcal{B}_c$ and $\bar{\mathcal{B}}_c$), obtained with the fits shown in

	signal fit	BELLE MC VALUE
stream 0	$(3.48 \pm 0.11)\%$	$(3.439 \pm 0.003)\%$
stream 1	$(3.54 \pm 0.11)\%$	$(3.431 \pm 0.003)\%$
stream 2	$(3.48 \pm 0.11)\%$	$(3.433 \pm 0.003)\%$
stream 3	$(3.47 \pm 0.11)\%$	$(3.433 \pm 0.003)\%$
stream 4	$(3.34 \pm 0.11)\%$	$(3.434 \pm 0.003)\%$
stream 5	$(3.45 \pm 0.11)\%$	$(3.436 \pm 0.003)\%$
average	$(3.46 \pm 0.05)\%$	$(3.434 \pm 0.001)\%$

Table (6.3) Measured branching fraction values obtained using only unmixed events for the six different streams (only statistical uncertainties are displayed) and its average.

	signal fit	BELLE MC VALUE
stream 0	$(1.35 \pm 0.09)\%$	$(1.300 \pm 0.002)\%$
stream 1	$(1.30 \pm 0.07)\%$	$(1.299 \pm 0.002)\%$
stream 2	$(1.31 \pm 0.07)\%$	$(1.300 \pm 0.002)\%$
stream 3	$(1.31 \pm 0.07)\%$	$(1.299 \pm 0.002)\%$
stream 4	$(1.22 \pm 0.09)\%$	$(1.295 \pm 0.002)\%$
stream 5	$(1.30 \pm 0.07)\%$	$(1.298 \pm 0.002)\%$
average	$(1.30 \pm 0.03)\%$	$(1.299 \pm 0.001)\%$

Table (6.4) Measured branching fraction values obtained using only unmixed events for the six different streams (only statistical uncertainties are displayed) and its average.

The good agreement within statistical errors of the obtained branching fractions excluding mixing events guarantees that the procedure to obtain the branching fractions is correct and any possible discrepancy observed when mixing is included does not depend on it.

568 . . . Adding mixed events:

569

570 • Neutral correlated decays

	total fit	signal fit	BELLE MC VALUE
stream 0	$(3.74 \pm 0.26)\%$	$(3.41 \pm 0.14)\%$	$(3.439 \pm 0.003)\%$
stream 1	$(3.69 \pm 0.26)\%$	$(3.65 \pm 0.14)\%$	$(3.431 \pm 0.003)\%$
stream 2	$(3.53 \pm 0.25)\%$	$(3.42 \pm 0.14)\%$	$(3.433 \pm 0.003)\%$
stream 3	$(3.58 \pm 0.25)\%$	$(3.43 \pm 0.14)\%$	$(3.433 \pm 0.003)\%$
stream 4	$(3.58 \pm 0.25)\%$	$(3.34 \pm 0.14)\%$	$(3.434 \pm 0.003)\%$
stream 5	$(3.57 \pm 0.26)\%$	$(3.37 \pm 0.14)\%$	$(3.436 \pm 0.003)\%$
average	$(3.58 \pm 0.10)\%$	$(3.44 \pm 0.06)\%$	$(3.434 \pm 0.001)\%$

Table (6.5) Measured branching fraction values obtained using the results listed in for the six different streams (only statistical uncertainties are displayed) and its average.

571 XX

572

573 • Neutral anticorrelated decays

	total fit	signal fit	BELLE MC VALUE
stream 0	$(1.13 \pm 0.13)\%$	$(1.22 \pm 0.07)\%$	$(1.300 \pm 0.002)\%$
stream 1	$(1.13 \pm 0.13)\%$	$(1.27 \pm 0.07)\%$	$(1.299 \pm 0.002)\%$
stream 2	$(1.20 \pm 0.13)\%$	$(1.28 \pm 0.08)\%$	$(1.300 \pm 0.002)\%$
stream 3	$(1.67 \pm 0.16)\%$	$(1.29 \pm 0.07)\%$	$(1.299 \pm 0.002)\%$
stream 4	$(1.33 \pm 0.15)\%$	$(1.20 \pm 0.07)\%$	$(1.295 \pm 0.002)\%$
stream 5	$(1.24 \pm 0.13)\%$	$(1.30 \pm 0.08)\%$	$(1.298 \pm 0.002)\%$
average	$(1.28 \pm 0.06)\%$	$(1.26 \pm 0.03)\%$	$(1.299 \pm 0.001)\%$

Table (6.6) Measured branching fraction values obtained using the results listed in for the six different streams (only statistical uncertainties are displayed) and its average.

574 6.3 Systematics

575 6.3.1 Continuum background

576 2 sources of systematics:

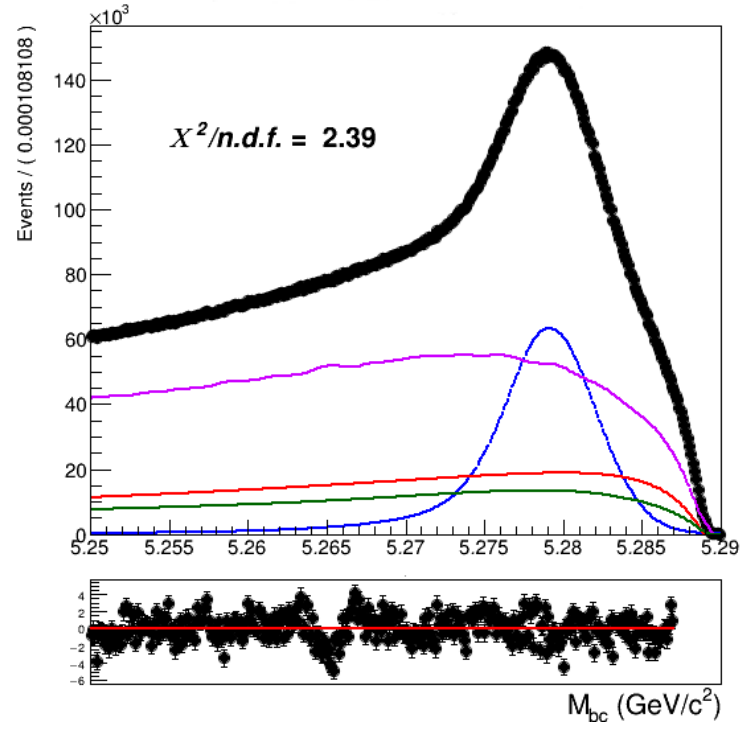
577 • difference in the shapes

578 • different impact of the Continuum Suppression cut

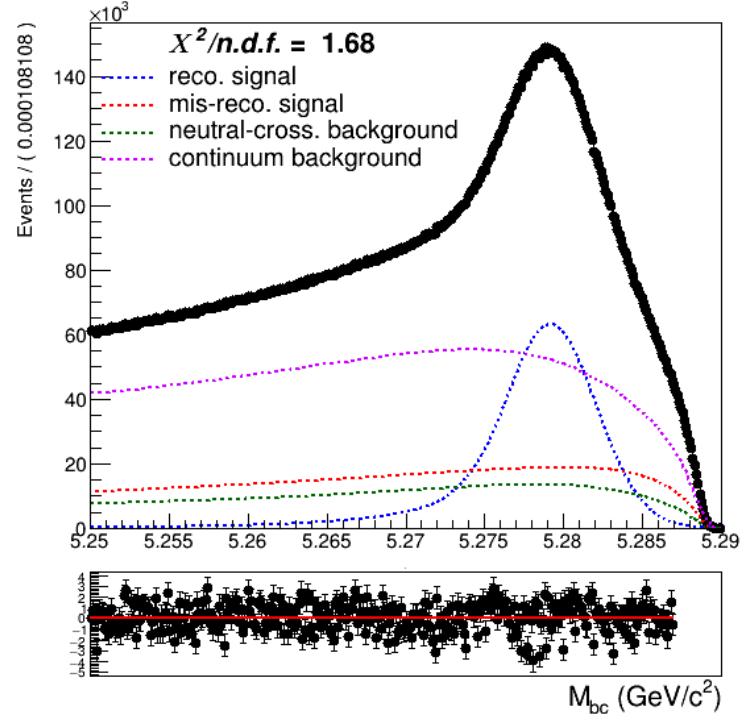
579 To estimate the first type of uncertainty two-dimensional fits with varied parameters'
580 values by their uncertainties (a fit with $+\text{err}$ and $-\text{err}$) were performed.

581 The rescaling procedure used to model the on-resonance continuum distribution is far
582 from perfect, especially in the case of the B_{tag} fit. As one can see from Fig. 6.5a:

583



(a)



(b)

Figure (6.5) Above: Fitted M_{bc} of charged correlated decays with scaled off-resonance continuum distribution. Below: same M_{bc} distribution with the true continuum distribution.

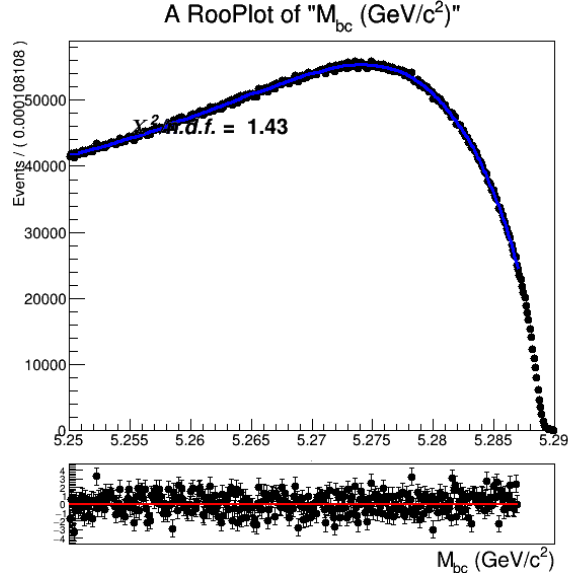


Figure (6.6) M_{bc} on-resonance MC distribution of charged correlated decays.

584 The bumps visible for $M_{bc} < 5.27 \text{ GeV}/c^2$ in Fig. 6.5a are a consequence of the
 585 imperfect scaling procedure used to model the continuum background. In fact, they
 586 disappear completely (in Fig. 6.5b) when the same fit is performed using the pdf obtained
 587 directly from the on-resonance distribution (Fig. 6.6) There is a significant impact on the
 588 fit results, the values from the B_{tag} fit in the charged correlated channel are reported here
 589 below:

	Fit with scaled continuum	Fit with true continuum
N_{recSig} yields	$4.7787 \cdot 10^6 \pm 6.75 \cdot 10^3$	$4.7474 \cdot 10^6 \pm 8.01 \cdot 10^3$

Table (6.7) Signal yields obtained in B_{tag} fits shown in Fig. 6.5a and Fig. 6.5b.

590 The yields are to be compared with the values obtained when fitting the total signal
 591 distribution shown in Fig. 4.1: $N_{recSig} = 4.7571 \cdot 10^6 \pm 3.23 \cdot 10^3$. The continuum modeling
 592 alone is responsible for a discrepancy of about 3.2σ on the yields. This discrepancy
 593 and the impact on the branching fraction are accounted in the systematic uncertainties
 594 derived from the continuum modeling. This source of systematics is responsible for
 595 0.02 % uncertainty on the branching ratio. The other type of systematic uncertainty in
 596 modeling the continuum is originated by the continuum suppression cut having a slightly
 597 different efficiency on data (as a consequence of the shift in off-resonance MC and data
 598 visible in the the *foxWolframR2* distribution in Fig. 3.19). As already discussed, it
 599 originates a possible discrepancy of about 2.25% in continuum background events in the
 600 two dimensional fit (and only 1.25% in the B_{tag} fit). The statistical uncertainty on this
 601 fraction of events can also be taken into account as systematics. Being the number of
 602 events in the off-resonance data sample without the continuum suppression applied is

very small, the uncertainty in the mentioned fraction of events is negligible compared to the statistical uncertainty on the on-resonance continuum background events in MC: it would account for 0.002% on the BR value. Therefore, this second source of uncertainty is not taken into account. There would be also a third of systematic uncertainty given by potential difference in the on-/off-resonance correction between data and MC, but there's no way one can estimate it properly.

6.3.2 Crossfeed ratio

The ratio crossfeed/misreconstructed signal is kept fixed in the B_{tag} fit to the MC value. This choice was made according to the fact that the two categories of events have a similar origin: in both cases the B mesons were not correctly reconstructed, either because of missing or wrongly added particles (misreconstructed signal events) or, in the case of crossfeed events because the tagged B meson was not the required one (B^0 meson instead of a $B^{+/-}$ meson). The ratio between these two categories of events is therefore expected to be very similar in MC and data, though there's no guarantee that the efficiency to reconstruct them is the same in data and consequently the above mentioned ratios could differ on data. Unfortunately there's no direct way to have an estimate of the possible discrepancy for them.

In [4] (and previously in [5]) it was found that there's a substantial difference in terms of tagging efficiency for FEI applied on Monte Carlo and on Belle data, being the discrepancy around $\sim 20\%$. We can assume that the efficiency for the two categories of events on data will both differ of that value and the ratio of the events being the same MC value, but in absence of any other method to estimate the uncertainty on it one can consider a maximal discrepancy of 20% between Monte Carlo and data to study the impact on the yields¹.

6.3.3 Crossfeed B_{tag} PDF

Since also the shapes of the PDFs describing the crossfeed background in the B_{tag} fit are fully fixed to the ones determined with the limited Monte Carlo statistics, also their statistical uncertainties need to be taken into account as possible source of systematics. To estimate this source of uncertainty fits with varied parameters' values by their uncertainties (a fit with +err and -err) were performed.

6.3.4 2DFit crossfeed normalization

As discussed in the 2D simultaneous fit section, in each decay channel the normalization of crossfeed events is estimated using the fitted reconstructed signal events of the corresponding crossfeeding channel in the as follows:

$$N_{cross} = N_{recSig}^{correlated} \left(\frac{\epsilon_{cross}}{\epsilon_{recSig}} \right)^{correlated} + N_{recSig}^{anticorrelated} \left(\frac{\epsilon_{cross}}{\epsilon_{recSig}} \right)^{anticorrelated} \quad (6.5)$$

¹This method was also validated with the control decay sample and the originated uncertainty is well within the PDG reported ones.

where ϵ_{recSig} and ϵ_{cross} are respectively the efficiency to reconstruct correctly signal events in the crossfeeding channel and the efficiency of reconstructing those events as crossfeed in the channel of interest.

The uncertainties of those two efficiencies may cause an overestimation or underestimation of the crossfeed in the channel of interest. To estimate the systematic uncertainty deriving from it, the simultaneous fits have been performed modifying the ratio of the efficiencies by the respective uncertainties maximising the shift, i.e. using $\frac{\epsilon_{cross}+\sigma}{\epsilon_{recSig}-\sigma}$ and $\frac{\epsilon_{cross}-\sigma}{\epsilon_{recSig}+\sigma}$

6.3.5 2DFit M_{bc} peaking/flat PDFs

As discussed in ?? the shapes of the PDFs describing the generic background (peaking or flat in M_{bc}) are kept fixed in the two dimensional fit. This source of systematics is handled in the same way as the crossfeed background in the B_{tag} : fits with varied parameters' values by their uncertainties were performed (procedure is repeated for M_{bc} peaking and flat background).

6.3.6 PID efficiency correction

The PID selection efficiency for the three charged particles in the signal decay needs to be corrected on MC due to various differences, when comparing to data. The Belle PID group has prepared a set of correction factors and tables of systematic uncertainties for PID efficiencies for all charged particles. The proton identification efficiency was studied in [6]. The inclusive Λ^0 decay $\Lambda^0 \rightarrow p\pi^-$ was used to examine the proton identification efficiency difference between data and MC in *Belle*. The datasets for the SVD1 and SVD2 periods are treated separately, and the efficiency ratio dependence on proton charge, momentum and polar angle is considered. The study is done for the proton ID cut values 0.6, 0.7, 0.8 and 0.9². The binning on the momentum starts at 0.2 GeV. The proton ID efficiency is defined as

$$\epsilon_{PID} = \frac{\text{number of } p\text{ tracks identified as } p}{\text{number of } p\text{ tracks}}$$

and the comparison between MC efficiency and data efficiency by a double ratio defined as

$$R_p = \epsilon^{data} / \epsilon^{MC}$$

The average proton ID correction is estimated to be: $R_p = 0.969 \pm 0.003$.

The kaon identification efficiency was studied in detail in Belle Note 779 [7] (http://belle.kek.jp/secured/belle_note/gn779/bn779.ps.gz). The decay $D^{*+} \rightarrow D^0\pi^+$

²Here, proton ID cut value X means $\mathcal{L}_{p/K} > X$ and $\mathcal{L}_{p/\pi} > X$

671 followed by $D^0 \rightarrow K^- \pi^+$, was used to examine it. As for the proton identification efficiency
 672 it considers the dependence on Kaon charge, momentum and polar angle and same ID cut
 673 values³

674 **6.3.7 Tracking efficiency**

675 The tracking efficiency correction for tracks with $p > 0.2$ GeV is studied in [?] . The track
 676 finding efficiency is measured by comparing the number of partially and fully reconstructed
 677 D^* decays. Based on this analysis, the measured difference in the tracking efficiency in
 678 data and MC is $R = (-0.13 \pm 0.30 \pm 0.10)\%$ per track. Since the ratio is much smaller than
 679 the statistical uncertainty, this ratio is not applied as a correction, instead, as done in [?],
 680 a systematic uncertainty of 0.35% per track is applied. The total systematic uncertainty
 681 is the sum over the three charged tracks used to reconstruct the Λ_c baryon: 1.05%.

682 **6.3.8 $B^- \rightarrow \Lambda_c^+$ decays**

683 Regarding the systematic uncertainties deriving from the continuum modeling in the
 684 two-dimensional fit are estimated as already described in ?? one obtains an absolute value
 685 of 0.04% on the branching fraction.

686 Instead the imperfect modeling of the continuum in the B_{tag} fit is responsible for a 0.02 %
 687 on the branching fraction in absolute value.

688 Continuum modeling uncertainties account for 0.05% total uncertainties on the branching
 689 fraction in absolute value.

690 To estimate the systematics derived from the crossfeed/misreconstructed signal ratio,
 691 in the B_{tag} fit the number of crossfeed events were varied artificially in order to have $\pm 20\%$
 692 different ratio, keeping the previously determined Monte Carlo ratio fixed.

	$-\sigma$	$+\sigma$	$\pm\bar{\sigma}$
offset	20010	9330	14670

Table (6.8) Offsets on the N_{recSig} signal yields obtained varying of $\pm 20\%$ the ratio of crossfeed/misreconstructed events in the B_{tag} fit and mean deviations reported in the last column.

693 The offsets that a potential $\pm 20\%$ different crossfeed/misreconstructed signal ratio
 694 produce results in 0.01% systematic uncertainty on the branching fraction.

695 The systematic uncertainties deriving from the crossfeed background modeling in the B_{tag}
 696 fit is estimated following the same procedure described for continuum modeling systematics
 697 in the two dimensional fit.

³Here, Kaon ID cut value X means $\mathcal{L}_{K/\pi} > X$, so for values below X the tracks are identified as pions

698 In the table below the signal yields' offsets are listed changing the parameters within their
699 uncertainties, and the mean offsets value used to calculate the expected uncertainty on the
700 BR value. The resulting absolute systematic uncertainty is about 0.01% on the BR value.

	$-\sigma$	$+\sigma$	$\pm\bar{\sigma}$
offset	14810	17280	16045

Table (6.9) Offsets on the N_{recSig} signal yields obtained varying the parameters of crossfeed background PDFs within their uncertainties in the B_{tag} fit and mean deviations reported in the last column.

701 The ratio between the two FEI efficiencies is: $\frac{\epsilon_{FEI,sig}^+}{\epsilon_{FEI}^+} = 0.965 \pm 0.009$
702 The uncertainty on this value originates a systematic uncertainty of 0.01% on the Br
703 value. The Λ_c reconstruction efficiency is determined to be $\epsilon_{\Lambda_c} = 37.78 \pm 0.45\%$. The
704 systematic uncertainty originated by its uncertainty is 0.04% on the Br value.
705
706 For Kaons and Pions the average ID correction is estimated to be $R_K = 0.853 \pm 0.010$
707 and $R_\pi = 0.983 \pm 0.008$ respectivley.
708 overall correction factor: 0.813 ± 0.013 The final PID efficiency systematic error is
709 determined to be 0.05% on the branching fraction value.
710 The systematic uncertainty of 0.35% per track for the tracking efficiency instead results in
711 0.01% uncertainty on the branching fraction value.

source	relative (%)	absolute (%)
Continuum modeling	1.77	0.05
B_{tag} fit Crossfeed fraction	0.35	0.01
B_{tag} fit Crossfeed PDF shape	0.35	0.01
2DFit crossfeed normalization	0.70	0.02
M_{bc} peaking PDF shape	0.35	0.01
M_{bc} flat PDF shape	0.35	0.01
$\epsilon_{FEI,sig}^+/\epsilon_{FEI}^+$	0.35	0.01
ϵ_{Λ_c}	1.41	0.04
PID	1.75	0.05
Tracking efficiency	0.35	0.01
Total	3.18	0.09

Table (6.10) Systematic uncertainties in the determination of the $B^+ \rightarrow \bar{\Lambda}_c^- X$ branching fraction in %.

6.3.9 $B^+ \rightarrow \Lambda_c^+$ decays

Regarding the systematic uncertainties deriving from the continuum modeling in the two-dimensional fit are estimated one obtains an absolute value of 0.01% on the branching fraction.

Instead the imperfect modeling of the continuum in the B_{tag} fit is responsible for a 0.02 % on the branching fraction in absolute value.

Continuum modeling uncertainties account for 0.02% total uncertainties on the branching fraction in absolute value.

Regarding the crossfeed/misreconstructed signal ratio in the B_{tag} fit, the produced offsets are displayed here below.

	$-\sigma$	$+\sigma$	$\pm\bar{\sigma}$
offset	38890	21220	30055

Table (6.11) Offsets on the N_{recSig} signal yields obtained varying of $\pm 20\%$ the ratio of crossfeed/misreconstructed events in the B_{tag} fit and mean deviations reported in the last column.

The offsets that a potential $\pm 20\%$ different crossfeed/misreconstructed signal ratio produce results in 0.01% systematic uncertainty on the branching fraction, as in the charged correlated decays.

The systematic uncertainties deriving from the crossfeed background modeling in the B_{tag} fit is about 0.02%. In the table below the signal yields' offsets are listed changing the parameters within their uncertainties, and the mean offsets value used to calculate the expected uncertainty on the BR value.

	$-\sigma$	$+\sigma$	$\pm\bar{\sigma}$
offset	17960	68700	43330

Table (6.12) Offsets on the N_{recSig} signal yields obtained varying the parameters of crossfeed background PDFs within their uncertainties in the B_{tag} fit and mean deviations reported in the last column.

source	relative (%)	absolute (%)
Continuum modeling	1.61	0.02
B_{tag} fit Crossfeed fraction	0.12	0.01
B_{tag} fit Crossfeed PDF shape	1.93	0.02
2DFit crossfeed normalization	3.23	0.04
M_{bc} peaking PDF shape	0.80	0.01
M_{bc} flat PDF shape	-	-
$\epsilon_{FEI,sig}^+/\epsilon_{FEI}^+$	0.81	0.01
ϵ_{Λ_c}	1.61	0.02
PID	1.26	0.01
Tracking efficiency	0.92	0.01
Total	4.80	0.06

Table (6.13) Systematic uncertainties in the determination of the $B^+ \rightarrow \bar{\Lambda}_c^+ X$ branching fraction.

Chapter 7

Control channel

7.1 ...

The procedure used to discriminate correctly reconstructed signal events from mis-reconstructed ones and described in 2.3 was validated on the flavor correlated $B \rightarrow D^0$ channel.

More precisely ...

$$B^+ \rightarrow D^0 X, D^0 \rightarrow K^+ \pi^-$$

about 30 times more frequent than the corresponding decay into the Λ_c baryon. Due to the much larger statistics and its current relatively precise measured value (see PDG), it is a suitable decay channel to make sure that the method applied doesn't create any bias.

7.1.1 Dataset used

For this analysis the amount of data and Monte Carlo simulated data used was restricted to the SVD2 period: experiments ranging from 31 to 65. This choice was made to save processing time, anyway most of the $B\bar{B}$ meson pairs were produced in this range of experiments (620×10^6 out of almost 800×10^6).

7.1.2 Event selection and reconstruction

The approach used for the inclusive decays reconstruction is the same as for the $B \rightarrow \Lambda_c$ analysis. The same FEI training was used, though excluding the signal decay $D^0 \rightarrow K^+ \pi^-$ from the decay chains used by the FEI to reconstruct the B_{tag} . Same preliminary selection criteria were applied to the tag-side B meson candidates as well.

In the *rest of event* (ROE) of the reconstructed B_{tag} meson, to select $D^0 \rightarrow K^+ \pi^-$ signal candidates, the following event selection criteria are applied:

- $dr < 2$ cm and $|dz| < 4$ cm

756 • $\frac{\mathcal{L}_K}{\mathcal{L}_K + \mathcal{L}_\pi} > 0.6$

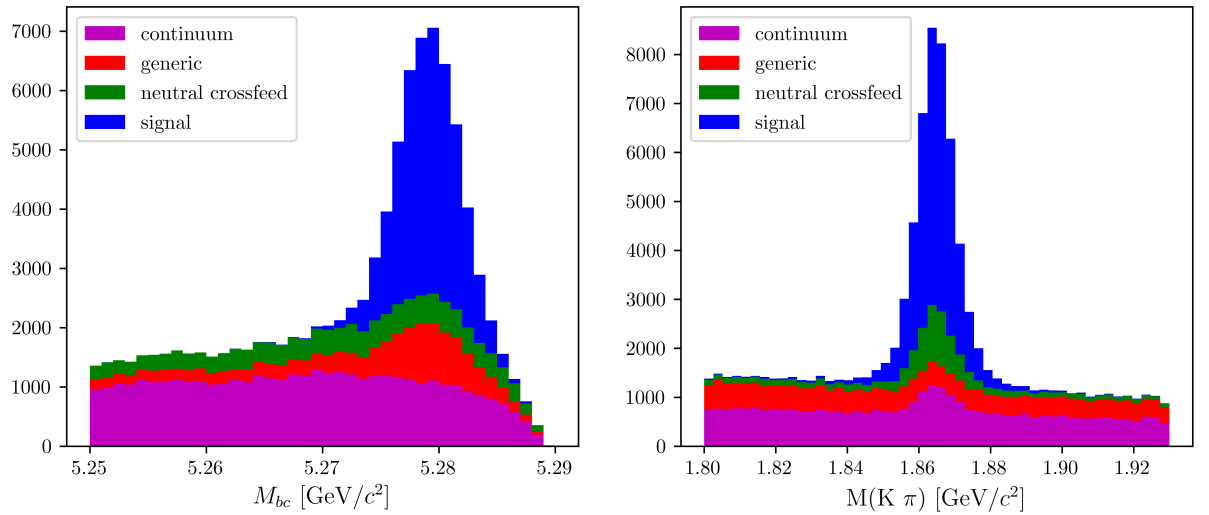
757 For the D^0 candidates a vertex fit is performed with **TreeFitter**, requiring it to converge.
 758 If there are more than one D^0 combination, then the best candidate based on the χ^2
 759 probability is chosen. The D^0 signal region is defined to be $|M_{D^0} - m_{D^0}| < 30 \text{ MeV}/c^2$
 760 ($\sim 3\sigma$), where m_{D^0} is the nominal mass of D^0 .
 761

7.1.3 Signal selection optimization

763 Following the same procedure as for the $B \rightarrow \Lambda_c$ analysis, the optimized selection cuts
 764 obtained for the event based ratio of the 2-nd to the 0-th order Fox-Wolfram moments,
 765 the B_{tag} signal probability and the momentum of the D^0 candidates in the center of mass
 766 system are¹

- 767 • $foxWolframR2 < 0.3$
 768 • $SignalProbability > 0.004$
 769 • $p_{CMS}^{D^0} > 1 \text{ GeV}/c^2$

770 Figure ?? shows the distributions of M_{bc} and invariant mass in the signal region² for the
 771 $B^- \rightarrow D^0 X$ reconstructed events after the selection cuts were applied.



¹illustrative plots can be found in Appendix

²signal region: $M_{bc} > 5.27 \text{ GeV}/c^2$ and $|M_{D^0} - m_{D^0}| < 30 \text{ MeV}/c^2$

7.1.4 Probability Density Functions (PDFs) for two dimensional fit

As already said the main goal of the control sample analysis is to ensure that the method used to extract the signal yields discriminating the correctly reconstructed from the misreconstructed signal events by fitting is valid. The reconstructed events in M_{bc} are fitted with a Crystal Ball, the misreconstructed signal with a Novosibirsk function. As in the $B \rightarrow \Lambda_c$ analysis both components have a correspondent peak in the D^0 mass which is fitted with a sum of three gaussians with a common mean. The fitted distribution of M_{bc} and $M(\pi K)$ are shown in Fig. 7.1 with signal MC sample.

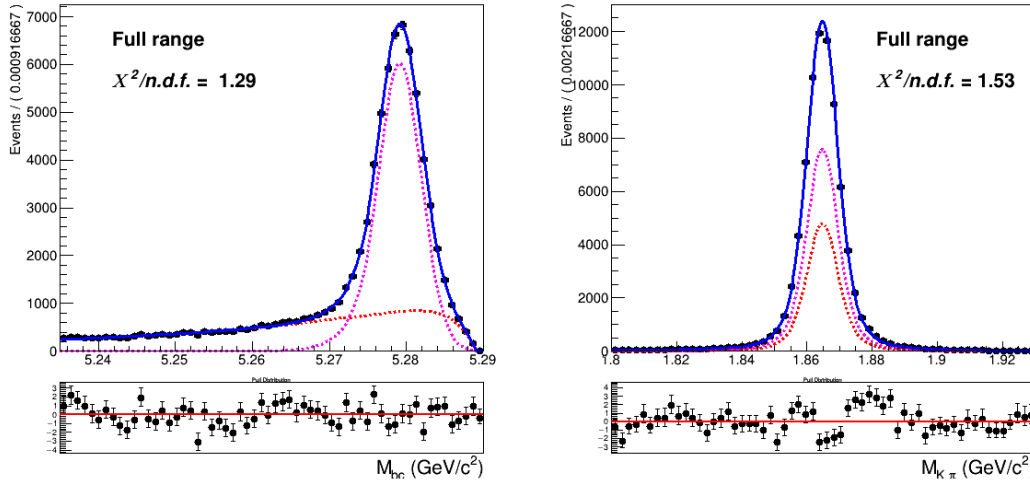


Figure (7.1) Two dimensional fit of total signal events in M_{bc} and $M(pK\pi)$ (in magenta reconstructed signal PDF, misreconstructed signal PDF in red)

Besides the misreconstructed signal the other background components are:

- **generic** (charged B) background
- **crossfeed** (neutral B) background
- **continuum** background

Generic background

The generic background deriving from other B^+B^- events presents a similar shape in M_{bc} : it is fitted again with a sum of Crystal Ball and Novosibirsk function. Instead the distribution in the D^0 mass is fitted with a sum of first order Polynomial function and a small gaussian peak, which is due to the small amount of flavor anti-correlated $B^+ \rightarrow D^0$ reconstructed events (see Fig. 7.2). The total two-dimensional PDF is a product of the one-dimensional PDFs in M_{bc} and D^0 mass:

$$P_{B,D^0}^{GenBkg}(M_{bc}, M(K\pi)) = [\Gamma_{CB}(M_{bc}) + \Gamma_{Nov}(M_{bc})] \times [\rho_{pol1}(M(K\pi)) + \rho_G(M(K\pi))] \quad (7.1)$$

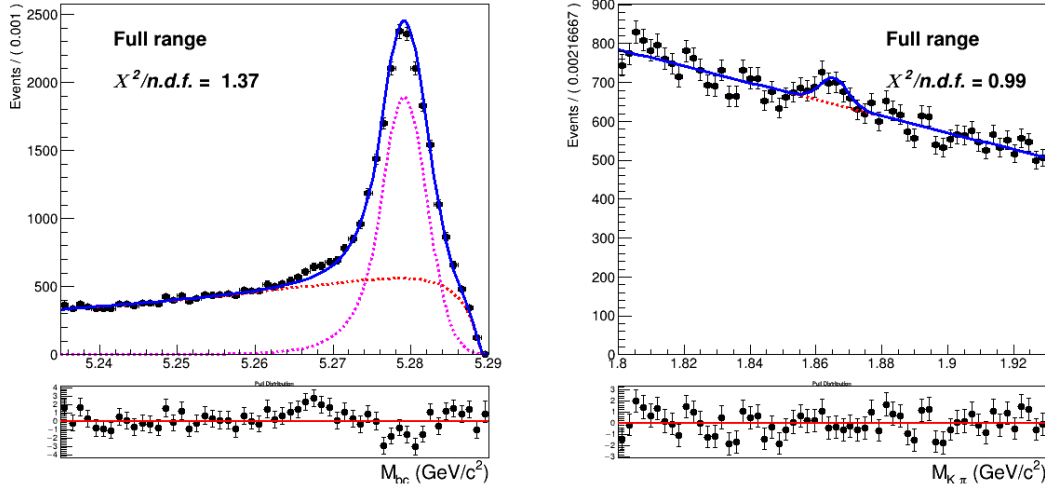


Figure (7.2) Two dimensional fit of generic (B^+B^-) background events in M_{bc} and $M(K\pi)$.

792 **Crossfeed background** The crossfeed background deriving from $B^0\bar{B}^0$ events is shown
 793 in Fig. 7.3 The M_{bc} distribution is fitted with a sum of Novosibirsk and Argus functions.
 794 The distribution in the D^0 mass is fitted with a first order Chebyshev polynomial and the
 795 D^0 mass peak is fitted with the same sum of three gaussians used to describe the signal
 796 peak (same parametrization used already in $B \rightarrow \Lambda_c$ analysis).

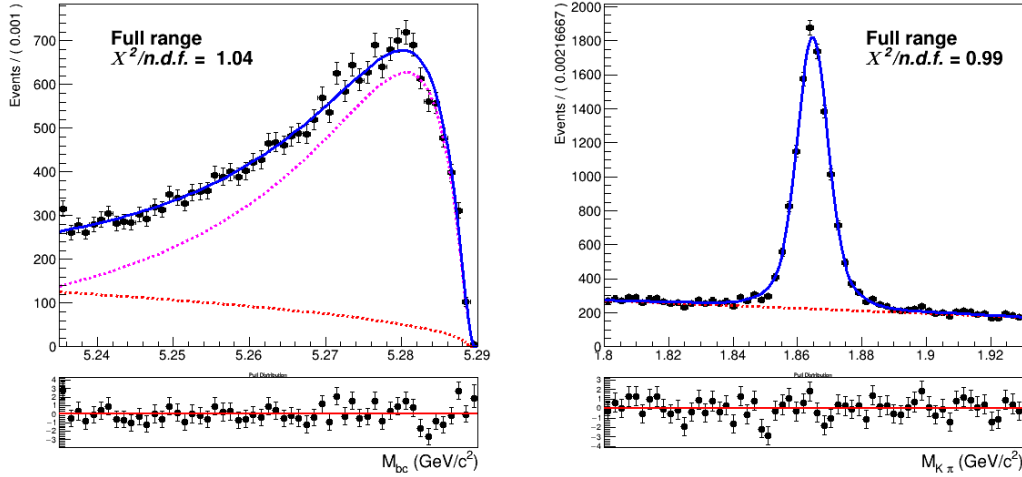


Figure (7.3) Two dimensional fit of crossfeed ($B^0\bar{B}^0$) events in M_{bc} and $M(K\pi)$.

797 Continuum background

798 The procedure adopted to model the continuum background is the same used for the
 799 $B \rightarrow \Lambda_c$ decays, but in this case the available statistics is enough to perform the scaling
 800 with all the selection cuts also in the case of the two-dimensional fit (not removing the
 801 continuum suppression).

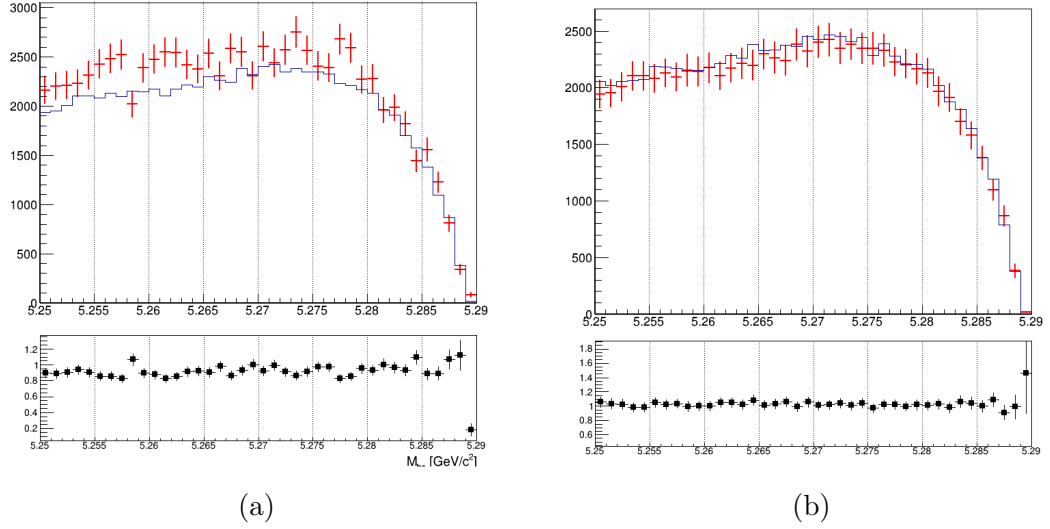


Figure (7.4) In Fig. 7.4a M_{bc} distributions of the MC (scaled) off-resonance sample (in red) and on-resonance (in blue). In Fig. 7.4b M_{bc} distributions of the corrected scaled off-resonance and on-resonance MC continuum.

For each bin a correction factor is calculated, in order to have a reasonable match with the expected continuum background. Fig. 7.4b shows the applied correction on an independent MC sample. As in the case of $B \rightarrow \Lambda_c$ analysis, then the resulting M_{bc} distribution is fitted with a Novosibirsk function, whereas the D^0 mass distribution is fitted with a sum of first order Chebyshev polynomial and the sum of three gaussians used to describe the signal peak (as shown in Fig. 7.5). The fraction of events in the peak is the same in on- and off-resonance MC. This method is applied also to scale the off-resonance data.

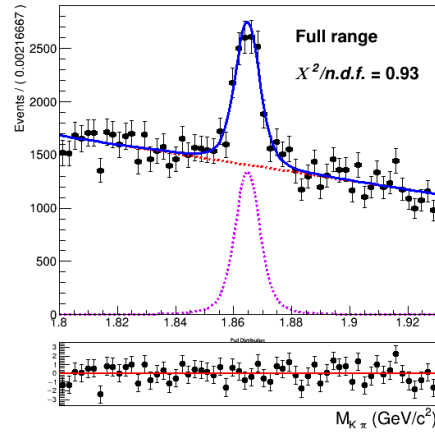


Figure (7.5) D^0 mass fit of scaled off-resonance Monte Carlo

7.1.5 2D Fit on Monte Carlo simulated data

As in the $B \rightarrow \Lambda_c$ study, five streams of Monte Carlo simulated data have been used to get values for the shaping parameters for the individual components described in the previous section and the fit model is tested on an independent stream.

For the six fits, the same conditions were applied to the widths:

- σ_{G1} : the width of the wider of the three Gaussian functions in $\rho_G(M(K\pi))$
- σ_{CB} parameter for the Crystal Ball describing the signal
- the σ_{CB} parameter for the Crystal Ball describing the generic background is expressed as function of the signal σ_{CB} with a ratio fixed from the MC.

For the fits on Monte Carlo simulated data the crossfeed normalization is expressed as ratio of its contribution and the misreconstructed signal as found in MC and kept fixed (to see how the parametrized form impacts a fit on data is then performed). Exemplary, the distributions of stream 0 overlaid by the fitted PDF are depicted in Fig. 7.6

In Table 7.1 the yields for reconstructed and misreconstructed signal are listed for each stream.

stream	0	1	2	3	4	5
NrecSig	56986 ± 400	57766 ± 437	55607 ± 426	57068 ± 372	58385 ± 369	57501 ± 437
NmisSig	31453 ± 321	30513 ± 350	32580 ± 350	33340 ± 399	29966 ± 390	32012 ± 355

Table (7.1) reconstructed and misreconstructed signal yields obtained fitting 6 independent streams

To be sure that the PDFs enables us to extract the signal yield in an unbiased way, the sum of reconstructed and misreconstructed signal yields, i.e. total signal, from the fits are compared to the true values of each stream (Table 7.2). There are quite some differences between the fitted signal yield and the true values in individual streams. However, all these deviations are within statistical expectations.

streams	fit	MC truth	fit - MC truth	
stream 0	88439 ± 340	88144	+ 295	(+0.33%)
stream 1	88279 ± 361	88551	-272	(- 0.31%)
stream 2	88187 ± 360	88487	-300	(- 0.34%)
stream 3	90408 ± 372	90149	+ 259	(+ 0.29%)
stream 4	88351 ± 383	87981	+ 370	(+ 0.42%)
stream 5	89513 ± 366	89710	-197	(- 0.22%)
sum	533177	533022	+155	(+0.03%)

Table (7.2) Comparison of fitted and truth-matched total signal events in each stream.

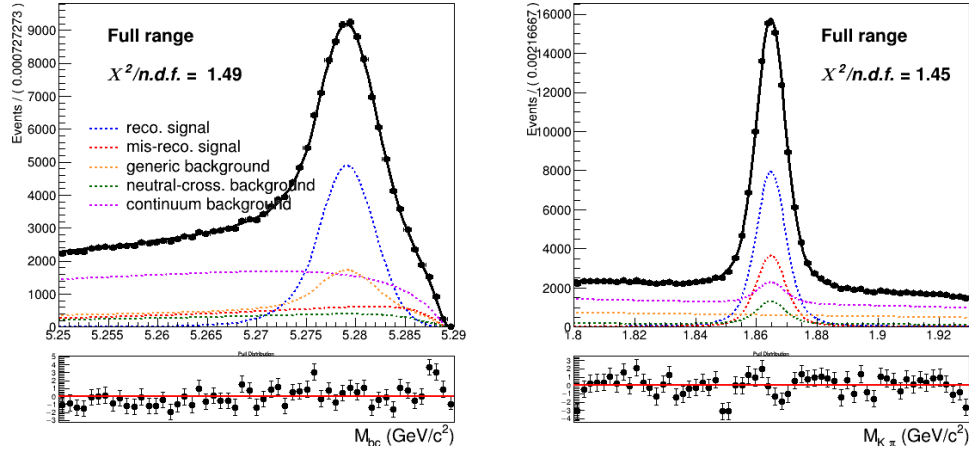


Figure (7.6) Two dimensional fit on stream 0 Monte Carlo simulated data.

7.1.6 2D Fit on data

After obtaining the model for the continuum background scaling and correcting the M_{bc} distribution of the off-resonance data, the model tested on Monte Carlo simulated data is applied on data with same free parameters and yields. Fig. 7.7 shows the projections of the two dimensional fit (see Appendix ?? for the projections of signal regions and sidebands).

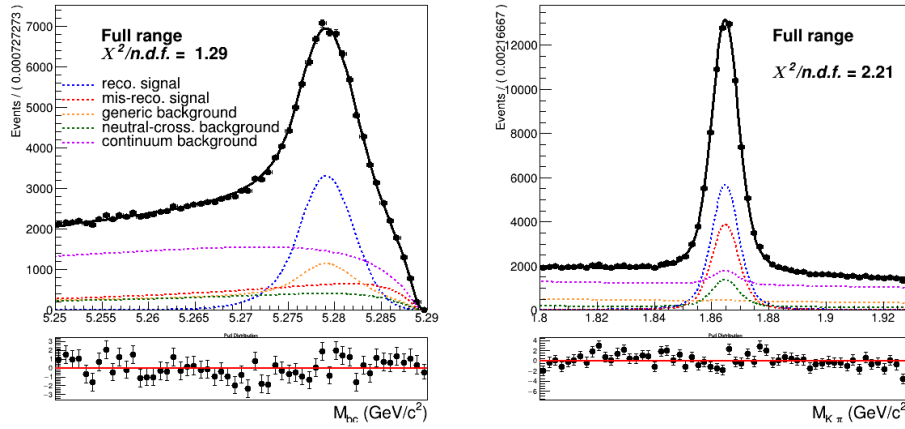


Figure (7.7) Two dimensional fit on Data (same conditions applied as in Fig. 7.6)

Yields for the reconstructed and misreconstructed signal and for generic background are obtained from the fit:

NrecSig	35629 ± 368
NmisSig	24425 ± 311
Generic	24596 ± 407

ratio	MC	DATA
NmisSig/NrecSig	0.56 ± 0.01	0.68 ± 0.01
NmisSig/Generic	0.90 ± 0.02	0.99 ± 0.02
Generic/NrecSig	0.62 ± 0.01	0.69 ± 0.02

Table (7.3) Comparison of ratios of yields from the two dimensional fits on Monte Carlo simulated data and on Data.

841 The total normalization from the fit is 174230 ± 407 (to be compared with the total data
842 events: 173964).

843

844 The crossfeed normalization was parametrized in the form described by Eq. (3.7) in
845 the case of the 2D fit shown above. The signal reconstruction efficiency that enters the
846 formula to estimate the true number of signal events ($N_{sig} = N_{recSig}/\epsilon$) is now the signal
847 reconstruction efficiency on data: ϵ_{data} . It can be estimated scaling the one found on
848 Monte Carlo by a correction factor that takes into account the different FEI efficiency
849 on data and the signal-side reconstruction corrected for the different PID efficiency (see
850 ??-??), $\epsilon_{data} = \epsilon_{MC} \cdot c_{FEI} \cdot c_{D^0} = (0.216 \pm 0.016)\%$.

851

852 where $c_{FEI} = 0.810^{+0.013}_{-0.012} \pm 0.054$ is the correction factor for the FEI efficiency determined
853 in [5]

854 whereas the factor c_{D^0} is the PID correction reported in ??.

855 Fig. 7.8 shows this second fit.

ratio	MC	DATA
NmisSig/NrecSig	0.56 ± 0.01	0.66 ± 0.01
NmisSig/Generic	0.90 ± 0.02	0.93 ± 0.02
Generic/NrecSig	0.62 ± 0.01	0.71 ± 0.02

Table (7.4) Comparison of ratios of yields from the two dimensional fits on Monte Carlo simulated data and on Data (from the fit shown in Fig. 7.8).

In the following table yields for the reconstructed and misreconstructed signal and for generic background obtained from this second fit are reported:

NrecSig	36553 ± 360
NmisSig	24115 ± 283
Generic	25900 ± 409

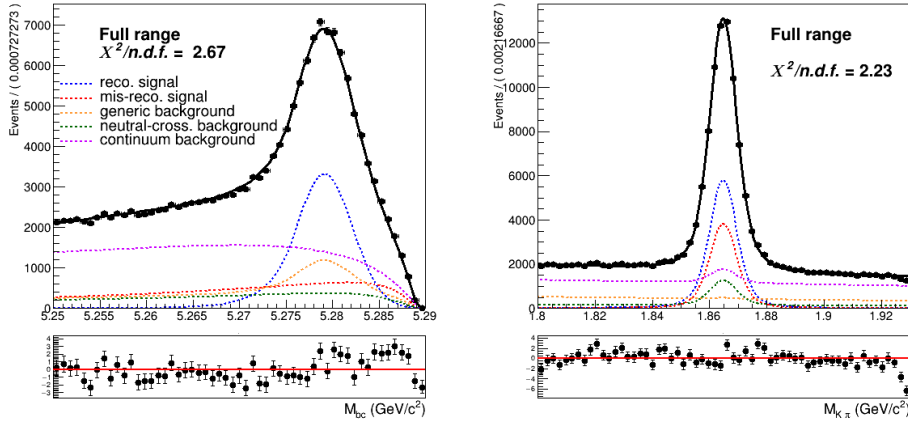


Figure (7.8) Two dimensional fit on Data with parametrized normalization of crossfeed background

860 Appendices

861 **.0.7** $B^- \rightarrow \Lambda_c^+$ decays: additional plots

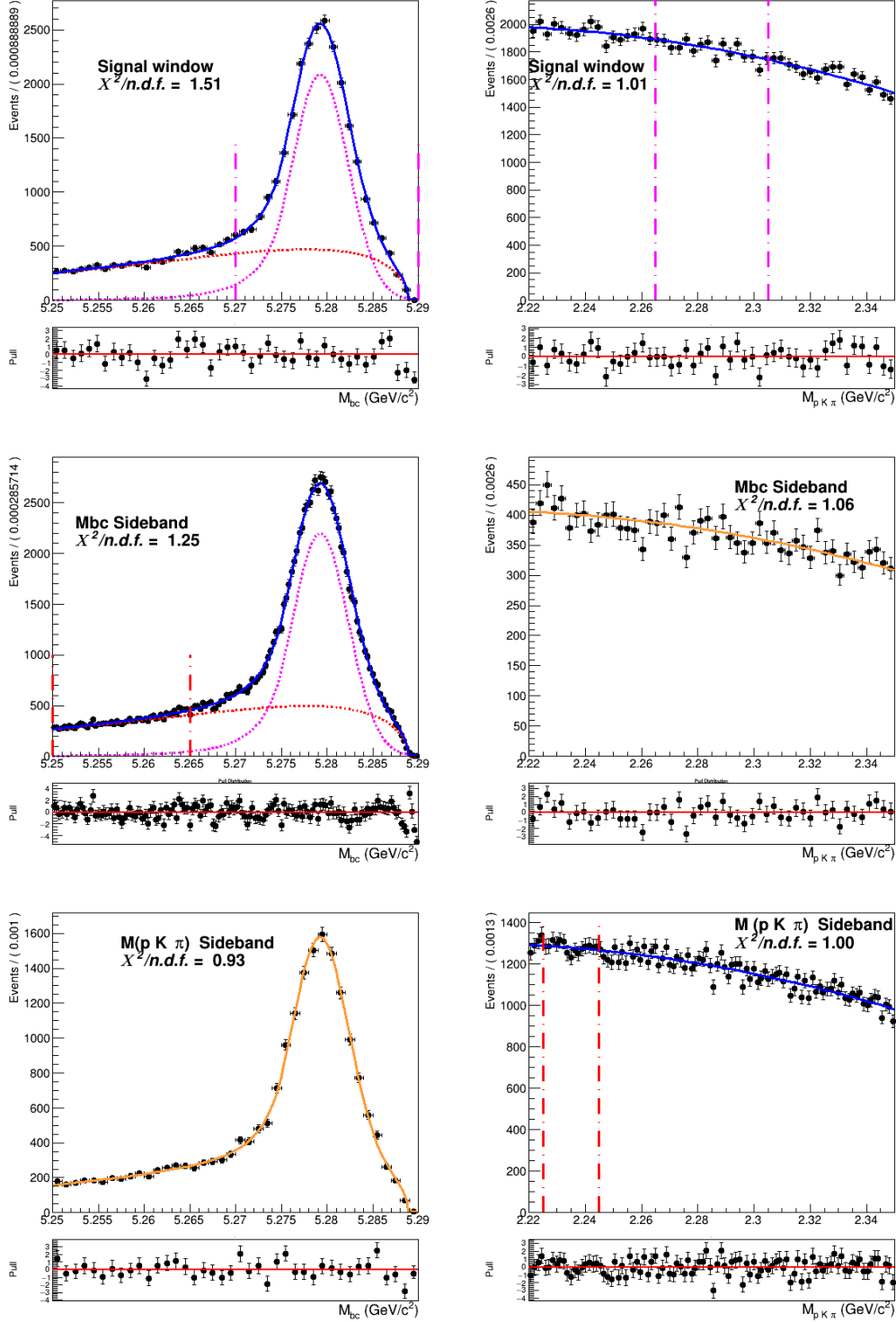


Figure (9) Signal region and sidebands of the two dimensional fit of generic background shown in ??

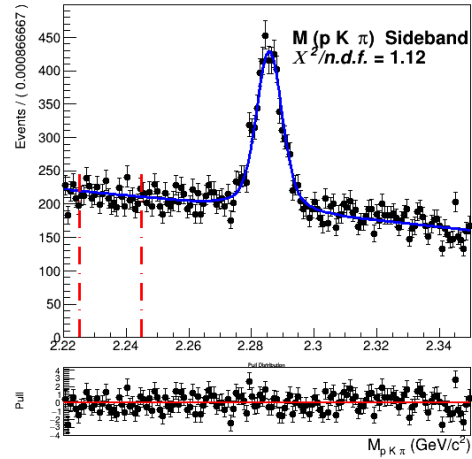
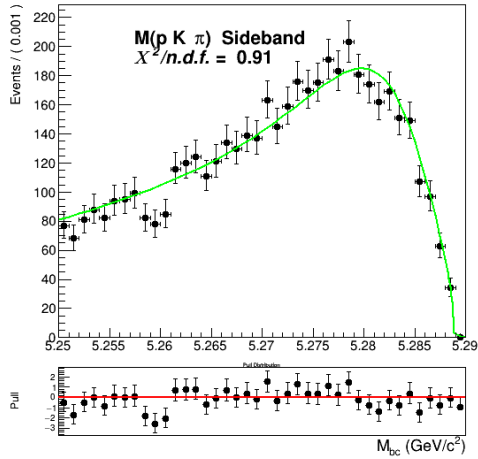
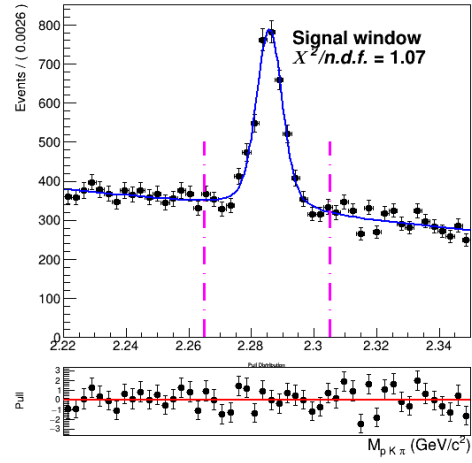
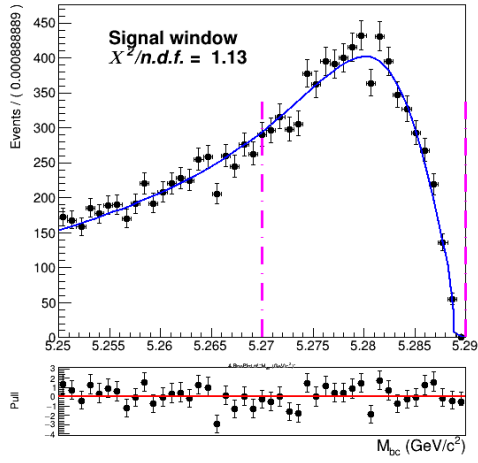
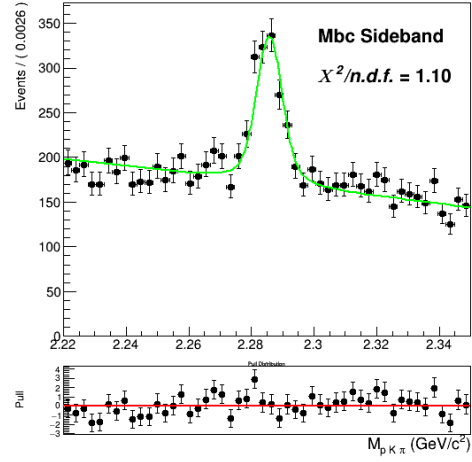
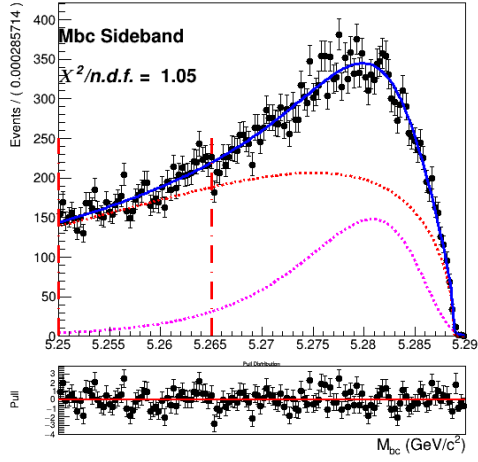


Figure (10) Signal region and sidebands of the two dimensional fit of crossfeed background after parametrization

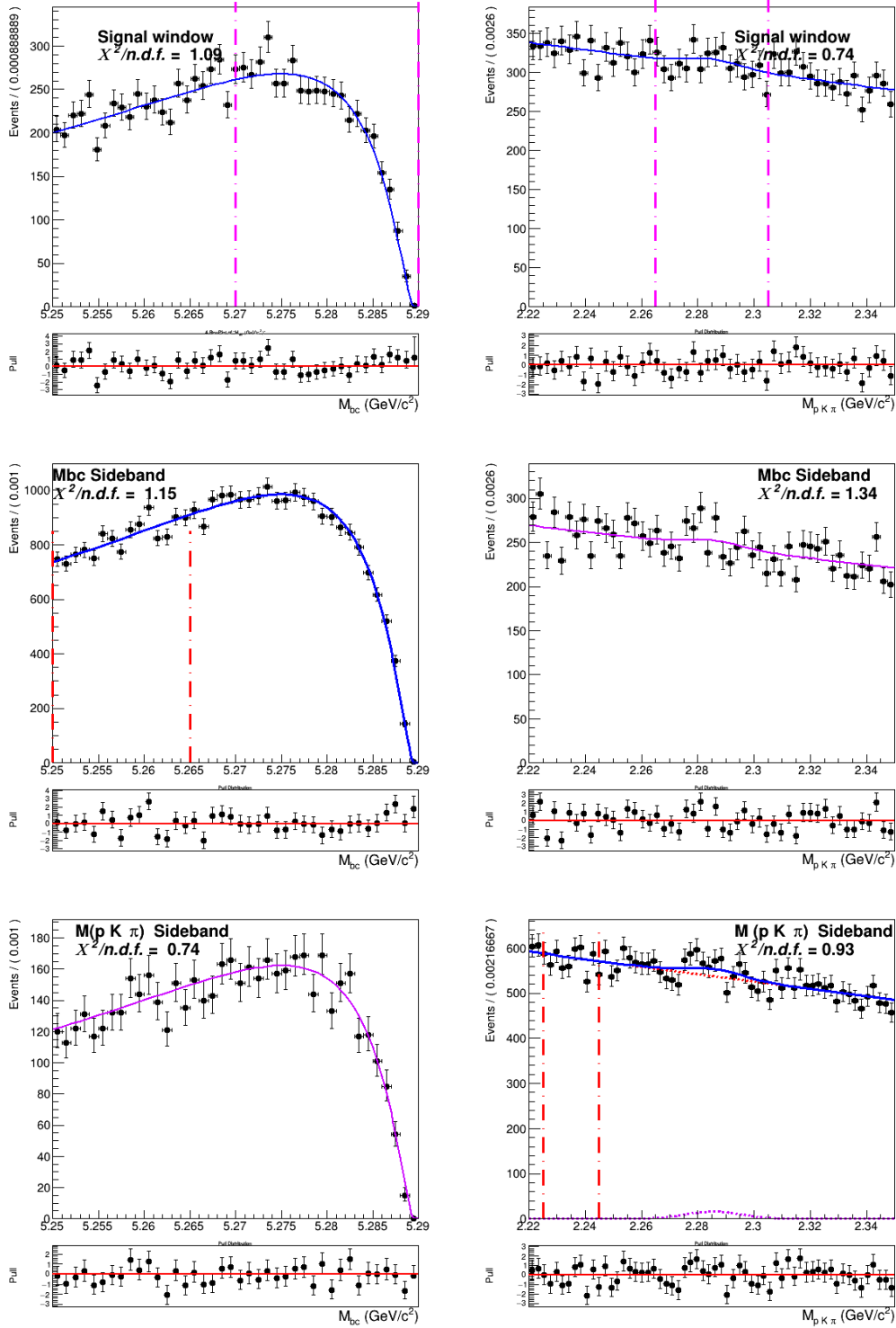


Figure (11) Signal region and sidebands of the two dimensional fit of continuum back-ground shown in Fig. 3.22

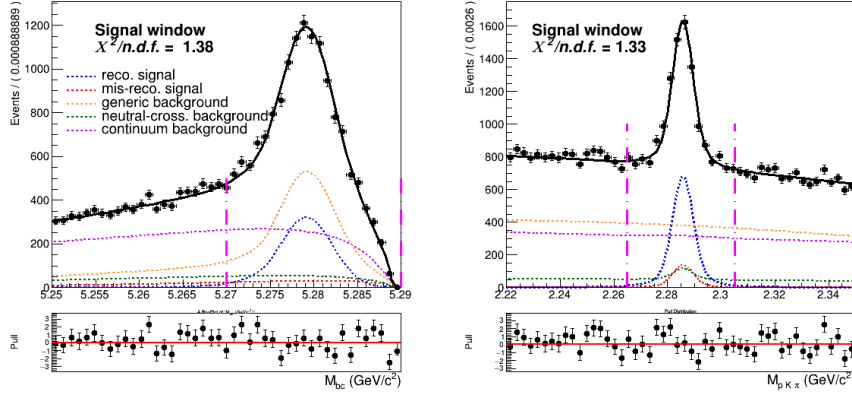


Figure (12) Signal region ($2.22 < M(pK\pi) < 2.35 \text{ GeV}/c^2$ and $5.27 < M_{bc} < 5.29 \text{ GeV}/c^2$) projections of the dimensional fit on stream 0 Monte Carlo simulated data.

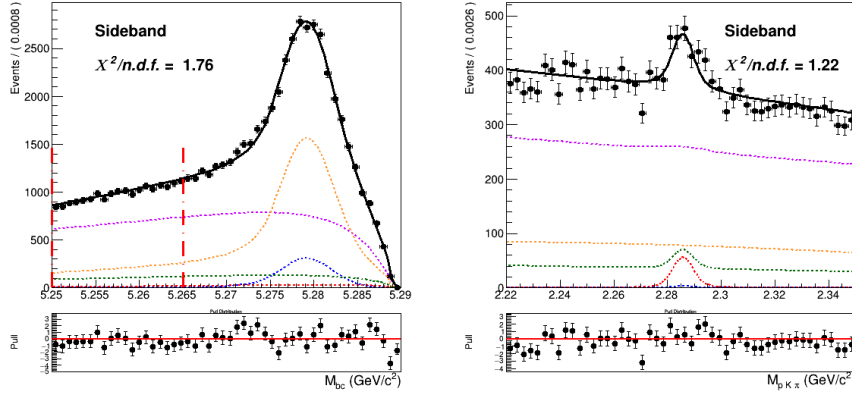


Figure (13) Sideband region of $5.25 < M_{bc} < 5.265 \text{ GeV}/c^2$ projection of the two dimensional fit on stream 0 Monte Carlo simulated data.

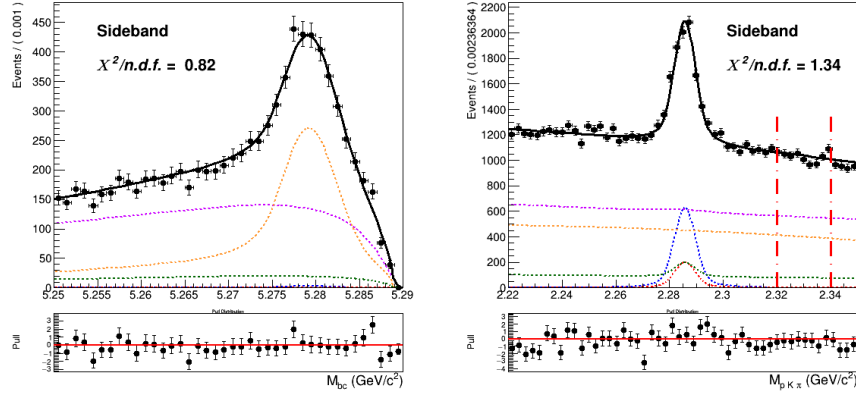


Figure (14) Sideband region of $2.22 < M(pK\pi) < 2.35 \text{ GeV}/c^2$ projection of the two dimensional fit on stream 0 Monte Carlo simulated data.

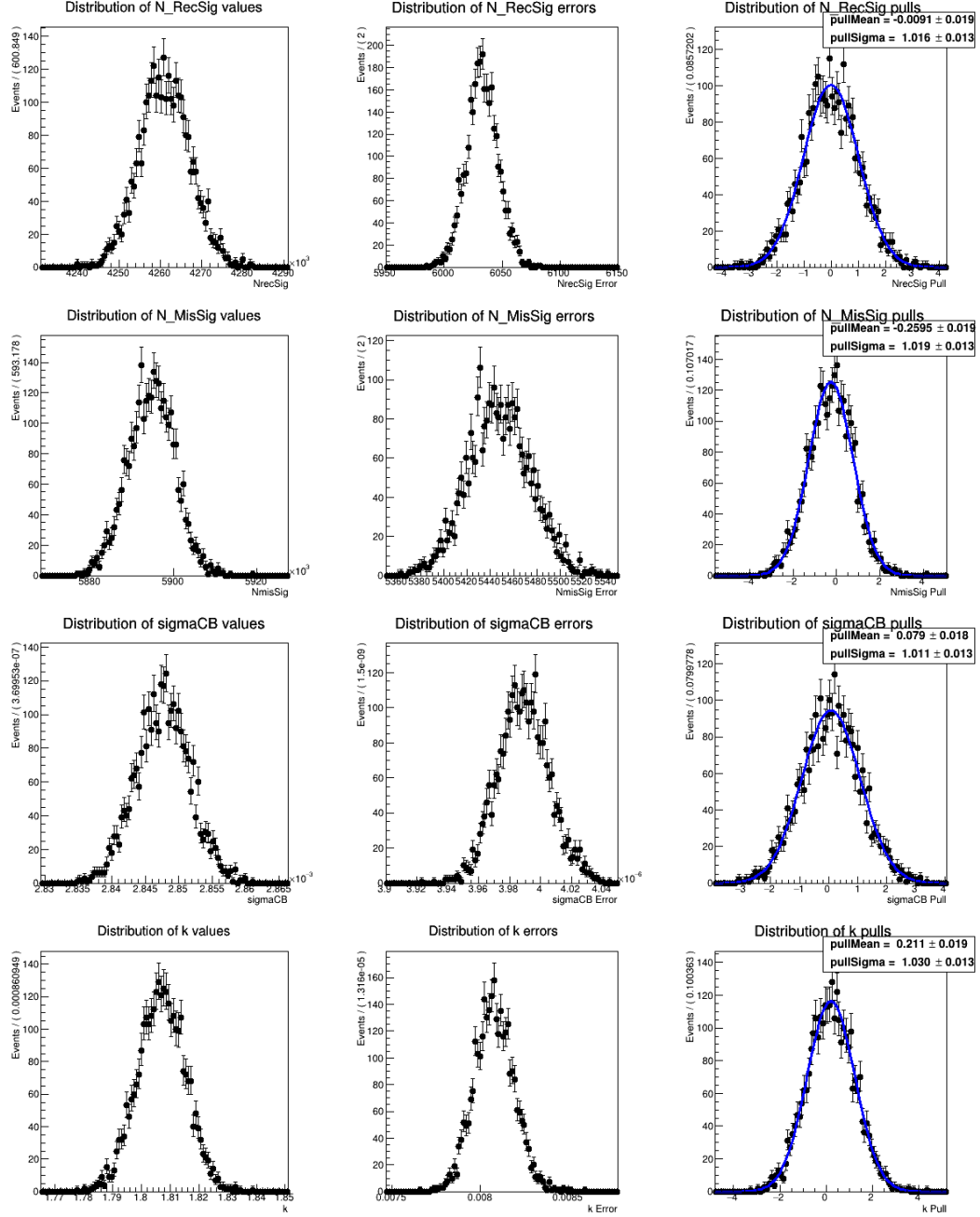


Figure (15) Toy MC study for the B_{tag} fit model described in Sec. ??

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